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MODEL-BASED PREDICTION, FAULT ESTIMATION AND OPTIMAL CONTROL:

Application on Wine Fermentation and Time-Delay Systems

Florianópolis, Santa Catarina – Brazil
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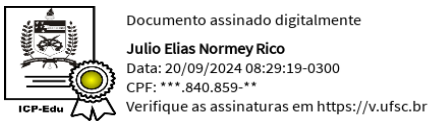
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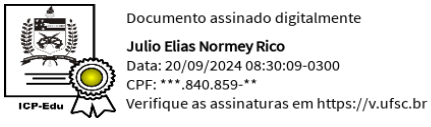
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RESUMO

Dentre os algoritmos avançados de controle de processos, o Controle Preditivo baseado em Modelo (do inglês, Model Predictive Control, MPC) é o mais utilizado na indústria. Um elemento crucial do MPC é o modelo, uma vez que é usado para prever o comportamento futuro do sistema. Independentemente do tipo de modelo, incerteza nos parâmetros e fenômenos não modelados, normalmente presentes na prática, podem deteriorar o desempenho e a confiabilidade. Uma maneira de lidar com isso é o uso de estimadores e preditores on-line para detectar e antecipar distúrbios e falhas. O objetivo desta tese é aprimorar o monitoramento e controle de processos industriais através de estimação baseada em modelos, predição e controle ótimo, com foco na fermentação do vinho tinto e preditores para sistemas com atraso. Esta tese pode ser dividida em quatro partes. Primeiro, uma estrutura Observador-Preditor para sistemas com atraso chamada Preditor de Kalman PK foi comparada com o consolidado Preditor de Smith Filtrado (do inglês, Filtered Smith Predictor, FSP). É mostrado que o PK define implicitamente um FSP e, por isso, todas as ferramentas no domínio da frequência utilizadas para analisar as propriedades de malha fechada do FSP podem ser utilizadas. Primeiramente são obtidos resultados para sistemas SISO, que são então estendidos para sistemas MIMO quadrados. Argumenta-se que o PK tem muitas vantagens sobre o FSP na fase de projeto, especialmente para sistemas não estáveis. Na segunda parte, foi proposta um Estimador de Horizonte Deslizante (do inglês, Moving Horizon Estimation, MHE), que utiliza um horizonte de medições e atuações passadas, para estimação de falhas em uma planta de produção de cana-de-açúcar. Este sistema é composto por inúmeras partes sujeitas a falhas durante a operação que foram modeladas como termos multiplicativos. A eficácia e robustez do MHE foram discutidas num estudo de simulação. As duas últimas partes da tese se concentram no processo de vinificação. Para monitoramento, foi proposta uma abordagem de estimativa do estado do Filtro de Kalman Estendido (do inglês, Extended Kalman Filter, EKF) para detecção precoce de fermentação de vinho parada e lenta. Com base no modelo, o EKF é capaz de estimar variáveis não medidas, como a biomassa. Ao mesmo tempo, esta informação é utilizada para prever a probabilidade de ocorrência de uma falha no final da fermentação. O estudo de simulação investiga a sensibilidade do processo de fermentação a diversos fatores, como parâmetros do modelo e condições iniciais, especialmente nitrogênio. O algoritmo nas simulações de Monte Carlo foi capaz de prever 95 % da fermentação problemática nos primeiros dias, fornecendo informações precisas aos enólogos durante a fermentação, o que leva a uma melhor tomada de decisões. Por fim, foi desenvolvido um problema de controle ótimo relacionado ao MPC para maceração de vinho tinto, onde o suco de uva é fermentado em contato com as cascas e sementes. A principal preocupação é gerir o ponto de drenagem considerando dois objetivos associados à produtividade aos custos. Foi encontrado um bom compromisso entre os objetivos através da aplicação de técnicas de tomada de decisão multicritério.

Palavras-chaves: Controle Preditivo. Controle Ótimo. Filtro de Kalman. Estimação de Horizonte Deslizante. Detecção de falhas. Sistemas com Atraso. Fermentação de Vinho Tinto.

ABSTRACT

Among the advanced process control algorithms, Model Predictive Control (MPC) is the most used in the industry. A crucial element of MPC is the model since it is used to predict the future behavior of the system. Regardless of the type of model, uncertainty in the parameters and unmodeled phenomena, usually present in practice, might deteriorate the performance and reliability. One way to cope with this is to use online estimators and predictors to detect and anticipate disturbances and faults. The goal of this thesis is to improve the monitoring and control of industrial processes through model-based estimation, prediction and optimal control, with a focus on red wine fermentation and predictors for time-delay systems. This thesis can be divided into four parts. First, an Observer-Predictor structure for time-delay systems called Kalman Predictor (KP) was compared to the well studied Filtered Smith Predictor (FSP). It is shown that the KP implicitly defines an FSP and, because of that, all the frequency-domain tools used to analyze the closed-loop properties of the FSP can be used. First, results for SISO systems are obtained, which are then extended for square MIMO systems. It is argued that the KP has many advantages over the FSP in the design phase, especially for non-stable systems. In the second part, a Moving Horizon Estimation (MHE), which uses a horizon of past measurements and inputs, was proposed for fault estimation in a sugarcane production plant. This system is composed of numerous parts subject to faults during operation which were modeled as multiplicative terms. The effectiveness and robustness of the MHE was discussed in a simulation study. The last two parts of the thesis are focused on the winemaking process. For monitoring, an Extended Kalman Filter (EKF) state estimation approach was proposed for early detection of stuck and sluggish wine fermentation. Based on the model, the EKF is able to estimate unmeasured variables such as biomass. At the same time, this information is used for predicting the likelihood of a fault occurring at the end of the fermentation. The simulation study investigates the sensitivity of the fermentation process to various factors such as model parameters and initial conditions, especially nitrogen. The algorithm under Monte Carlo simulations was able to predict 95% of the problematic fermentation within the first few days, providing accurate information to enologists during fermentation which leads to better decision-making. Finally, an optimal control problem related to MPC was developed for red wine maceration, where the grape juice is fermented in contact with the skins and seeds. The main concern is to manage the digging-out point considering two objectives associated with process efficiency and costs. A good compromise between the objectives was found by applying multi-criteria decision-making techniques.

Keywords: Model Predictive Control. Optimal Control. Kalman Filter. Moving Horizon Estimation. Fault Detection. Time-Delay Systems. Red Wine Fermentation.

RESUMO EXPANDIDO

Introdução

Dentre os algoritmos avançados de controle de processos, o Controle Preditivo baseado em Modelo (do inglês, Model Predictive Control, MPC) é o mais utilizado na indústria. Um elemento crucial do MPC é o modelo, uma vez que é usado para prever o comportamento futuro do sistema. Independentemente do tipo de modelo, incerteza nos parâmetros e fenômenos não modelados, normalmente presentes na prática, podem deteriorar o desempenho e a confiabilidade. Uma maneira de lidar com isso é o uso de estimadores e preditores on-line para detectar e antecipar distúrbios e falhas. Nesse trabalho, essas técnicias são aplicadas ao processo de fermentação de vinho e para sistemas com atraso.

Objetivos

O objetivo desta tese é aprimorar o monitoramento e controle de processos industriais através de estimação baseada em modelos, predição e controle ótimo, com foco na fermentação do vinho tinto e preditores para sistemas com atraso.

Metodologia

Na literatura de controle, os estados são frequentemente considerados conhecidos e determinísticos, mas normalmente este não é o caso. Como os estados são frequentemente utilizados pelo controlador, é necessário estimá-los de alguma forma. Para esta tarefa foram utilizados o Filtro de Kalman e o Estimador de horizonte deslizante (do inglês, Moving Horizon Estimation). O primeiro é um algoritmo recursivo baseado em um modelo estocástico do processo, enquanto o segundo tem a mesma formulação, mas resolve um problema de otimização a cada tempo de amostragem. Para o controle é considerado o MPC e o controle ótimo. Esta classe de algoritmo utiliza diretamente um modelo do processo para prever seu comportamento em um horizonte finito. Então, a otimização numérica é usada para calcular uma sequência de controle que minimiza a distância entre a previsão e a referência.

Resultados e Discussão

Esta tese pode ser dividida em quatro partes. Primeiro, uma estrutura Observador-Preditor para sistemas com atraso chamada Preditor de Kalman PK foi comparada com o consolidado Preditor de Smith Filtrado (do inglês, Filtered Smith Predictor, FSP). É mostrado que o PK define implicitamente um FSP e, por isso, todas as ferramentas no domínio da frequência utilizadas para analisar as propriedades de malha fechada do FSP podem ser utilizadas. Primeiramente são obtidos resultados para sistemas SISO, que são então estendidos para sistemas MIMO quadrados. Argumenta-se que o PK tem muitas vantagens sobre o FSP na fase de projeto, especialmente para sistemas não estáveis. Foram feitas comparações com outras estruturas de preditores para ilustrar as vantagens do PK.

Na segunda parte, foi proposta um Estimador de Horizonte Deslizante (do inglês, Moving Horizon Estimation, MHE), que utiliza um horizonte de medições e atuações passadas, para estimação de falhas em uma planta de produção de cana-de-açúcar. Este sistema é composto por inúmeras partes sujeitas a falhas durante a operação que foram modeladas como termos multiplicativos. A eficácia e robustez do MHE foram discutidas num estudo de simulação.

As duas últimas partes da tese se concentram no processo de vinificação. Para monitoramento, foi proposta uma abordagem de estimativa do estado do Filtro de Kalman

Estendido (do inglês, Extended Kalman Filter, EKF) para detecção precoce de fermentação de vinho parada e lenta. Com base no modelo, o EKF é capaz de estimar variáveis não medidas, como a biomassa. Ao mesmo tempo, esta informação é utilizada para prever a probabilidade de ocorrência de uma falha no final da fermentação. O estudo de simulação investiga a sensibilidade do processo de fermentação a diversos fatores, como parâmetros do modelo e condições iniciais, especialmente nitrogênio. O algoritmo nas simulações de Monte Carlo foi capaz de prever 95 % da fermentação problemática nos primeiros dias, fornecendo informações precisas aos enólogos durante a fermentação, o que leva a uma melhor tomada de decisões.
Por fim, foi desenvolvido um problema de controle ótimo relacionado ao MPC para maceração de vinho tinto, onde o suco de uva é fermentado em contato com as cascas e sementes. A maceração na produção de vinho é realizada até o mosto atingir uma concentração específica de açúcar (ponto de drenagem). A principal preocupação é gerir o ponto de drenagem considerando dois objetivos associados à produtividade aos custos. Duas estratégias de controle foram comparadas: nutrição livre e nutrição tradicional. Foi encontrado um bom compromisso entre os objetivos através da aplicação de técnicas de tomada de decisão multicritério. A opção preferida foi a adição de nitrogênio somente no início da fermentação e uma alta temperatura de fermentação, resultando em um ponto de drenagem com um pequeno erro.

Considerações Finais
Essa tese apresenta contribuições em preditores para sistemas com atraso, estimação de falhas e controle ótimo, com aplicação em processos industriais como geração de energia renovável e fermentação de vinho. O tema geral é o uso de algoritmos para monitoramento e operação ótima de processos industriais. A partir das análises dos resultados, são sugeridos trabalhos futuros. Os resultados na área de preditores para sistemas com atraso podem ser estendidos para sistemas MIMO gerais e sistemas não lineares. O estimador para predição de falhas em fermentação de vinho foi provado extensivamente em simulações, porém, é importante analisar o desempenho com dados reais. No caso do controle de maceração, é necessário continuar avançando no desenvolvimento do controlador para então provar o algoritmo em vinificações piloto.

Palavras-chaves: Controle Preditivo. Controle Ótimo. Filtro de Kalman. Estimação de Horizonte Deslizante. Detecção de falhas. Sistemas com Atraso. Fermentação de Vinho Tinto.

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1 INTRODUCTION

The continuous technological advances of the last decades were mainly motivated by the demands of the consumers for products and services that are increasingly customized and being delivered quickly. This made the industries develop more efficient production methods and better products while always aiming to increase their profits and improve their position in the market. Another factor to be considered is population growth, which requires an increase in the production of goods and, as a consequence, increases the demand for resources and raises important environmental and social issues.

Among the technological advances observed, the process control area has aroused strong interest both from industry and from academia due to its ability to help processes meet these new challenges imposed by the consumer market. Advanced control and optimization techniques have been applied in the process industry since the 70s, mainly in some particular sectors such as the oil and gas industry (1). Model Predictive Control (MPC) is the class of advanced control algorithms that has the greatest impact in the industrial environment where practically all refineries in the world use MPC in their units with good results (2). Thanks to the success, many other industrial sectors have adopted this methodology or are analyzing the possibility of adopting it, as in the case of the renewable energy industry and the food and beverage industry which are of particular interest of this thesis.

MPC is not a specific control strategy but it is the name given to a large set of control methods with some common ideas such as the prediction through a process model and a moving horizon (3). The petrochemical industry has enabled the development of practical knowledge and validation of the theoretical developments of MPC algorithms since their emergence and continues to contribute to the evolution of these controllers. The great acceptance in the industry caught the attention of the academy and in the 80s the first academic works appeared. The main reason for the large impact of MPC in the industry is, perhaps, the ability of the MPC to: be easily applied to Multiple-Input Multiple-Output (MIMO) systems; directly include system feedback and feed-forward control in its formulation; include input and output constraints in the optimization; intrinsically compensate transport delays.

MPC relies heavily on model-based strategies, making the quality of the model and the availability of process variable measurements crucial. An integral component of an advanced control and optimization structure is the estimator, which serves multiple essential functions. Firstly, in many cases, certain states of the system necessary for control, monitoring, or prediction may not be directly measurable. In such instances, employing an estimator becomes imperative. This approach can prove cost-effective by either reducing the number of sensors or substituting expensive measurements with more economical alternatives. Furthermore, even when all required variables are measurable, sensor data

is often corrupted by noise, necessitating filtering. Additionally, the estimator plays a vital role in detecting disturbances and uncertainties. This information is invaluable for updating the process model used in optimization or MPC, thereby enhancing overall system performance.

For many years, two optimal estimators have received attention: the Kalman Filter (KF) and the Moving Horizon Estimator (MHE). KF has a stochastic formulation, can be either continuous or discrete, can estimate states and parameters, allows sensor fusion, is recursive and is not particularly hard to implement. There are extensions to the KF for nonlinear systems, such as the Extended Kalman Filter (EKF) or the Unscented Kalman Filter (UKF) (4). Nevertheless, it has been shown that both of them might fail by not considering process constraints and could lead to worse estimation performance (5). On the other hand, MHE considers a backward estimation horizon comprised of past input and output information and, with this data, it is able to estimate the current and past states based on a model of the plant (6). Since the MHE uses multiple past measurements at each sampling time, it is expected to outperform a recursive observer by considering more information in the estimation. This estimator is formulated so to minimize the quadratic estimation error, thereby, it solves an optimization problem similar to the KF. The main difference is that the KF is an unconstrained and recursive optimization problem while the MHE allows to directly consider constraints and more complex models such as nonlinear differential-algebraic systems (DAE). The downside is that MHE needs to solve an optimization problem at each sampling time which can be time consuming. Nevertheless, due to their similarities, MHE and MPC often use the same numerical software which might facilitate the implementation when both are used.

Another important problem in many industrial processes is the negative effect of the dead time in the closed-loop behavior. Because of the implicit nature of the predictor in MPC, there are not many works that pay attention to the impact of the predictor structure on the closed-loop properties in the MPC context. In some recent works, the authors have shown that the predictor design is important in the robustness behavior of the whole control strategy (7). The study separates the prediction and the control computation stage of the MPC. In the former, a model of the plant is used to predict the future behavior of the process where the delays are compensated. In the latter, the predictions are used to compute the control action based on the optimization of a desired cost function. The way these algorithms compensate the dead time affect the final results even if the control computation stage is not modified.

Since the control and prediction stages can be separated, different predictor structures can be used in the MPC. It was shown that structures based on the Filtered Smith predictor (FSP) can be advantageous when trying to achieve a good compromise between performance and robustness (8). Although these strategies offer better results than the original MPC strategies based on optimal predictors, the tuning simplicity can be lost on some complex cases, for example, when the order of the system increases or when some

model disturbances have to be taken into account in the MIMO case. Thus, there are some research opportunities to develop different approaches for the predictor in dead-time compensation that can be adopted by the MPC.

Part of this work is the application of MHE for hybrid energy plants based on different primary energy sources such as biomass, biogas, solar and wind power energy. Nowadays, they are interesting alternatives to achieve more sustainable energy production. To adequately operate such type of microgrids, it is necessary to use Energy Management Systems (EMS) that have to consider future predictions of the energy generation, on-off operation of the different energy sources and multi-objective specifications. Some recent results in these types of systems show that operation and energy production can be improved when MPC is used as a control strategy for these plants (9). However, as already mentioned, closed-loop performance can deteriorate when the model used in the MPC structure is not appropriately designed or changes in process operation (e. g. faults or loss of efficiency) causing the system to work in a different operating point where the model is no longer adequate. This situation can be avoided if an appropriate estimation of the process is done and the MPC strategy is re-adapted to maintain the expected performances. As shown in Chapter 5, an MHE scheme was used to estimate faults in a simulated microgrid case study.

A portion of the thesis was supported by the Center for Research and Innovation (CRI) of Concha y Toro Winery in Talca, Chile, shown in Figure 1. There, the engineering research group focus on several aspects of red wine production including the automation and optimal control of the maceration, where the grape juice is fermented in contact with the skins and seeds. The maceration is crucial because, (a) most of the fermentation is done at the same time, (b) it extracts important chemicals responsible for the organoleptic characteristics of the final bottle, (c) it is processed on fermentation tanks in batches with variable and uncertain initial conditions, (d) it is difficult to model the chemical and physical phenomena involved on the fermentation and extraction, (e) it is a logistic challenge for the winery since it can only be done on a couple of months during harvest.

Figure 1 – Center for Research and Innovation located near Talca, Chile.



Source: Center for Research and Innovation Website.

In the maceration, the winemakers aim to optimize the wine quality, the fermentative maceration length, and the fermentation tanks productive cycle simultaneously. Maceration in high volume wine production is carried out until a specific sugar content (digging-out point) is attained, usually before the end of the alcoholic fermentation. To

achieve the desired result, the enologist has some tools available such as temperature control of the tank and nutrient addition in the form of diammonium phosphate which affects the rate of fermentation. With so much factors influencing the productivity and quality of the batch, it is a challenge to find the operational conditions which will optimize the process.

Model-based control and monitoring are seen as promising methods for improving wine fermentation since they could lower energy costs and assist with tank management, improving productivity (10). At first, the use of novel sensors for density and CO_2 allows online monitoring and estimation of non-measurable variables while feedback control can take into account the variability of the must composition. But with the development of more complex models which could include, for instance, production of aromas, it would be possible to better control the organoleptic characteristics of the must with the use of optimization and advanced control.

It is clear that important research opportunities exist in theoretical and application aspects of estimation and control of industrial processes. Thus, the main objective of this thesis is to improve the monitoring and control of industrial processes through model-based estimation, prediction and optimal control, with a focus on red wine maceration and predictors for time-delay systems.

This thesis is organized as a compendium of publications resulting from two main lines of research conducted in parallel: predictors for time-delay systems based on the Kalman filter, and estimation and control of wine fermentation. With the exception of Chapter 2, where some preliminaries topics and methods are reviewed, each chapter corresponds to a published article which are presented out of chronological order to enhance the document's flow:

a) Chapter 3: D. M. Lima, B. M. Lima, and J. E. Normey-Rico, "A predictor for dead-time systems based on the Kalman Filter for improved disturbance rejection and robustness," *J. Process Control*, 2021 (11) .

b) Chapter 4: D. M. Lima, B. M. Lima, and J. E. Normey-Rico, "A predictor for square multivariable dead-time systems with multiple delays based on the Kalman filter," *J. Process Control*, 2023 (12) .

c) Chapter 5: B. M. Lima, M. M. Morato, P. R. C. Mendes, and J. E. Normey-Rico, "Moving Horizon Estimation of Faults in Renewable Microgrids," *IFAC-PapersOnLine*, 2019 (13).

d) Chapter 6: B. M. Lima, R. Luna, D. M. Lima, J. E. Normey-Rico, and J. R. Pérez-Correa, "Advancing Wine Fermentation: Extended Kalman Filter for Early Fault Detection", *Food and Bioproducts Processing*, 2024 (**submitted**).

e) Chapter 7: R. Luna, B. M. Lima, J. Cuevas-Valenzuela, J. E. Normey-Rico, and J. R. Pérez-Correa, "Optimal Control Applied to Oenological Management of Red Wine Fermentative Macerations," *Fermentation*, 2021 (14).

Finally, Chapter 8 concludes with comments on the results as a whole and suggestions of future works.

2 PRELIMINARIES AND LITERATURE REVIEW

This chapter briefly review concepts on red wine production, optimal estimation and MPC. Before that, some definitions and notation which will be used are presented.

The continuous dynamic nonlinear model used in this work is given by

$$\begin{aligned} \dot{x} &= f_c(x,\, u,\, q,\, p), \\ y &= h(x,\, u,\, q,\, p), \end{aligned}$$

which has dimensions $x \in \mathbb{R}^{n_x}$ for the states, $u \in \mathbb{R}^{n_u}$ for the control inputs, $q \in \mathbb{R}^{n_q}$ for the disturbances, $p \in \mathbb{R}^{n_p}$ for the parameters and $y \in \mathbb{R}^{n_y}$ for the measured outputs. $f_c(\cdot)$ and $h(\cdot)$ represent nonlinear functions for the states and outputs, respectively. This model can be transformed to discrete time by any method such as Runge-Kutta or collocation, where the result is given by:

$$\begin{aligned} x^+ &= f(x,\, u,\, q,\, p), \\ y &= h(x,\, u,\, q,\, p), \end{aligned} \tag{1}$$

with x^+ denoting the state at the next time step. For estimation, noises can be included in the state and output equations as in

$$\begin{aligned} x^+ &= f(x,\, u,\, q,\, p) + w, \\ y &= h(x,\, u,\, q,\, p) + v, \end{aligned} \tag{2}$$

with $w \in \mathbb{R}^{n_x}$ and $v \in \mathbb{R}^{n_y}$ being random variables modeled by zero-mean Gaussian noises with covariance matrices Q and R, respectively.

For variables at a specific time k, the notations $x[k]$ and x_k can be used. In the estimation context, $\hat{x}$ represents the estimate of x and $\hat{x}_{k|k\text{-}1}$ represents the estimate of x_k given the information at time k-1.

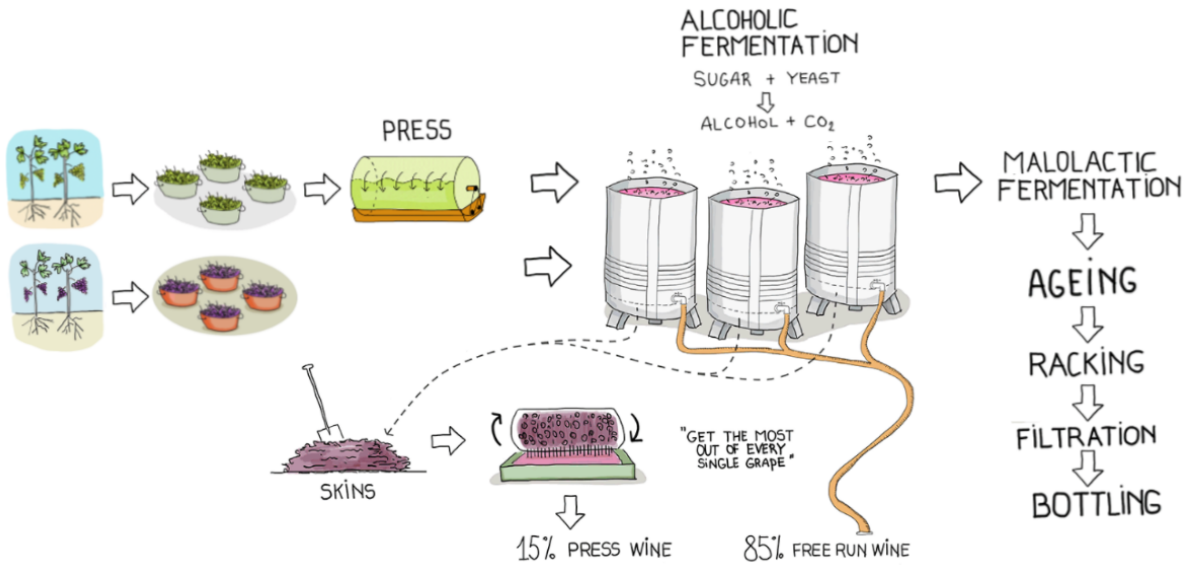
To avoid too many equations in optimization problems, constraints on the variables are abbreviated. The term $x \in \mathbb{X}$ is a general notation for a vector x constrained in a subset of $\mathbb{X} \subseteq \mathbb{R}^{n_x}$. In the simplest case, $\mathbb{X}$ defines box constraints on each element of x such that $x_{\text{lb}} \leq x \leq x_{\text{ub}}$. But this notation also includes a general polytope defined as $A_x x \leq b_x$ or ellipsoids $x'\, P\, x \leq 1$ with P being positive definite.

Squared Euclidean norms commonly found on objective functions are abbreviated as $\|z\|_P^2 = z'\, P\, z$ where P is usually a diagonal matrix.

2.1 RED WINE PRODUCTION

Winemaking requires several steps, as summarized in Figure 2. At first, the grapes are monitored for sugar concentration which indicates the ideal time for harvesting. Sugar content is important because this is the nutrient which is transformed in alcohol in the

Figure 2 – Production steps in wine production.



Source: Adapted from Wine Folly Website.

fermentation. If the fruit is not ripe enough, the final wine will not have enough alcohol. Then, the fruits are harvested mechanically or by hand and brought to the cellar.

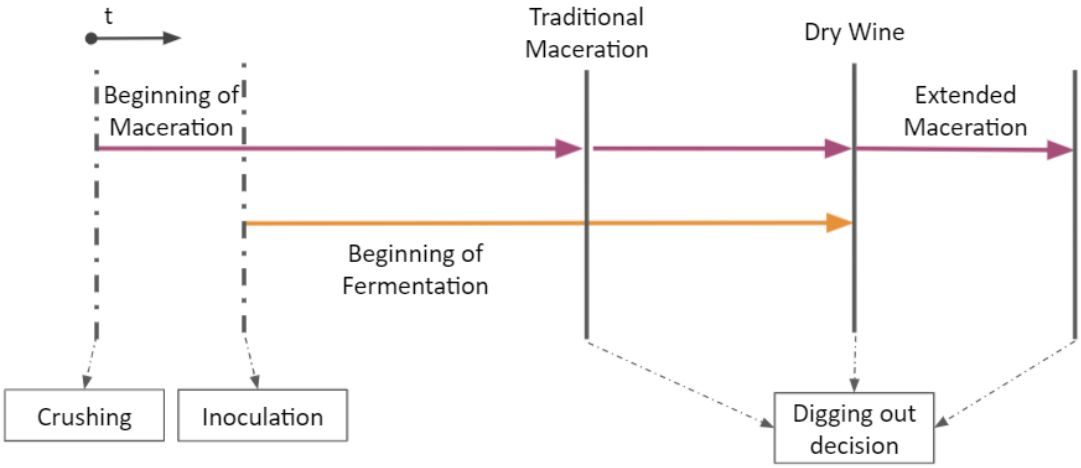
At this point, a machine separates the grapes from the leaves and stems. For white wines, the juice is pressed from the grape and the solids are discarded. Then, the fermentation proceeds without the skins and seeds. Meanwhile, for red wines, the grapes are only lightly crushed to release the juice and the whole content is transferred to the fermentation tanks. The solids are important for red wines since they carry chemicals responsible for the color and mouthfeel. The red wine currently is the focus of the CRI since it is the most produced and consumed, specially the Cabernet Sauvignon variety.

The vintage, the period of wine production, consists of a couple of months of the year when the fruits become ideally ripe for harvesting and fermentation. In this period, the cellars where the grapes are processed operate in full capacity, including at nights and weekends. The vintage occurs around the Autumn season, so in the southern hemisphere it starts in the first semester whereas in the northern it starts in the second. Since the vintage period is limited, it is important to efficiently use the resources available. One important bottleneck related to this thesis is the amount of tanks available on the cellars. This is one of the motivations in automating the maceration process to increase productivity.

Here, it is important to differentiate between two processes visualized in Figure 3: the maceration and alcoholic fermentation. Maceration is the contact of the solids and the juice, whereas the fermentation is the transformation of sugar in alcohol. The maceration starts as soon as the grapes are crushed and transferred to the tanks. Then, some time passes while samples of the juice are analyzed. The fermentation starts when the tank is inoculated with yeast, usually *Saccharomyces cerevisiae*, and ends when there is no more sugar left, i. e. the wine is dry. This takes around 7 days for red wines while it takes longer

white, since the latter are fermented in lower temperatures. Meanwhile, the maceration ends when the must is pressed and the solids are separated from the juice, also called the *digging-out* point.

Figure 3 – Fermentation vs. maceration and digging-out point.



Source: the author.

The maceration duration is an important decision to be made. For high valued wines, the maceration can be extended so that there is a higher extraction of chemicals presented on the skins and seeds. Meanwhile, in high volume wines, the maceration is ended before the fermentation so that the tank can be used in another batch. Here, a trade-off between quality and productivity is sometimes present.

These cited chemicals found in high concentration in red grapes belong to a group called polyphenols. They have a big influence on the color, aroma and mouthfeel of the wine. Because of that, it is important to properly manage their extraction during the red wine production since they affect the final quality and price of the bottle.

The focus of this thesis ends in the maceration. Nevertheless, there are other steps necessary in wine production such as the malolactic fermentation, ageing, racking, filtration and bottling.

The wine fermentation is a batch bioprocess which has specific challenges such as: high variability of initial conditions, complex biological systems which are difficult to model, few measurable variables, nonlinear dynamics, time-varying parameters, among others (15). Automation and control can provide consistency, productivity, better monitoring and fault detection.

For monitoring, most industrial cellars have online temperature sensors but density, which is proportional to alcohol content, usually is manually measured by hydrometer, but online sensors are currently being tested and adopted by the CRI engineering group. Meanwhile, automatic control is not commonly found in the industry at this time. Usually, only the tank temperature is controlled in a closed-loop where the setpoint is defined by the enologist.

2.2 OPTIMAL ESTIMATION

In the control literature, the states are often assumed to be known and deterministic but usually this is not the case. In industrial processes, not all states are directly measured either because it is impossible to do so or because the sensors required are expensive. Since the states are often used by the controller, it is necessary to estimate them somehow. Beside that, it is important in estimation to deal with uncertainties. One of them is noise which is common in sensors and must be filtered. Another uncertainty is disturbance which is an unwanted input that affects the controlled variables. Another problem that appears often in the literature is parameter estimation. Parameter can be loosely defined as a variable that is considered to be constant when compared to the time-scale of the dynamic system.

Just like in control, it is interesting to perform the estimation by minimizing some sort of criteria or index. This leads to optimal estimation which is a broad term to describe techniques that have an objective function. This includes the Kalman Filter (KF) and Moving Horizon Estimation (MHE) which are reviewed in this section. They have a stochastic framework where the models are disturbed by random variables and a quadratic objective function which minimizes the prediction error.

In general, estimation and system identification are closely related concepts where the first is basically an online version of the second. Both of them attempt to infer information about the system by observing the measured output and most of them are based on a least-squares formulation (16). The main difference is that system identification is also concerned with the choice of model to be used on the problem while in estimation and control the model structure (equations and parameters) are usually given.

2.2.1 Kalman Filter and its Extensions

Kalman filter (KF), also known as linear quadratic estimator, is a class of recursive algorithms that uses the past measurements to estimate the current state based on statistical knowledge. The filter was named after Rudolf E. Kalman which published important results in this field (17). This algorithm has been widely successful in the industry with applications in state and parameter estimation, sensor fusion, signal processing and much more (18).

To understand the algorithm, it is necessary to have some background in statistics. First, consider a linear state-space model with disturbances[1] given by

$$\begin{aligned} x^+ &= A\,x + B\,u + w \\ y &= C\,x + v \end{aligned} \tag{3}$$

with w and v being zero-mean Gaussian noise. Gaussian noise are represented as a mean

[1] The modeling of the disturbance is omitted for brevity, but it can be easily extended for this case.

and a covariance matrix with the notation being $w \sim \mathcal{N}(0,\, Q)$ and $v \sim \mathcal{N}(0,\, R)$. Usually Q and R are modeled by covariance matrices that are diagonal, for example:

$$Q = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}, \quad R = \begin{bmatrix} \delta_1^2 & 0 \\ 0 & \delta_2^2 \end{bmatrix} \tag{4}$$

with σ_i and δ_i being the standard deviation of each variable. In $\mathbb{R}^{n_y}$, this matrix defines an ellipsoid ($v'\, R^{-1}\, v \leq 1$) with its semi-axes having lengths equal to δ. The noise v has a 68% chance to be inside this region.

In an optimization point of view, the Kalman filter solves the following unconstrained weighted least squares problem (6):

$$V_{\text{KF}} = \min_{\hat{x}(0),\, w_i} \|\hat{x}_0 - \bar{x}_0\|^{\,2}_{P_0^{-1}} + \sum_{i=0}^{k-1} \|w_i\|^2_{Q^{-1}} + \sum_{i=0}^{k} \|v_i\|^2_{R^{-1}} \tag{5}$$

which minimizes the estimated disturbance sequence w_i and the prediction error $v_i = y_i\text{-}C\hat{x}_i$, according to the positive definite matrices Q, R and P_0, and an initial estimate $\bar{x}_0$.

For unconstrained linear systems, this optimization problem has a recursive solution which will be described here. The filter is initialized by two components: the expected mean $\hat{x}_0$ and the expected covariance $P_{0|0}$. This matrix represents the confidence in the state estimate since it defines an ellipsoidal region around $\hat{x}_0$. After the filter is initialized, it performs two steps at each sampling time: the prediction and update. At first, an *a priori* state estimate is calculated by integrating the system. At the same time, the state noise is propagated through the model resulting in an *a priori* covariance $P_{k|k-1}$. The prediction equations are given by:

$$\begin{aligned} \hat{x}_{k|k-1} &= A\, \hat{x}_{k-1|k-1} + B\, u_{k-1} \\ P_{k|k-1} &= A\, P_{k-1|k-1}\, A' + Q \end{aligned} \tag{6}$$

Notice that in the second equation, $P_{k-1|k-1}$ is propagated through the model by where Q is added. Since Q is positive definite, $P_{k|k-1}$ will always be "bigger" than $P_{k-1|k-1}$ which means that the estimate $\hat{x}_{k|k-1}$ becomes more uncertain.

Next, it is the update step. First, the Kalman gain K_k is calculated using P, C and R. Then, using K, the estimates $\hat{x}_{k|k-1}$ and $P_{k|k-1}$ are updated according to the following equations:

$$\begin{aligned} K_k &= P_{k|k-1}\, C' \left(C\, P_{k|k-1}\, C' + R\right)^{-1} \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k \left(y_k - C\, \hat{x}_{k|k-1}\right) \\ P_{k|k} &= (I - K_k\, C)\; P_{k|k-1} \end{aligned} \tag{7}$$

It is interesting to notice that $P_{k|k}$ is updated by subtracting the term $K_k C P_{k|k-1}$ which makes it "smaller". This means that the uncertainty of $\hat{x}_{k|k}$ decreases.

The KF have some interesting properties. If w and v are zero-mean, uncorrelated and Gaussian, then the KF is the best linear estimator for the optimization problem defined in (5) (19). If the linear system and the noise covariance matrices are time-invariant, the filter converges to a "steady-state" gain K_∞ which can be calculated using any standard software. Then, the update equation is given by

$$\begin{aligned} \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_\infty \left(y_k - C\,\hat{x}_{k|k-1}\right) \\ &= (I - K_\infty\, C)\ \hat{x}_{k|k-1} + K_\infty\, y_k \end{aligned} \tag{8}$$

The underlying algorithm solves a discrete algebraic Riccati equation that results in a pair $(K_\infty,\ P_\infty)$, as explained in more details in Simon (19, Chapter 7.3). The steady-state KF is not optimal because it does not use the optimal gain at each time step, although its performance approaches the ideal when $k \to \infty$. There are systems where the algorithm does not converge, which means that there is no solution for K_∞. Nevertheless, if the system is detectable, there is at least one solution given by the pair $(K_\infty,\ P_\infty)$.

The resulting filter is very similar to the Luenberger observer since both have a constant gain and are linear. They differ, though, in the design and tuning (20). The Luenberger observer uses pole placement to design a stable filter for the linear system. It is easy to make the connection between frequency response of the filter and the chosen poles but it is not clear how each pole affects each state. Meanwhile, the steady-state KF has tuning parameters for each noise variable which might facilitate the design for large systems.

The KF has some adaptations for nonlinear systems with the Extended Kalman Filter (EKF) being one of them (21). The challenge in nonlinear systems is that there is no easy way to propagate the covariance matrix P through the model (22). The EKF solves this using the nonlinear model in Eq. 1 for the estimate of the mean $\hat{x}$ and using a linearized model to calculate P and the Kalman gain K at each sampling time. The algorithm is very similar to the linear case with a few modifications.

First, it is necessary to initialize the filter in the same way as done in the linear KF. Then, it is necessary to linearize the system in Eq. 1 around $\hat{x}_{k\text{-}1|k\text{-}1}$ as defined by:

$$A_{k-1} = \left.\frac{df}{dx}\right|_{\hat{x},\,u}.$$

With this information, the update equations are given by:

$$\begin{aligned} \hat{x}_{k|k-1} &= f(\hat{x}_{k-1|k-1},\ u_{k-1}) \\ P_{k|k-1} &= A_{k-1}\, P_{k-1|k-1}\, A'_{k-1} + Q \end{aligned}$$

Next, the output equation is linearized using $\hat{x}_{k|k-1}$:

$$C_k = \left.\frac{dh}{dx}\right|_{\hat{x}}.$$

Then, the update step is given by:

$$\begin{aligned} K_k &= P_{k|k-1}\, C_k' \left(C_k\, P_{k|k-1}\, C_k' + R\right)^{-1} \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k \left(y_k - h(\hat{x}_{k|k-1})\right) \\ P_{k|k} &= \left(I - K_k\, C_k\right) P_{k|k-1} \end{aligned} \tag{9}$$

Notice that the estimate of the state and the output are propagated through the nonlinear models while the gain and covariance matrix are calculated with the linearized model. Because of this, EKF might have poor performance for highly nonlinear systems (5).

EKF is a very practical solution and only recently its stability and convergence were proven. This requires strict conditions such as small initial estimation error, small noise terms and no model mismatch. Furthermore, it does not incorporate physical state constraints which may cause an estimation failure in some cases. In Haseltine; Rawlings (5), the authors show some situations where the EKF have an inadequate performance. If the nonlinear model has multiple states that satisfy the steady-state measurements, the filter might converge to the incorrect state with a poor initial guess. The state constraints can be applied to $x_{k|k}$ in a "clipping strategy" but this can cause a poor estimation performance. This motivates the use of a better algorithms for nonlinear systems.

2.2.2 Moving Horizon Estimation

The Moving Horizon Estimation (MHE) attempts to solve the minimization problem given by Equation 5 at each sampling time, using all past information available. Since the number of measurements grows indefinitely, the computational complexity scales at least linearly with the horizon length k. As stated before, this problem only has a recursive solution if the system is linear and unconstrained in which it becomes the standard Kalman Filter. What MHE does is to optimize only in a finite horizon T, approximating the past information in a term called *arrival cost*. The optimization problem is written as

$$\begin{aligned} \min \quad & \ell_{ac}\left(\hat{x}[k-T],\, \bar{x}\right) + \sum_{i=k-T}^{k-1} \|w[i]\|^2_{Q^{-1}} + \sum_{i=k-T}^{k} \|v[i]\|^2_{R^{-1}} \\ \text{s.t.} \quad & \hat{x}[i+1] = f(\hat{x}[i],\, u[i]) + w[i] \\ & y[i] = h(\hat{x}[i],\, u[i]) + v[i] && \forall i \in \mathbb{Z}_{k-T:k-1} \\ & \hat{x}[j] \in \mathbb{X} && \forall j \in \mathbb{Z}_{k-T:k} \end{aligned} \tag{10}$$

with u and y being known variables. $\ell_{ac}(\cdot)$ is the *arrival cost* which approximately summarizes past data.

The simplest option for $\ell_{ac}(\cdot)$ is to ignore the arrival cost altogether, i. e. zero prior weighting. Analyzing the objective function of MHE one can see that if a big horizon is used, $\ell_{ac}(\cdot)$ becomes small compared to the stage cost and it could be discarded. Nevertheless,

zero prior weighting might require a big horizon to achieve stability and performance which requires a high computational cost (23, Chapter 4).

Another option is to approximate the arrival cost by a KF since it is a recursive algorithm. If no constraints are active and the model is linear, the KF is the exact solution for $\ell_{ac}(\cdot)$. If the model is not highly nonlinear, EKF might be a good approximation for it. This filtering update is implemented by applying the EKF equations in parallel to MHE at the beginning of the estimation horizon. This means that before MHE is solved, the filter estimate $\bar{x} = \hat{x}_{k-T|k-T}$ and $P_{k-T|k-T}$. Then, the arrival cost is given by

$$\ell_{ac}\left(\hat{x}[k-T],\, \bar{x}\right) = \|\hat{x}[k-T] - \bar{x}\|^2_{P^{-1}_{k-T|k-T}}\,. \tag{11}$$

Given a fixed horizon T, the filtering update gives a better estimation performance compared to zero prior weighting because it better approximates optimization problem with growing horizon. Because of that, a smaller horizon is necessary to achieve a good performance, decreasing the computational cost. Also, it is important to notice that when $k \leq T$, the MHE has an increasing horizon until it starts moving.

The disturbances were ignored in the explanation for brevity, but the MHE easily allows the use of more advanced models. For instance, it is possible to implement general Differential-Algebraic Equations (DAE) of the form:

$$\begin{aligned} \dot{\hat{x}} &= f_c(\hat{x},\, z,\, u,\, \hat{q},\, \hat{p}) \\ 0 &= g(\hat{x},\, z,\, u,\, \hat{q},\, \hat{p}) \\ y &= h(\hat{x},\, u,\, z,\, \hat{q},\, \hat{p}) \end{aligned} \tag{12}$$

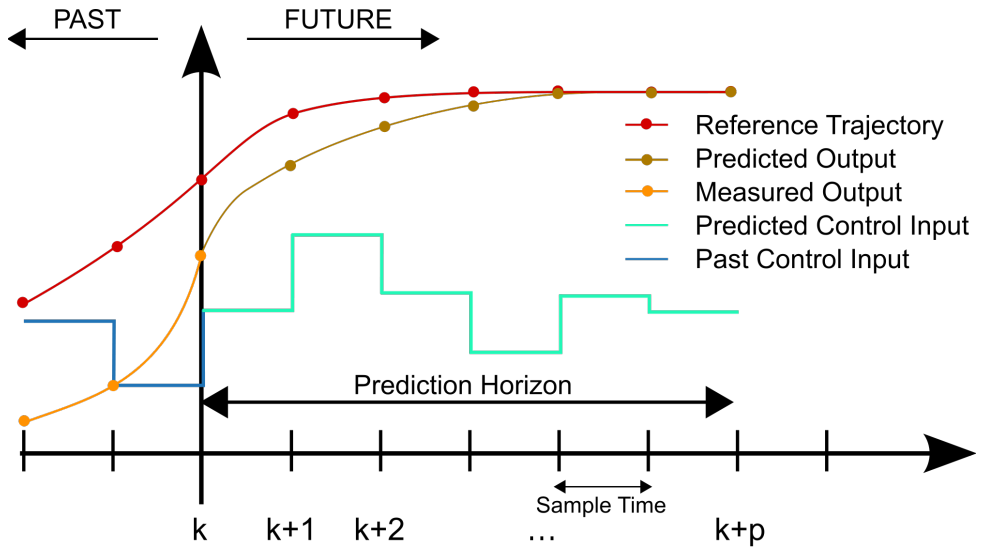
where z are algebraic variables, $g(\cdot)$ are equations that define them and $\hat{p}$ are time-invariant parameters. Furthermore, it is possible to add any kind of nonlinear constraints along the horizon. This kind of flexibility is not present in EKF.

2.3 MODEL PREDICTIVE CONTROL

The MPC uses in its formulation a prediction horizon of the system, as visualized in Figure 4. Using the current state of the plant, the algorithm predicts the future output based on a model of the process. With this information, the controller calculates a control sequence that minimize the distance between the prediction and the reference. To limit the computational cost, the optimization is only performed in a finite prediction horizon.

Many control algorithms can be classified as MPC since they have the same basic elements, even though they seem very different from each other. A sufficiently general

Figure 4 – Basic elements of MPC.



Source: Wikimedia Commons.

description of MPC is given by

$$\begin{aligned} \min \quad & \sum_{i=0}^{N} \ell\left(x[i],\, u[i]\right) \\ \text{s.t.} \quad & x[i+1] = f(x[i],\, u[i],\, \hat{q}[i]) \\ & y[i] = h(x[i],\, u[i],\, \hat{q}[i]) \\ & x[i] \in \mathbb{X}, \quad u[i] \in \mathbb{U} \qquad \forall i \in \mathbb{Z}_{0:N} \\ & x[0] = \bar{x}_0 \end{aligned} \tag{13}$$

with $\bar{x}_0$ as initial condition for the states (known or estimated), $\ell(\cdot)$ is a general stage cost, N the prediction horizon and $\hat{q}$ are the estimated disturbances. The prediction of $\hat{q}$ depends on how the disturbances of the system are modeled. Furthermore, specific terminal cost and constraint on the last predicted state can be added to improve or guarantee stability as advocated in Mayne (24).

The most common formulation is called tracking MPC where a quadratic cost is used for the stage cost given by

$$\ell_{\text{tr}}(x,\, u) = \|x - x_{\text{SP}}\|_Q^2 + \|u - u_{\text{SP}}\|_R^2 ,$$

where $(x_{\text{SP}},\, u_{\text{SP}})$ are setpoints for the variables and $(Q,\, R)$ are positive diagonal weighing matrices. Usually the setpoint is an equilibrium which optimizes the steady-state operation of the plant. Through Q and R, the designer can penalize each variable to achieve a desired performance. Penalties in x leads to faster tracking while higher values for R restricts the movement of the input u.

In the last decade or so, there has been an increased interest in cost functions for MPC which directly optimizes the economic aspects of the plant (25, 26). These so called

Economic MPCs (EMPC) are characterized by general $\ell_e(\cdot)$ which defines economic costs or profits (27, 28). This can provide an optimal operation of the plant, for instance, while changing setpoints or rejecting disturbances. Nevertheless, it becomes harder to guarantee the stability of the closed-loop system and even the optimization problem can become too nonlinear and incapable of converging to a solution (29).

2.4 FINAL COMMENTS

This chapter introduced some of the basic concepts needed to understand the developments that are presented in the following chapters. The Kalman filter is used in next two chapters on predictors for time-delay systems, whereas the nonlinear EKF and MHE are used in chapters 5 and 6. Meanwhile, MPC and optimal control is applied to red wine maceration in Chapter 7.

3 KALMAN PREDICTOR FOR SISO TIME-DELAY SYSTEMS

Context. This work continues and improves the research topic on time-delay systems which started on the master's dissertation (30). The first results on the proposed predictor were presented in a congress in 2018 (7). Then, the research continued improving the analysis and discussion about the topic. This chapter summarizes the following article:

- (11) D. M. Lima, B. M. Lima, and J. E. Normey-Rico, "A predictor for dead-time systems based on the Kalman Filter for improved disturbance rejection and robustness," *J. Process Control*, vol. 105, pp. 108–116, Sep. 2021.

The effects of dead-time in closed-loop systems have been a subject of study for many decades now and it still attract the attention of researchers (8). In linear time-invariant systems with delay, the closed-loop characteristic equation is not polynomial anymore in the continuous-time case and, in the discrete-time case, the order of the system increases proportionally to the size of the dead-time. In both cases, the presence of delay makes the system harder to control with traditional techniques (31).

Nevertheless, dead time remains an essential part of process dynamics modeling, since it can represent naturally occurring phenomena such as information, energy or mass transportation, and it can also be used to approximate high-order systems with low-order ones. Intuitively, the problems caused by dead time in the closed loop are easy to understand: (a) when the input of the process changes, its effect will be felt in the output only after the dead time; (b) disturbances are not felt immediately; (c) because of (a) and (b), the controller tries to correct something that happened in the past (32).

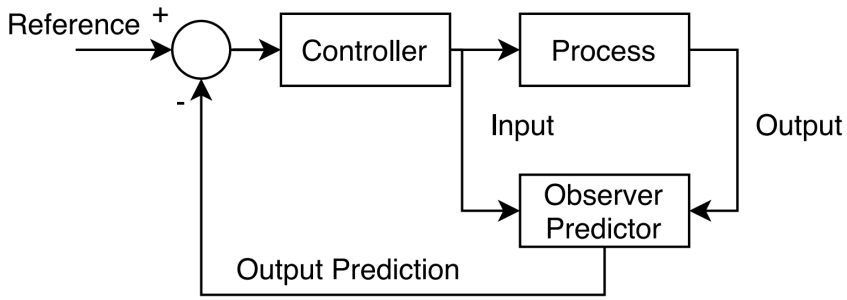
In the seminal work of O. J. M. Smith, a predictor, which is now called Smith Predictor (SP), was proposed to cope with the effects of dead time (33). This structure uses information from the input and output of the process to predict the system behavior after the dead time. Then, this signal can be used by a primary controller as if the system did not have dead time. In this case, the controller only "sees" the predicted output. Because of that, when a predictor such as the SP is used, the dead time does not appear in the closed-loop characteristic equation in the nominal case which greatly simplify the design of the closed-loop system (34). This also makes the design of both the controller and predictor independent, though the closed-loop dynamics and robustness varies with different tuning and predictor structure.

Throughout the past half-century, more predictor structures were proposed in the literature where many of them involve modifications of the original SP. Although the SP solved many problems related to dead-time systems, it could not be applied to open-loop unstable and integrative processes and had slow disturbance rejection. An important modification of the SP is the Filtered Smith Predictor (FSP) (35), which introduces a filter in the predictor structure that can, if properly tuned, deal with any type of process. Also, it gives an extra degree of freedom to deal with disturbances and robustness. However,

as the order of the system increases its tuning guidelines become more complicated, specially when MIMO systems are considered. New works related to the FSP generally deals with some sort of simplification on the proposition of tuning guidelines to achieve better disturbance rejection or robustness (36, 37).

Regardless of the predictor, they can all be analyzed through an observer-predictor (O&P) structure illustrated in Figure 5 (38). In the O&P, an observer is used to estimate the plant states and disturbances and then a predictor is used to predict the system output d-steps ahead where d is the discrete delay in number of samples.

Figure 5 – Observer-predictor structure.



Source: Diagram from Lima; Lima; Normey-Rico (11) which is based on Palmor (32).

In Garcia et al. (39), the robustness of a dead-time compensation scheme based on the O&P was analyzed for delay, sampling period and plant parameters uncertainties. The authors showed how a stable observer leads to a stable predictor using an augmented model and a robust Linear Matrix Inequalities (LMI) controller on a quadrotor. A robust H_2 controller is designed in Mirkin; Raskin (38). An O&P based on the Kalman filter is used in Sanz; Castillo; Albertos (40) to control a quadrotor but robustness properties were not analyzed. On all these works, frequency domain analysis is overlooked in favor of LMI design framework.

In Hao; Liu; Zhou (41), an O&P is proposed where the observer is tuned with a pole-placement method and a secondary frequency-domain filter is used to adjust disturbance rejection and robustness. It considers step and sinusoidal disturbances in the estimation. Although the authors analyze the output error bounds and closed-loop stability, the effects of the predictor on closed-loop robustness are not addressed.

In Cuenca et al. (42), a predictor scheme is proposed using a Extended Kalman filter (EK) but no disturbance is estimated making the predictor unable generate an offset-free prediction of the system. The authors improve this predictor in Alcaina et al. (43), where an Extended State Observer (ESO) is used to estimate disturbance effects. However, the analysis of the predictor properties is not complete, specially in regards to its tuning parameters and its effects on the closed-loop properties.

It is worth mentioning that, even though the KP proposed here considers a fixed delay in its formulation, it can still deal with a time-varying delay. In some works, control

schemes are proposed to deal with this type of delay. For instance, an Internal Model Control (IMC) based formulation was used to control a nonlinear static process under model uncertainties, including a variable delay (44). In Gonzalez Sorribes; García Gil (45), a novel observer-predictor control is proposed to stabilize uncertain systems subject to unknown but bounded arbitrarily-fast time-varying input and output delays. Nonetheless, it has been shown, using different methods, that a predictor with fixed delay can still be robust even with a time-varying delay (46, 47, 48).

Here, it is proposed a modification of the Kalman Predictor (KP), which is a Kalman filter based O&P with fixed delay which simplifies the one in (7). Some characteristics of the proposed KP are:

a) It supports to the estimation of different disturbance types which confers interesting disturbance rejection and robustness capabilities;

b) It inherently copes with systems of any order and dynamics, such as non-minimum phase and unstable or integrative poles without any additional procedure;

c) It has only one tuning parameter that can be adjusted for better disturbance rejection or robustness.

It also worth noting that the KP implicitly defines an FSP, whose properties are well known in the literature. Hence, all the tools used to interpret the closed-loop characteristics of a dead-time compensation scheme with an FSP can be used for the KP, specially the frequency-domain ones. This work will focus only on SISO systems whereas the MIMO case will be addressed in future works.

The remaining of this chapter is divided in the following way: preliminaries and notation are presented next. The KP is introduced in Section 3.2, the frequency domain analysis in Section 3.3 and tuning guidelines of the KP are in given in Section 3.3. In Section 3.4 simulation results are shown and analyzed.

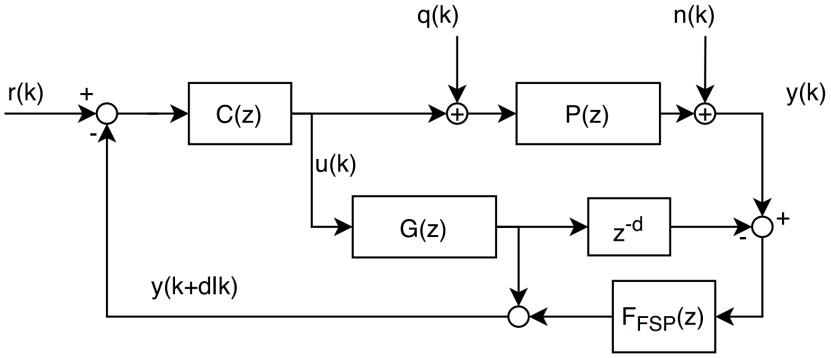
3.1 FILTERED SMITH PREDICTOR

The FSP, shown in Figure 6, uses a dead-time free model of the process $G(z)$ to compute the predicted value of the output after the delay. This predicted value is sent via feedback to a primary controller $C(z)$. This prediction must be corrected in some way to take into account disturbances and model mismatch. In the FSP, this is done by calculating the prediction error, i. e. the difference between the real output of the process and the nominal output of the process, which is then filtered by $F_{FSP}(z)$ and finally added to the nominal prediction.

The FSP characteristics are listed below (31):

a) Dead-time compensation, i. e. the characteristic equation of the closed-loop transfer functions are not influenced by the dead time in the nominal case and the primary controller $C(s)$ can be tuned disregarding the delay. The transfer func-

Figure 6 – FSP block diagram where q is an input disturbance, n is an output disturbance or noise, r is the reference, u, y, $y(k+d|k)$ are the process input, output and output prediction after the delay, respectively.



Source: Lima; Lima; Normey-Rico (11).

tions are given by:

$$\frac{Y(z)}{R(z)} = \frac{C(z)G(z)}{1+C(z)G(z)}z^{-d} \tag{14a}$$

$$\frac{Y(z)}{Q(z)} = P(z)\left(1 - \frac{C(z)G(z)z^{-d}}{1+C(z)G(z)}F_{FSP}(z)\right) \tag{14b}$$

$$\frac{Y(z)}{N(z)} = 1 - \frac{C(z)G(z)z^{-d}}{1+C(z)G(z)}F_{FSP}(z) \tag{14c}$$

b) The closed-loop set-point tracking is not affected by the prediction-error filter $F_{FSP}(z)$ in the nominal case.

c) The closed-loop disturbance rejection response can be adjusted through the prediction-error filter $F_{FSP}(z)$. Depending on the filter properties, this response can be accelerated or slowed down.

d) For unstable and integrative systems, the open-loop poles of the process in Equation 14b must be removed to achieve stability and zero-error disturbance rejection. To do this, the following condition must be satisfied for every undesirable p_i open-loop pole of $P(z)$:

$$\left(1 - z^{-d}F_{FSP}(z)\right)\Big|_{z=p_i} = 0, \quad \forall\, i \tag{15}$$

e) Given a model mismatch that is represented by the multiplicative error of a family of models in the frequency domain $\delta P_i(z)$, with $P_i(z) = P(z)(1+\delta P_i(z))$, the closed-loop system remains stable for all $P_i(z)$ if the following robustness condition, which is based on the Nyquist stability criterion, is true:

$$\left|\frac{C(z)G(z)}{1+C(z)G(z)}z^{-d}\right| |F_{FSP}(z)| < \frac{1}{|\delta P_i(z)|}, \quad \forall\, z = e^{\pm wj},\ w \in [0,\ \pi/T_s]\,,\ \forall\, i. \tag{16}$$

Hence, the closed-loop system can become more robust if $F_{FSP}(z)$ has low-pass characteristics, and vice-versa.

f) For a time-varying delay, the condition above changes to the one below (48):

$$\left|\frac{C(z)G(z)}{1+C(z)G(z)}z^{-d}\right| |F_{FSP}(z)| < \frac{1}{|\delta P_i(z)| + (1+|\delta P_i(z)|)\,|1-z|\,N}, \quad \forall\, z = e^{\pm wj},\ w \in [0,\ {}^{\pi}/_{T_s}]\,,\ \forall\, i, \quad (17)$$

where the delay is now given by $d(k) = d_n + \delta d(k)$, with d_n and $\delta d(k)$ being, respectively, the time-invariant and time-variant part of the delay. Also, $0 \leq d(k) \leq N\,T_s$, with $N \in \mathbb{R}^+$, $\tilde{N} = \sqrt{\lfloor N \rfloor^2 - 2\,g\lfloor N \rfloor + g}$ and $g = N - \lfloor N \rfloor$. Notice that if $N = 0$, i. e. no time-varying delay, Equation 17 becomes Equation 16.

The following discrete SISO state-space model will be used to represent the system:

$$\begin{aligned} x(k+1) &= A\,x(k) + B\,u(k-d) + B_\phi\,\phi(k) \\ y(k) &= C\,x(k) \end{aligned} \quad (18)$$

with $x \in \mathbb{R}^n$ for the states, $u \in \mathbb{R}$ for the control action, $\phi \in \mathbb{R}^q$ for the disturbance and $y \in \mathbb{R}$ for the measured output. A unique known input delay d is considered. In the frequency domain, using the Z-Transform, this model is given by

$$Y(z) = P(z)\,U(z) + P_\phi(z)\,\phi(z),$$

where

$$P(z) = G(z)\,z^{-d} = C\,(zI - A)^{-1}B\,z^{-d},$$
$$P_\phi(z) = G_\phi(z)\,z^{-d} = C\,(zI - A)^{-1}B_\phi.$$

3.2 THE KALMAN PREDICTOR

The idea behind the Kalman Predictor is to use an observer, which in this case is the Kalman filter (KF), to estimate the current state of the system and then use the system equations to compute the state d-steps ahead. By doing this, a Dead-Time Compensation (DTC) scheme is generated which is similar to the FSP. The KP algorithm is simple and has the following steps (7):

a) Use the KF to estimate the current state of the process taking into account possible disturbances;

b) Obtain the prediction of the system after the dead time by integrating the system model;

c) Use this state prediction in the feedback controller;

d) Repeat after each sampling time.

In this work, some modifications will be introduced in relation to previous works that will make the KP more general by making it able to cope with different types of disturbances

and simpler to be tuned. One interesting observation is that any state observer could be used instead of the KF. However, the KF has some advantages: i) it is not necessary to think in terms of eigenvalues or poles of the observer; ii) the KF automatically takes into account the system dynamics (unstable, non-minimum phase, among others) (49, Chapter 13).

Firstly, the user need to obtain a state-space model of the system which will be used as an intermediary step to obtain the predictor. Then, the disturbances have to be modeled to represent both model mismatch and external signals. Their estimation is essential to obtain an offset-free predictions of the system states or output. Each particular problem might have different types of disturbances but, in general, it is reasonable to assume that the disturbance is constant (50). But in some problems a sinusoidal, or ramp dynamics, might represent the disturbance better and, in some cases, a combination of types might be necessary. Then, the user has to create an augmented state-space representation of the system where the disturbances are part of the state equations.

For example, consider the case where a constant disturbance ϕ_1 and a sinusoidal one ϕ_ω are present, they can be represented by the following difference equations, respectively:

$$\begin{aligned}\phi_1(k+1) - \phi_1(k) &= 0\\ \phi_\omega(k+2) - 2\cos(\omega\, T_s)\,\phi_\omega(k+1) + \phi_\omega(k) &= 0\end{aligned}$$

where T_s is the sampling time and ω the disturbance frequency. The augmented model can be written, for this particular case, as

$$\begin{aligned}\bar{x}(k+1) &= \bar{A}\,\bar{x}(k) + \bar{B}\,u(k-d) + H\,w\\ y(k) &= \bar{C}\,\bar{x}(k) + v\end{aligned} \tag{19}$$

with matrix $H \in \mathbb{R}^{(n_x+n_q)\times n_w}$, n_q being the number of disturbance states and

$$\bar{x}(k) = \begin{bmatrix} x(k)\\ \phi_1(k)\\ \phi_\omega(k)\\ \phi_\omega(k-1)\end{bmatrix}, \quad \bar{A} = \begin{bmatrix} A & B_\phi & B_\phi & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 2\cos(\omega\, T_s) & \text{-}1\\ 0 & 0 & 1 & 1\end{bmatrix}, \quad \bar{B} = \begin{bmatrix} B\\ 0\\ 0\\ 0\end{bmatrix},$$
$$\bar{C} = \begin{bmatrix} C & 0 & 0 & 0\end{bmatrix}.$$

For the proposed KP algorithm, the disturbances affect the system through the input, i. e. $B_\phi = B$, this is done because disturbances due to model mismatch are dependent on control input, hence they are better represented with this consideration.

The matrix H selects where the noise w appears in the state equations. In Lima; Lima; Normey-Rico (7), an assumption was made that there was an individual noise signal w_i for each state of Equation 19, i. e. $n_w = n + q$ for SISO systems, which is the standard approach when designing a KF as a state observer.

However, in this paper, it is proposed that the number of noise signals will always be one independent of the number of states or types of disturbances considered in the

augmented model. Furthermore, by using only one noise signal, the tuning becomes considerably easier because the number of tuning parameters of the KF is directly related to the number of noise signals v and w. For SISO systems, there will only be two tuning parameters, Q and R, which are scalar numbers in this case. In this example, H is then given by:

$$H = \begin{bmatrix} \mathbf{0}_{n_x \times 1} \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

where $\mathbf{0}_{n_x \times 1}$ is a zeros matrix with dimension (n_x, 1).

It is important to notice that the user can follow the procedure explained here to estimate any type and number of disturbances. The only restriction is that the system 19 must be detectable. The detectability of pair ($\bar{A}$, $\bar{C}$) guarantees the existence of a steady-state KF that is marginally stable (19, Theorem 26). With the augmented system, the user can use any software tool to compute the steady-state KF which generally return L and $M \in \mathbb{R}^{(n+q)\times 1}$ which are, respectively the filter gain and the innovation gain, with $L = \bar{A}\, M$.

The procedure explained is done off-line in a design phase. The first on-line step of the algorithm is to use the steady-state KF to estimate the augmented state $\bar{x}(k|k)$ which has the following dynamics:

$$\bar{x}(k+1|k) = \left(\bar{A} - L\,\bar{C}\right) x(k|k-1) + \begin{bmatrix} L & \bar{B} \end{bmatrix} \begin{bmatrix} y(k) \\ u(k-d) \end{bmatrix} \tag{20a}$$

$$\bar{x}(k|k) = \left(I - M\,\bar{C}\right) \bar{x}(k|k-1) + \begin{bmatrix} M & 0 \end{bmatrix} \begin{bmatrix} y(k) \\ u(k-d) \end{bmatrix} \tag{20b}$$

where the equations represent, respectively, the filter dynamics and its output. To obtain the system output prediction $y(k|k)$, it is necessary to multiply Equation 20b by $\bar{C}$.

As a result of the previous step, the estimated state $\bar{x}(k|k)$ is now available. Next, it is necessary to obtain the predicted state after the dead time at $k+d$ as a function of $\bar{x}(k|k)$, which is done by integrating the system (20a) d steps in the future. It is possible to combine the estimation of the Kalman filter with the d-steps ahead prediction in a single dynamic system (7). Then, the Kalman Predictor is then given by:

$$\bar{x}(k+1|k) = \left(\bar{A} - L\,\bar{C}\right) \bar{x}(k|k-1) + \begin{bmatrix} L & 0 & \dots & 0 & \bar{B} \end{bmatrix} \bar{u}(k) \tag{21a}$$

$$\begin{aligned} y(k+d|k) = \bar{C}\,\bar{A}^d \left(I - M\,\bar{C}\right) \bar{x}(k|k-1) + \\ + \bar{C} \begin{bmatrix} \bar{A}^d M & \bar{B} & \bar{A}\bar{B} & \dots & \bar{A}^{d-1}\bar{B} \end{bmatrix} \bar{u}(k) \end{aligned} \tag{21b}$$

with $\bar{u}(k) = [y(k),\, u(k-1), \dots,\, u(k-d)]'$. To finalize the algorithm, the information predicted can be used by the primary controller and the input signal is sent to the plant.

3.3 FREQUENCY DOMAIN ANALYSIS

As already stated, the FSP has many interesting properties that can be analyzed in the frequency domain. This section shows how the KP implicitly defines an FSP and, because of that, the analysis tools for FSP can be used for the KP.

In the SISO case, the KP in Equation 21b has the prediction $y(k+d|k)$ as output, and two inputs: (i) $u(k)$, the manipulated variable, and its delayed values; and (ii) $y(k)$, the measured output. Applying the Z-Transform in Equation 21b:

$$Z\left\{y(k+d|k)\right\} = Y_d(z) = F_{KP}(z)\,Y(z) + S_{KP}(z)\,U(z) \tag{22}$$

with

$$F_{KP} = \bar{C}\left[\bar{A}^d\left(I - M\,\bar{C}\right)\left(z\,I - \bar{A} + L\,\bar{C}\right)^{-1} L + \bar{A}^d M\right],$$
$$S_{KP} = \bar{C}\left[\bar{A}^d\left(I - M\,\bar{C}\right)\left(z\,I - \bar{A} + L\,\bar{C}\right)^{-1}\bar{B}\,z^{-d} + \sum_{j=1}^{d}\bar{A}^{j-1}\bar{B}\,z^{-j}\right],$$

where the last equation indicates that the inputs of the KP, $Y(z)$ and $U(z)$ are filtered by $F_{KP}(z)$ and $S_{KP}(z)$, respectively, to generate the estimated output after the dead-time.

For unstable and integrative processes, to guarantee the internal stability of the predictor, the FSP must be implemented with the structure shown in Figure 7, where

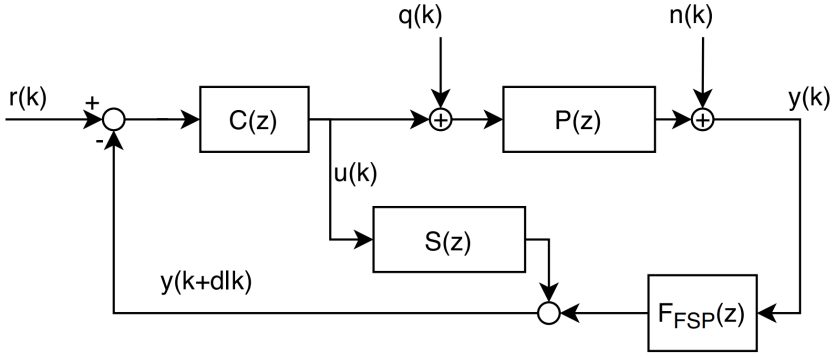
$$S_{FSP}(z) = G(z)\left[1 - z^{-d}F_{FSP}(z)\right], \tag{23}$$

and $S(z)$ should be a stable transfer function for any $G(z)$. From the figure, it can be easily shown that the output prediction of the FSP can be represented in the frequency domain by:

$$Y_d(z) = F_{FSP}(z)\,Y(z) + S_{FSP}(z)\,U(z), \tag{24}$$

with $S_{FSP}(z)$ as defined in Equation 23.

Figure 7 – Implementation structure of the FSP for unstable and integrative systems.



Source: Lima; Lima; Normey-Rico (11).

Comparing Equation 22 and the FSP prediction (24), it is clear that the KP at least have some similarities with the FSP since they both have the same inputs which are then filtered to generate the output prediction after the dead-time. Investigating further, it has been shown the theorem below.

Theorem 3.1. *The KP defined by (21a) and (22) implicitly defines a FSP described by (24), where the prediction-error filter $F_{FSP}(z)$ is equal to $F_{KP}(z)$ from (22).*

Proof. See Lima; Lima; Normey-Rico (7). ■

It has been shown in a previous work that the KP defined by (21a) and (22) implicitly defines a FSP described by (24), where the prediction-error filter $F_{FSP}(z)$ is equal to $F_{KP}(z)$ from (22) (7). This makes the closed-loop analysis of the KP in the frequency domain much more simple. Since the FSP and the KP are equivalent, all the properties of the FSP given in Section 3.1 are valid for the KP. All the user needs to do is to obtain $F_{KP}(z)$, the implicit prediction-error filter of the KP given by (22).

One interesting fact is that the FSP allows the correction of one important flaw of the original Smith Predictor where it can not be used with integrative or unstable systems. This happened for two reasons, internal stability of the predictor and unstable closed-loop in the unstable case. The latter can easily be checked through Equation 14b. Notice that the transfer function has the system model $P(z)$ as a factor, hence, if the system is open-loop unstable, (14b) is also unstable. Regarding the settling time of the disturbance rejection, this poses a problem even if the system is stable because the response will always have its open-loop poles.

Also, note that $S_{FSP}(z)$, given by (23), has the system model as a factor causing the same instability problem of (14b). However, in Normey-Rico; Eduardo F. Camacho (31), the authors showed that if $F_{FSP}(z)$ is properly tuned, it is possible to remove the undesired poles from $S_{FSP}(z)$, solving this particular flaw of the SP. The condition that must be satisfied for this to happen is (15), which will make $S_{PSF}(z)$ have zeros at the same place of the undesired poles, forcing a cancellation, thus eliminating the undesired dynamics.

The KP, however, inherently does this independent of tuning as explained in the following corollary:

Corollary 3.1. *Given the detectability of system (19), Theorem 3.1 and the KP defined by (21b) and (22), independent of tuning, the resulting implicit $S_{KP}(z)$ will never have any open-loop poles of $P(z)$, i. e.*

$$\left(1 - z^{-d} F_{KP}(z)\right)\Big|_{z=p_i} = 0, \ \forall\, i \tag{25}$$

where p_i represents the i-th pole of $P(z)$.

Proof. From (22) notice that only the poles of $F_{KP}(z)$ that come from the term $(z\,I - \bar{A} + L\,\bar{C})^{-1}$ appear on $S_{KP}(z)$. But, by (23), the open-loop poles given by $(z\,I - \bar{A})^{-1}$ should

Table 1 – Benchmark models.

Model	Transfer Function	T_s	Parameters
$P_1(s)$	$\frac{1}{(s+1)^3}e^{-5s}$	0.1	–
$P_2(s)$	$\frac{1}{(s+1)(1+\alpha s)(1+\alpha^2 s)(1+\alpha^3 s)}e^{-3s}$	0.1	$\alpha = 0.5$
$P_3(s)$	$\frac{1-\alpha s}{(s+1)^3}e^{-2s}$	0.1	$\alpha = 0.1$
$P_4(s)$	$\frac{100}{(s+10)^2}\left(\frac{1}{s+1} + \frac{0.5}{s+0.05}\right)e^{-10s}$	2.0	–
$P_5(s)$	$\frac{(s+6)^2}{s(s+1)^2(s+36)}e^{-7s}$	0.5	–
$P_6(s)$	$\frac{\omega_0^2}{(s+1)\left(s^2+2\zeta\,\omega_0 s+\omega_0^2\right)}e^{-15s}$	1.0	$\omega_0 = 1$, $\zeta = 0.1$
$P_7(s)$	$\frac{1}{s^2-1}e^{-5s}$	0.5	–

Source: Models from Lima; Lima; Normey-Rico (11) based on Åström; Hägglund (56).

also appear on $S_{KP}(z)$. Since they do not, by Theorem 3.1, this would only happen if (25) is true, or if $L = 0$, which can not happen, otherwise the observer would be open-loop. ■

Since the KP removes all open-loop poles of $S_{KP}(z)$, the problems described earlier do not happen. $S_{KP}(z)$ is always stable and the open-loop poles of $P(z)$ do not appear in disturbance rejection responses. Hence, one great advantage of the proposed KP is that it can be used for any input-time-delay process of any order with no need to design specific guidelines for unstable, integrative or non-minimum phase processes as some of the ones studied in the literature (51, 52, 53, 54, 55).

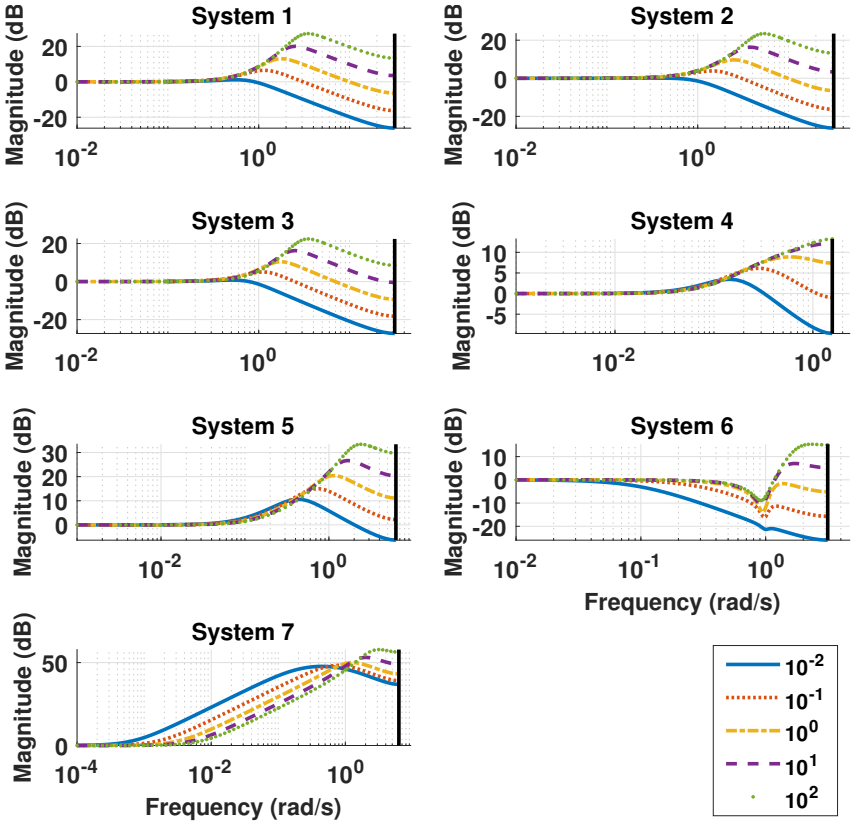
The proposed KP has two tuning parameters, the variance matrices Q and R of the disturbance state and of the output, respectively. Changing their values will affect the gains of the KF which, in turn, modify the prediction-error filter $F_{KP}(z)$. The relation between the filter characteristics such as bandwidth and the KP tuning parameters are hard to obtain algebraically because the relation between the system model and the tuning parameters are nonlinear. However, one interesting characteristic was inferred that helps the tuning procedure.

Before discussing the proposed guideline, the frequency characteristics of the F_{KP} will be shown for different systems and tuning parameters. In Åström; Hägglund (56), the authors give a list of systems to benchmark PID controllers which will be partially used here. The systems considered are shown in Table 1. Note that each system has different poles and zeros which cover most types of system characteristics: non-minimum phase, integrative, stable, unstable, multiple poles, high order dynamics, and oscillatory dynamics. Their discrete state-space models were obtained considering the respective sampling time.

To make the analysis simpler, the output variance matrix R will be set to 1 for all cases. This is done because the variances values are relative to each other for the steady-state KF and a SISO system, i. e. if all variances are multiplied by a fixed number the result is the same.

In Figure 8 the bode-plot of $F_{KP}(z)$ for different values of Q are shown for each system considering only an integral disturbance model. Notice that as the variance of

Figure 8 – Frequency domain magnitude plot of filter $F_{KP}(z)$ for varying values of Q and different systems for integral disturbance.



Source: Lima; Lima; Normey-Rico (11).

the disturbance-state increases the magnitude response of the filter at medium and high frequencies also increases for all systems. This indicates that $F_{KP}(z)$ will have faster poles which, in turn, indicates a faster disturbance rejection since the closed-loop disturbance rejection response in Equation 14b is dependent of $F_{KP}(z)$. On the other hand, the closed-loop robustness will diminish according to the Nyquist stability criterion (16), since higher bandwidth and magnitude makes it easier to violate the criterion for a certain model mismatch, which is a natural trade-off.

A more intuitive interpretation for this is that, when the variance Q has higher values, the KP must try to extract more information from the measured output to properly estimate the disturbance state which is varying too much, and vice-versa, i. e. if the variance is low, less information is needed for the estimation. This will make the predictor converge more rapidly but will make it less robust.

An interesting case is $P_7(s)$, which has unstable poles. The frequency characteristics have high amplification at medium and high frequencies independently of Q. This happens because the predictor must guarantee a certain feedback speed to cope with the unstable dynamics of the process. This behavior is well documented for the FSP applied to unstable systems (31).

Taking these observations into account, the following guidelines can be used:

a) For SISO systems and $R = 1$, the higher the value of the variance of the disturbance state Q, the higher the bandwidth of $F_{KP}(z)$ and its infinity norm. Hence, higher values of Q will help with disturbance rejection, whereas lower values will increase robustness, which is given by (16).

b) For stable and unstable processes, a good initial value is a weight of 1 normalized by the process gain $K_e = P(1)$: $Q = 1/K_e^2$. If the process is integrative, use the velocity gain.

3.4 ILLUSTRATIVE EXAMPLES

In this section, some illustrative examples will be used to present the applicability of the KP. In all cases, the primary controller will be the same, that is, only the predictor will be changed. However, ideally, the designs of the primary controller and of the predictor are done in parallel to achieve adequate tracking and robustness capabilities (35).

In Example 1, which show a stable case with sinusoidal disturbances, the following dead-time system is considered:

$$P(s) = \frac{1}{3s+1}e^{-2s}$$

The primary controller was designed considering the delay-free nominal model, aiming to achieve an offset free steady-state for step references and sinusoidal disturbances with frequency $w = 0.7\,\mathrm{rad/s}$ and have a nominal closed-loop settling time of 4.5 s, without the delay for reference step changes. This resulted in the following controller:

$$C(s) = 4.3423\frac{(s+1)\,(s+0.5)^2}{s\,(s^2+0.49)}$$

which is discretized using the Tustin method with sampling time $T_s = 0.1$s.

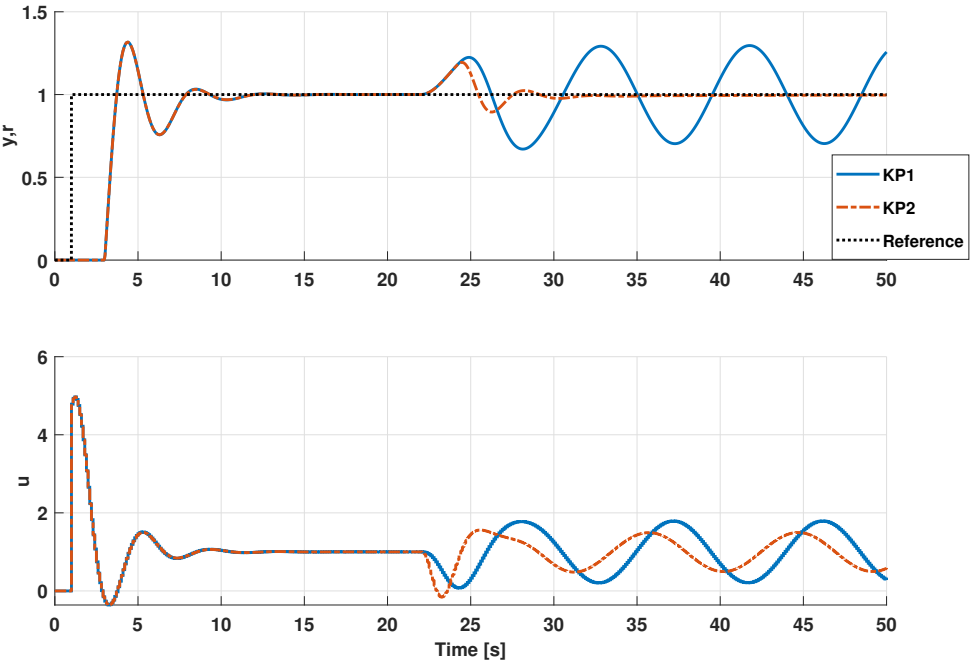
For the predictor structure, two KPs will be used, one that considers only a constant disturbance in the model, and another that considers a step and sinusoidal disturbance with frequency $\omega = 0.7\,\mathrm{rad/s}$. In both cases, $Q = 1$ was used, resulting in the following prediction error filters:

$$F_{KP1} = \frac{0.53871\,(z-0.9457)\,z}{z^2 - 1.741\,z + 0.7705}$$
$$F_{KP2} = \frac{4.9238\,(z-0.9929)\,(z^2-1.93z+0.9329)\,z}{(z-0.9951)\,(z-0.7377)\,(z^2-1.619\,z+0.7105)}$$

Notice that in the second case the filter is more complex because of the additional dynamics imposed by the sinusoidal dynamics inserted in the augmented system of the KF.

In Figure 9, the closed-loop responses are presented. A reference change from 0 to 1 is made at $t = 5$, and an input sinusoidal disturbance with frequency $\omega = 0.7\,\mathrm{rad/s}$ and 0.5 amplitude is added at $t = 20$. For step reference changes, both cases have the

Figure 9 – Example 1: closed-loop responses.



Source: Lima; Lima; Normey-Rico (11).

exact same result. However, they have different behaviors in the presence of sinusoidal disturbances. Even though the primary controller $C(s)$ is the same, and one that was designed to reject sinusoidal disturbances, only the KP2 case achieves this specification. This happens because $y(k+d|k)$ is incorrectly estimated by KP1, which is incapable of correct estimation in the presence of disturbances that are not of the constant kind. Hence, the controller makes the wrong action to correct the disturbance resulting in a steady-state error. This illustrates the importance of including the expected disturbance dynamics in the predictor.

Example 2, with an integrating system and time-varying delay, will be used to compare the proposed KP with two other strategies: the FSP with tuning guidelines from Normey-Rico; Camacho (35); and the Simplified Generalized Predictor (SGP) from Liu et al. (57). Keep in mind that the predictor structures that will be shown here are interchangeable, i. e. with proper tuning they can yield the same results since they implicitly define the same FSP structure. However, this can be hard to achieve considering the tuning guidelines suggested in the works where they were presented. These structures will also be tested against a time-varying delay to show that, even though they consider fixed delays in their tuning procedures, they still manage to confer a degree of robustness against this type of delay. This is a common situation in industrial practice, when a predictor-based controller is tuned for a robust behavior of a plant with a delay in the interval $[d_{min}, d_{max}]$. In most cases, robustness is also obtained for a varying delay in the same interval.

Consider the following model which represents a chain of evaporators tanks usually

found in the food industry (58):

$$P_n(s) = \frac{-0.1}{s\ (5\,s+1)} e^{-5}, \tag{26}$$

where time constants an dead times are in minutes and the output level and input flow are in percentages. They represent the controlled and manipulated variables, respectively. As in the previous case, a primary controller and a reference filter were designed to obtain a nominal closed-loop settling time of approximately 30 minutes with at most 5% peak for step reference changes, resulting in

$$C(s) = \frac{-3.9872\,(z-0.9275)\,(z-0.8187)}{(z-1)\,(z-0.7)},$$
$$F(s) = \frac{0.453\,(z-0.84)}{(z-0.9275)}$$

with $T_s = 1\,\text{min}$.

The predictors were tuned to obtain a disturbance rejection with approximately the same settling time resulting in this transfer function for the FSP:

$$F_{FSP}(z) = \frac{0.5421\,(z-0.9592)\,(z-0.897)\,(z^2-1.766\,z+0.8166)}{(z-0.9275)\,(z-0.8)^4}$$

For the SGP, the tuning parameters were $\lambda = 0.85$, $m = 1$ and $n_h = 2$, resulting in the following filters:

$$S_{SGP,num}(z) = \frac{\text{-}0.00937(z+0.936)(z-1)(z^2+0.493\,z+0.146)(z^2-1.08\,z+0.443)}{z^3(z-0.85)^4}$$
$$F_{SGP}(z) = \frac{0.0675(z-0.959)(z-0.819)z^2}{(z-0.85)^4}$$

Finally, for the KP, the tuning parameter was $Q = 0.01$, resulting in the following equivalent FSP filter:

$$F_{KP1}(z) = \frac{0.1726\,(z-0.9454)\,(z-0.8204)\,z}{(z-0.8142)\,(z^2-1.868\,z+0.877)}$$

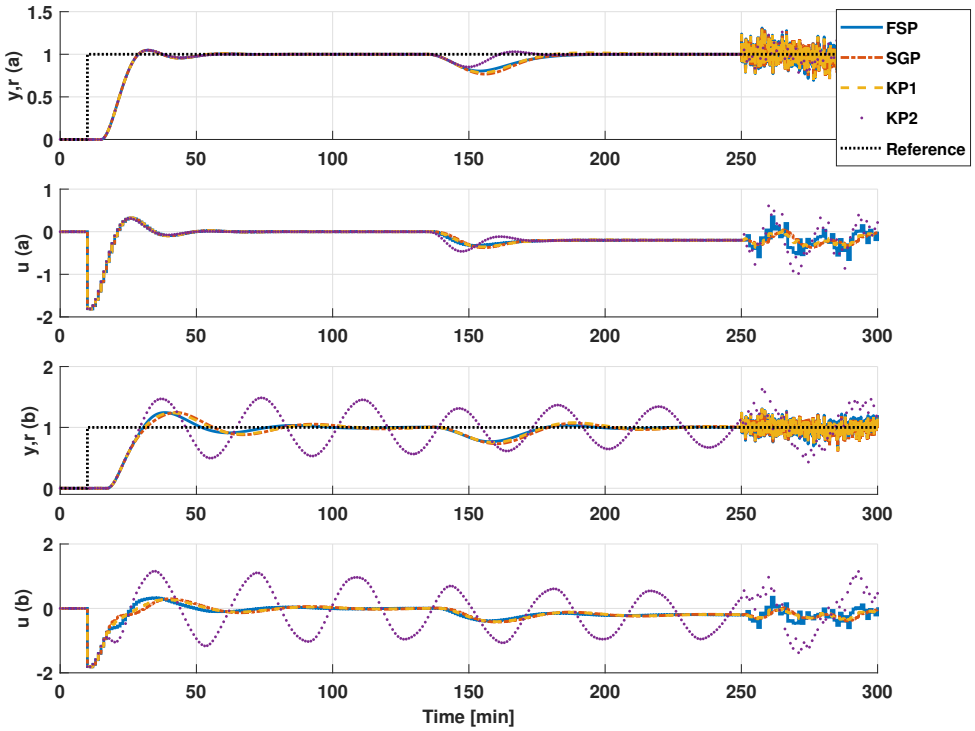
Furthermore, to show the improved disturbance rejection, a second KP was tuned with $Q = 1$ to decrease the settling time of the response to input disturbances, resulting in:

$$F_{KP2}(z) = \frac{0.6711\,z^3 - 1.145\,z^2 + 0.4887\,z}{z^3 - 2.432\,z^2 + 2.002\,z - 0.5548}$$

In this example, the advantages on tuning of the KP become more evident. Since the order and type of the system is higher than in the previous case, the tuning guidelines of the FSP and of the SGP change slightly, whereas nothing changes for the KP. Specially for the FSP, for higher order system or if the system has special dynamics (non-minimum phase, and unstable or integrator poles), the tuning procedure changes considerably (35).

The simulation results for the nominal case (a) and for the uncertain case (b) are presented in Figure 10. In both simulations there is a reference change at $t = 10$ from 0

Figure 10 – Example 2: closed-loop responses of the predictors for (a) nominal and (b) uncertain cases.



Source: Lima; Lima; Normey-Rico (11).

to 1, an input disturbance increase of 0.1 is introduced at $t = 130$ and at $t = 250$ noise is added to the output measurement.

In the nominal case, as indicated in the previous example, the reference tracking responses are the same, but this does not happen for the disturbance responses. Although they are very similar with the exception of KP2, which was tuned for a faster disturbance rejection. In case (b), there is an uncertainty in the delay in both its time-invariant and time-varying parts, which is now given by

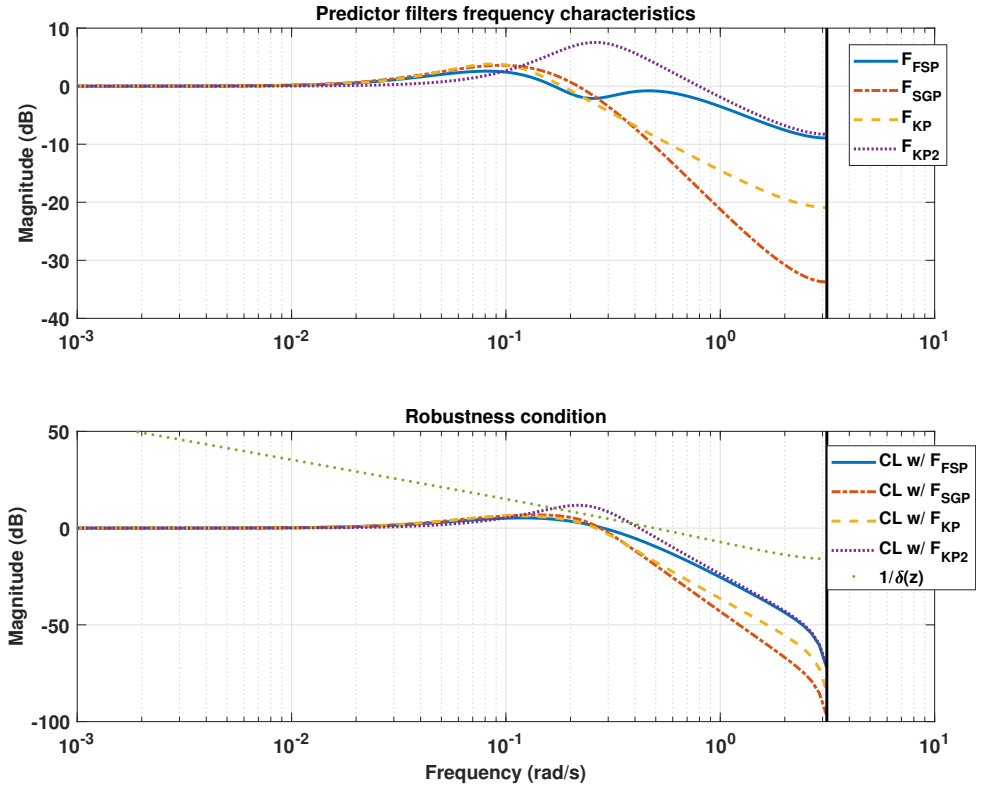
$$L(t) = 6.75 + 0.75\ \sin(2\,\pi\,0.1\,t) \rightarrow L \in [6,\ 7.5],$$

or, as defined in the discrete case in (17),

$$d(k) = 6 + \delta d(k) \rightarrow 0 \leq \delta d(k) \leq 1.5.$$

It can be seen from the results shown in Figure (10) that all predictors, with the exception of KP2, manage to maintain the system stability even in the face of a time-varying delay. With KP2 and this uncertainty, the closed-loop becomes unstable, showing the natural robustness/disturbance rejection trade-off. This stability issue of KP2 can be further analyzed with the robustness condition given by Equation 17, which is shown in Figure 11. Only KP2 do not satisfy the condition.

Figure 11 – Example 2. Top: frequency diagram of the predictors. Bottom: robustness condition taking into account the different filters and the nominal closed-loop (CL) transfer function.



Source: Lima; Lima; Normey-Rico (11).

3.5 CONCLUSION

In this chapter, a novel predictor for dead-time systems based on the Kalman filter was introduced. It was shown that the Kalman Predictor is equivalent to an FSP, which is an important discovery since every property of the FSP and every tool for its analysis can be readily applied to the KP. Some simulation examples were presented to corroborate the good performance and simple tuning of the proposed KP, showing its relevance even in the face of more complex problems, for example, dealing with sinusoidal disturbances and time-varying delays. In the next chapter, the MIMO case where different delays exist for each input is considered.

4 EXTENSION OF KALMAN PREDICTOR FOR MIMO SYSTEMS

Context. After the results from the last chapter, the efforts were put on extending the analysis to MIMO systems. This chapter summarizes the following article:

- (12) D. M. Lima, B. M. Lima, and J. E. Normey-Rico, "A predictor for square multivariable dead-time systems with multiple delays based on the Kalman filter," *J. Process Control*, vol. 124, pp. 105–117, Apr. 2023.

Dead time (or delay) is an important phenomena that is present in many processes and can be associated with information, energy or mass transportation, and it can also be used as a tool to approximate high-order systems with low-order ones (31). The delay usually complicates the feedback controller design and this issue becomes even more problematic in Multiple-Input-Multiple-Output (MIMO) systems due to the increased number of variables, multiple delays and coupling effects. The application of predictors to MIMO processes is more difficult than in the SISO case since the coupling effects and multiple delays make it harder to apply some of the techniques developed for the SISO case, and the tuning guidelines become more involved.

A unified approach (controller and predictor tuning) based on the FSP is provided in Flesch et al. (59) for MIMO square processes, and for MIMO non-square processes in Flesch; Santos; Normey-Rico (60). But, as the number of variables and order of the transfer functions increase, the tuning becomes more difficult. Something similar is done in Santos; Torrico; Normey-Rico (61), the controller is simplified to a gain and the MIMO FSP is tuned to achieve the desired closed-loop properties, but the conditions that the MIMO FSP must satisfy are more complex. In Giraldo et al. (62), the authors suggest the addition of a decoupler to the control structure, but to apply the proposed strategy the system has to satisfy certain conditions.

Some approaches use a state-space representation of the process to obtain the predictor, which is naturally MIMO. However, the prediction is usually obtained through the integration of the system model, which becomes complicated or non-causal in the presence of multiple delays, limiting the application of these techniques considerably. This happens with the approaches presented in (11, 63, 64, 41). A solution is shown in Albertos; Garcia (65) for multiple delays, but the predictor has a more complicated structure, it uses a model that always results in a large number of states, and it does not have degrees of freedom to adjust robustness and disturbance rejection, which are not thoroughly analyzed.

In this work, a modified version of the Kalman Predictor for SISO plants presented in Lima; Lima; Normey-Rico (11) is proposed for MIMO square processes that has the following advantages: (i) can deal with multiple delays; (ii) it can be applied to processes of any number of inputs/outputs; (iii) the predictor deals with systems of any order or dynamics (e. g. integrative, unstable and non-minimum phase) without particular rules for each case; (iv) it can be used with state-space or transfer functions models; (v) it does

not need to integrate the model to obtain the predictions after the delay; (vi) the tuning of the predictor parameters are intuitive and can be used to improve disturbance rejection or robustness; (vii) the predictor is defined in discrete-time, making the implementation straight-forward; (viii) it can be interpreted as a MIMO FSP, allowing the use of well known analysis tools, specially the frequency domain ones. However, it is important to note that, similarly as in other DTC structures, these advantages depend on how the delays are present in the MIMO process model, which will be discussed in the next section.

Furthermore, the properties of the predictor structure are analyzed independently of the design of the primary controller used in the MIMO DTC structure in order to understand how the predictor affects the closed-loop behavior. Different design techniques are used to compute the primary controller in each example, but the same primary controller is always used in such a way that the predictor properties are highlighted.

The remaining of this chapter is divided in the following way: the MIMO FSP is presented in Section 4.1; the model used with the proposed predictor is shown in Section 4.2; the Modified Kalman Predictor is presented in Section 4.3, and its analysis in Section 4.4; results and conclusions are discussed in Sections 4.5 and 4.6, respectively.

4.1 MIMO FILTERED SMITH PREDICTOR

The Filtered Smith Predictor (FSP) is an important predictor studied in various works (31, 59, 66), and is the basis for comparison in many others (67, 36, 57). Its properties are well defined mathematically and the other predictors structures usually can be interpreted as an FSP. Hence, all analysis tools available for the FSP can be used with new predictors, making the analysis of their properties more straight-forward.

First, consider a general MIMO process with multiple delays described by its discrete transfer function matrix $\mathbf{P}(z)$ in the following way:

$$\mathbf{P}(z) = \begin{bmatrix} G_{11}(z)z^{-d_{11}} & \cdots & G_{1m}(z)z^{-d_{1m}} \\ \vdots & \ddots & \vdots \\ G_{n1}(z)z^{-d_{n1}} & \cdots & G_{nm}(z)z^{-d_{nm}} \end{bmatrix} \tag{27}$$

where n, and m are the number of outputs and inputs, respectively. $G_{ij}(z)$ is a delay-free transfer function between output i and input j, while d_{ij} is the discrete delay between them. Since only square MIMO processes will be dealt with here, $n = m$. The system (27) can then be described as $\mathbf{P}(z) = \mathbf{L}(z)\mathbf{G}(z)$ where $\mathbf{L}(z)$ is a diagonal transfer function matrix that represents the minimal output delay of the system, and $\mathbf{G}(z)$ is called the fast model:

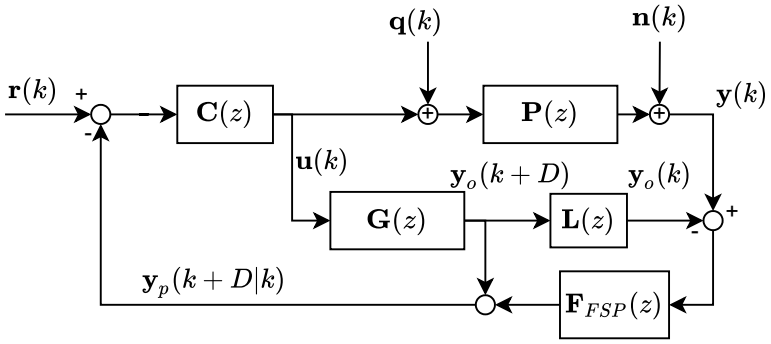
$$\mathbf{L}(z) = \begin{bmatrix} z^{-d_1} & 0 & 0 \\ 0 & \ddots & \vdots \\ 0 & 0 & z^{-d_n} \end{bmatrix} \tag{28}$$

$$\mathbf{G}(z) = \begin{bmatrix} G_{11}(z)z^{-(d_{11}-d_1)} & \cdots & G_{1n}(z)z^{-(d_{1n}-d_1)} \\ \vdots & \ddots & \vdots \\ G_{n1}(z)z^{-(d_{n1}-d_n)} & \cdots & G_{nn}(z)z^{-(d_{nn}-d_n)} \end{bmatrix} \tag{29}$$

where d_i is the minimal delay of output i or the minimum delay of the i-th line of $\mathbf{P}(z)$. The model output is defined as $\mathbf{y}_o(k)$ and the output of the fast model is $\mathbf{y}_o(k+D)$, which is the predicted value of the output before the minimal delays: $\mathbf{y}_o(k+D) = [y_1(k+d_1)\ y_2(k+d_2)\ \ldots y_n(k+d_n)]'$, and D is used to shorten the notation and to indicate that each output has a different delay.

In the MIMO case, the FSP (Figure 12) uses the fast model of the process $\mathbf{G}(z)$ to compute the predicted value of the output before the minimal delays, $\mathbf{y}_o(k+D)$. This predicted value is sent via feedback to a primary controller $\mathbf{C}(z)$, which has order $n \times n$. However, this prediction must be corrected in some way to take into account disturbances and model mismatch. In the FSP, this is done by calculating the prediction error, i. e. the difference between the real output of the process $\mathbf{y}(k)$ and the output model $\mathbf{y}_o(k)$, which is then filtered by $\mathbf{F}_{FSP}(z)$ (with dimension $n \times n$), and finally added to the nominal prediction. The output of the filter is defined as $\mathbf{y}_f(k)$ and the final corrected prediction used in the primary controller is $\mathbf{y}_p(k+D|k)$. Also, in Figure 12, $\mathbf{q}(k)$ is an input disturbance, $\mathbf{n}(k)$ is an output disturbance, or noise, $\mathbf{r}(k)$ is the reference and $\mathbf{u}(k)$ is the process input. Note that the corrected output prediction can be written as: $\mathbf{y}_p(k+D|k) = \begin{bmatrix} y_{p1}(k+d_1|k) & \cdots & y_{pn}(k+d_n|k) \end{bmatrix}^T$.

Figure 12 – MIMO Filtered Smith Predictor (FSP) block diagram.



Source: Diagram from Lima; Lima; Normey-Rico (12).

The main properties of the MIMO FSP are listed below (31):

a) The predictor output signal can be represented in the frequency domain as

$$\mathbf{Y}_p(z) = (\mathbf{I} - \mathbf{F}_{FSP}(z)\mathbf{L}(z))\,\mathbf{G}(z)\mathbf{U}(z) + \mathbf{F}_{FSP}(z)\mathbf{Y}(z) \tag{30}$$

with $\mathbf{Y}_p(z) = \mathcal{Z}\{\mathbf{y}_p(k+D|k)\}$.

b) The closed-loop matrix transfer functions $\mathbf{H}_r(z)$, $\mathbf{H}_q(z)$ and $\mathbf{H}_n(z)$, respectively

associated to the reference, disturbance and noise are:

$$\mathbf{H}_r(z) = \mathbf{P}(z)\mathbf{C}(z)(\mathbf{I} + \mathbf{G}(z)\mathbf{C}(z))^{-1}, \tag{31}$$

$$\mathbf{H}_q(z) = \left(\mathbf{I} - \mathbf{P}(z)\mathbf{C}(z)(\mathbf{I} + \mathbf{G}(z)\mathbf{C}(z))^{-1}\mathbf{F}_{FSP}(z)\right)\mathbf{P}(z), \tag{32}$$

$$\mathbf{H}_n(z) = \left(\mathbf{I} - \mathbf{P}(z)\mathbf{C}(z)(\mathbf{I} + \mathbf{G}(z)\mathbf{C}(z))^{-1}\mathbf{F}_{FSP}(z)\right), \tag{33}$$

and note that their characteristic equations are not influenced by the minimal output delays $\mathbf{L}(z)$ in the nominal case, thus, the tuning of the primary controller $\mathbf{C}(z)$ can be done taking into account only the fast model $\mathbf{G}(z)$, which can considerably simplify the design procedure. This happens because $\mathbf{G}(z)$ will always have a smaller order than $\mathbf{P}(z)$, except when the minimal output delays are all zero but, in this case, the advantages of any DTC structure are diminished. Also note that the difference between the delays in $\mathbf{P}(z)$ and $\mathbf{G}(z)$ depends on the problem. The most favorable condition, i.e. the greater delay reduction in the elements of a row of $\mathbf{G}(z)$, is achieved when the differences between the delays of the elements in the correspondent row of $\mathbf{P}(z)$ is small. The tuning of $\mathbf{C}(z)$ will not be the subject of this article, but tuning guidelines can be found in the literature (31, 68).

c) The reference tracking is not affected by the prediction-error filter $\mathbf{F}_{FSP}(z)$ in the nominal case.

d) The disturbance rejection can be adjusted by adequately tuning $\mathbf{F}_{FSP}(z)$. For example, in the case of input disturbances (32), the open-loop model $\mathbf{P}(z)$ appears as a factor of the matrix transfer function, making the disturbance rejection as slow as the open-loop dynamics. This issue can be avoided if the filter $\mathbf{F}_{FSP}(z)$ is diagonal and each of its non-null element satisfy the following condition (59):

$$\frac{d^\lambda}{dz^\lambda}\left(1 - F_{FSP,i}(z)z^{-d_i}\right)\Big|_{z=z_j} = 0,\ \lambda = 0, \ldots, m_j - 1,\ j = 1, \ldots, k \tag{34}$$

where $F_{FSP,i}(z)$ is the i-th diagonal element of $\mathbf{F}_{FSP}(z)$, z_j is the j-th undesirable pole of the i-th row of $\mathbf{G}(z)$, and m_j is the multiplicity of z_j.

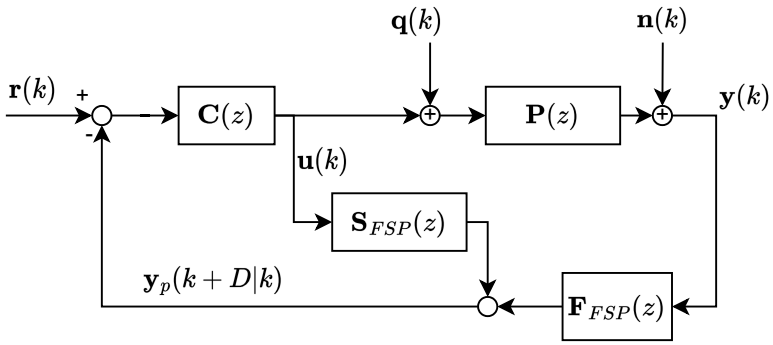
e) If the system $\mathbf{P}(z)$ has unstable or integrative poles, the MIMO FSP must be implemented using an alternative structure, shown in Figure 13, to guarantee the internal stability of the predictor, where:

$$\mathbf{S}_{FSP}(z) = \left(\mathbf{I} - \mathbf{F}_{FSP}(z)\mathbf{L}(z)\right)\mathbf{G}(z), \tag{35}$$

and $\mathbf{F}_{FSP}(z)$ must be tuned to obtain a stable transfer matrix $\mathbf{S}(z)$ by using (34) with any unstable or integrative poles of $\mathbf{G}(z)$. In this case, the predictor output can be represented in the frequency domain as:

$$\mathbf{Y}_p(z) = \mathbf{S}_{FSP}(z)\mathbf{U}(z) + \mathbf{F}_{FSP}(z)\mathbf{Y}(z) \tag{36}$$

Figure 13 – Implementation structure of the MIMO FSP for unstable and integrative systems.



Source: Diagram from Lima; Lima; Normey-Rico (12).

f) Consider an unstructured additive uncertainty $\mathbf{E}(z) = \mathbf{W}_2(z)\mathbf{\Delta}(z)\mathbf{W}_1(z)$ such that $\mathbf{P}(z) = \mathbf{P}_n(z) + \mathbf{E}(z)$, where $\mathbf{P}_n(z)$ is the nominal model and $||\mathbf{\Delta}(z)||_\infty \leq 1$. If the following sufficient condition is satisfied, the closed-loop system remains stable for $\mathbf{P}(z)$ (68, 59, 31),

$$\bar{\sigma}(\mathbf{W}_1(z)\mathbf{M}_0(z)\mathbf{W}_2(z)) \leq 1, \, \forall \, z = e^{\pm jwT_s}, \, w \in [0, \, \pi/T_s], \tag{37}$$

where $\mathbf{M}_0(z) = \mathbf{C}(z)\left(\mathbf{I} + \mathbf{G}(z)\mathbf{C}(z)\right)^{-1}\mathbf{F}_{FSP}(z)$ is stable. Hence, the closed-loop system can become more robust if $\mathbf{F}_{FSP}(z)$ has low-pass characteristics, and vice-versa. This condition is sufficient but not necessary for robustness, i.e the system might be stable even if the condition is not satisfied, hence, it is a conservative condition, but this usually is not an issue since, in practice, we tend to prefer a larger robustness margin. Also, in here, we consider an additive representation of the uncertainties, but there are more ways that the same uncertainty can be represented, as discussed in Skogestad; Postlethwaite (68). Each representation will have a different frequency domain characteristic which will give different robustness conditions for the same modeling error, thus, the user should be cautious about this choice and choose the error structure that has the least magnitude in the frequency domain, otherwise, there is a risk of being too conservative, which will limit, for example, settling time specifications.

4.2 SYSTEM MODEL

The predictor that will be proposed requires, as a first step, a specific state-space representation of the process, which can easily be obtained from the nominal transfer function matrix of the process.

As described earlier, the process transfer function is given by $\mathbf{P}(z) = \mathbf{L}(z)\mathbf{G}(z)$. In

state-space form, $\mathbf{G}(z)$ and $\mathbf{L}(z)$ can be written, respectively, as

$$\mathbf{G}(z) \rightarrow \begin{cases} \mathbf{x}(k+1) &= \mathbf{A}\,\mathbf{x}(k) + \mathbf{B}\,\mathbf{u}(k) \\ \mathbf{y}_o(k+D) &= \mathbf{C}\,\mathbf{x}(k) \end{cases} \tag{38}$$

$$\mathbf{L}(z) \rightarrow \begin{cases} \mathbf{x}_l(k+1) &= \mathbf{A}_l\mathbf{x}_l(k) + \mathbf{B}_l\mathbf{y}_o(k+D) \\ \mathbf{y}_o(k) &= \mathbf{C}_l\mathbf{x}_l(k) + \mathbf{D}_l\mathbf{y}_o(k+D) \end{cases} \tag{39}$$

where $\mathbf{y}_o$, $\mathbf{u} \in \mathbb{R}^n$, $\mathbf{x} \in \mathbb{R}^{n_{xG}}$, $\mathbf{x}_l \in \mathbb{R}^{n_{xL}}$, with n_{xL} and n_{xG} being, respectively, the number of states of $\mathbf{L}(z)$ and $\mathbf{G}(z)$ in state-space form. Both systems can be combined to generate a non-minimal state-space model of $\mathbf{P}(z)$ as shown below:

$$\begin{cases} \begin{bmatrix} \mathbf{x}(k+1) \\ \mathbf{x}_l(k+1) \end{bmatrix} &= \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{B}_l\mathbf{C} & \mathbf{A}_l \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}(k) \\ \mathbf{y}_o(k) &= \begin{bmatrix} \mathbf{D}_l\mathbf{C} & \mathbf{C}_l \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \end{bmatrix} \end{cases} \tag{40}$$

An important characteristic of this representation is that the output without the minimal output delay can be obtained directly from the states with

$$\mathbf{y}_o(k+D) = \begin{bmatrix} y_1(k+d_1) \\ y_2(k+d_2) \\ \vdots \\ y_n(k+d_n) \end{bmatrix} = \begin{bmatrix} \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \end{bmatrix}, \tag{41}$$

which can be used by the proposed predictor to directly obtain the predictions after the minimum output delays for processes with multiple dead-times. It is also important to emphasize that in a MIMO system we can consider measurement (output), control or internal delays, and each have a different physical meaning. However, they can all be reinterpreted as output delays to use the proposed formulation.

4.3 MODIFIED KP FOR MIMO SQUARE PROCESSES

The proposed Modified Kalman Predictor (MKP) for square MIMO systems with multiple delays uses a Kalman Filter to estimate the states of the system (40). While estimating the states, it is important to consider the effect of unknown disturbances which can represent both model mismatch and external signals. Their estimation is essential to obtain an offset-free prediction of the system states and/or outputs. A generic state-space model for disturbances is given below,

$$\begin{cases} \mathbf{x}_\phi(k+1) &= \mathbf{A}_\phi\mathbf{x}_\phi(k) \\ \mathbf{y}_\phi(k) &= \mathbf{C}_\phi\mathbf{x}_\phi(k) \end{cases} \tag{42}$$

where $n_{x\phi}$ is the number of disturbance states, $\mathbf{x}_\phi \in \mathbb{R}^{n_{x\phi}}$, $\mathbf{A}_\phi \in \mathbb{R}^{n_{x\phi}\times n_{x\phi}}$ represent the disturbance dynamics, for example, a constant or sinusoid, $\mathbf{C}_\phi \in \mathbb{R}^{n_{y\phi}\times n_{x\phi}}$, and $\mathbf{y}_\phi \in \mathbb{R}^{n_{y\phi}}$ is the actual disturbance that will affect the system.

The Kalman filter has an stochastic formulation in which zero-mean additive Gaussian noises are considered in the states ($\mathbf{w} \in \mathbb{R}^{n_w}$) and output ($\mathbf{v} \in \mathbb{R}^{n}$). They are defined by diagonal covariance matrices $\mathcal{Q}$ and $\mathcal{R}$, respectively, which means that the noises are uncorrelated. Then, the augmented model that considers the disturbance dynamics is obtained by combining (40) and (42), and assuming that $\mathbf{y}_\phi(k)$ is added to the control input $\mathbf{u}(k)$:

$$\begin{cases} \begin{bmatrix} \mathbf{x}(k+1) \\ \mathbf{x}_l(k+1) \\ \mathbf{x}_\phi(k+1) \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \mathbf{B}\mathbf{C}_\phi \\ \mathbf{B}_l\mathbf{C} & \mathbf{A}_l & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_\phi \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \\ \mathbf{x}_\phi(k) \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \mathbf{u}(k) + \mathbf{H}\mathbf{w}(k) \\ \mathbf{y}_o(k) = \begin{bmatrix} \mathbf{D}_l\mathbf{C} & \mathbf{C}_l & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \\ \mathbf{x}_\phi(k) \end{bmatrix} + \mathbf{v}(k) \end{cases} \tag{43}$$

or in compact form,

$$\bar{\mathbf{x}}(k+1) = \bar{\mathbf{A}}\bar{\mathbf{x}}(k) + \bar{\mathbf{B}}\mathbf{u}(k) + \mathbf{H}\mathbf{w}(k) \tag{44a}$$

$$\mathbf{y}_o(k) = \bar{\mathbf{C}}\bar{\mathbf{x}}(k) + \mathbf{v}(k) \tag{44b}$$

with

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \mathbf{B}\mathbf{C}_\phi \\ \mathbf{B}_l\mathbf{C} & \mathbf{A}_l & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_\phi \end{bmatrix}, \quad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \bar{\mathbf{C}} = \begin{bmatrix} \mathbf{D}_l\mathbf{C} & \mathbf{C}_l & \mathbf{0} \end{bmatrix},$$

and $\mathbf{H} \in \mathbb{R}^{n_H \times n_w}$, with $n_H = n_{xG} + n_{xL} + n_{x\phi}$. Matrix $\mathbf{H}$ selects where the noise $\mathbf{w}$ appears in the state equations. As stated before, for the proposed MKP algorithm, we consider that the disturbances affect the system as if they were inputs, this is done because disturbances due to model mismatch are dependent on control input, thus, they are better represented with this consideration. In this work, the focus will be only on square MIMO process with constant disturbances, thus, $n_\phi = n_{x\phi} = n$ and $\mathbf{C}_\phi = \mathbf{A}_\phi = \mathbf{I}$, but keep in mind that particular problems might be subject to different types of disturbances such as sinusoids or ramps.

When designing a Kalman filter, the standard approach is to define one noise signal per state, i. e. $n_w = n_{xG} + n_{xL} + n_{x\phi}$. However, this procedure makes the tuning procedure more involved because the number of states can be big for MIMO systems. Thus, following the suggestion in Lima; Lima; Normey-Rico (11), only one noise signal will be added per input. In this simplified case, the interest lies only in the system output estimation after the dead time, and not on the state estimation, hence, the noise in the states can be removed. With these considerations, matrix $\mathbf{H}$ has the following structure

$$\mathbf{H} = \begin{bmatrix} \mathbf{0}^T_{n_{xG}+n_{xL} \times n} & \mathbf{I}^T_{n \times n} \end{bmatrix}^T.$$

With the augmented system, any software tool can be used to compute the steady-state (fixed gain) Kalman filter to estimate the system states. These tools generally return

$\mathbf{L}$, $\mathbf{M} \in \mathbb{R}^{n_{xG}+n_{xL}+n_{x\phi} \times n}$ which are, respectively, the filter gain and the innovation gain with $\mathbf{L} = \bar{\mathbf{A}}\mathbf{M}$. This procedure can be done as long as the pair $(\bar{\mathbf{A}}, \bar{\mathbf{C}})$ is detectable, which guarantees that a marginally stable steady-state Kalman filter exists (Simon (19), Theorem 26).

Then, the steady-state Kalman Filter can be written as

$$\bar{\mathbf{x}}(k+1|k) = \left(\bar{\mathbf{A}} - \mathbf{L}\bar{\mathbf{C}}\right)\bar{\mathbf{x}}(k|k-1) + \begin{bmatrix}\mathbf{L} & \bar{\mathbf{B}}\end{bmatrix}\begin{bmatrix}\mathbf{y}(k)\\ \mathbf{u}(k)\end{bmatrix} \tag{45a}$$

$$\bar{\mathbf{x}}(k|k) = \left(\mathbf{I} - \mathbf{M}\bar{\mathbf{C}}\right)\bar{\mathbf{x}}(k|k-1) + \begin{bmatrix}\mathbf{M} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{y}(k)\\ \mathbf{u}(k)\end{bmatrix} \tag{45b}$$

$$\mathbf{y}_p(k|k) = \bar{\mathbf{C}}\bar{\mathbf{x}}(k|k) \tag{45c}$$

$$\mathbf{y}_p(k+D|k) = \begin{bmatrix}\mathbf{C} & \mathbf{0} & \mathbf{0}\end{bmatrix}\bar{\mathbf{x}}(k|k) \tag{45d}$$

where $\bar{\mathbf{x}}(k|k)$, $\mathbf{y}_p(k|k)$, and $\mathbf{y}_p(k+D|k)$ are, respectively, the updated estimation of the system states, the estimated system output and the estimated system output without the minimal output delays. Remember that the latter can be obtained simply by adequately selecting part of the augmented estimated states as explained in Section 4.2. Since the point of the predictor is to obtain the estimated output after the minimal delays, the Kalman Predictor is obtained from (45) after combining some of its terms:

$$\bar{\mathbf{x}}(k+1|k) = \left(\bar{\mathbf{A}} - \mathbf{L}\bar{\mathbf{C}}\right)\bar{\mathbf{x}}(k|k-1) + \begin{bmatrix}\mathbf{L} & \bar{\mathbf{B}}\end{bmatrix}\begin{bmatrix}\mathbf{y}(k)\\ \mathbf{u}(k)\end{bmatrix} \tag{46a}$$

$$\mathbf{y}_p(k+D|k) = \begin{bmatrix}\mathbf{C} & \mathbf{0} & \mathbf{0}\end{bmatrix}\left(\left(\mathbf{I} - \mathbf{M}\bar{\mathbf{C}}\right)\bar{\mathbf{x}}(k|k-1) + \begin{bmatrix}\mathbf{M} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{y}(k)\\ \mathbf{u}(k)\end{bmatrix}\right) \tag{46b}$$

or, in a compact form,

$$\bar{\mathbf{x}}(k+1|k) = \mathbf{A}_{MKP}\bar{\mathbf{x}}(k|k-1) + \mathbf{L}\mathbf{y}(k) + \bar{\mathbf{B}}\mathbf{u}(k) \tag{47a}$$

$$\mathbf{y}_p(k+D|k) = \mathbf{C}_{MKP}\bar{\mathbf{x}}(k|k-1) + \mathbf{D}_{MKP}\mathbf{y}(k) \tag{47b}$$

with

$$\mathbf{A}_{MKP} = \left(\bar{\mathbf{A}} - \mathbf{L}\bar{\mathbf{C}}\right), \quad \mathbf{C}_{MKP} = \begin{bmatrix}\mathbf{C} & \mathbf{0} & \mathbf{0}\end{bmatrix}\left(\mathbf{I} - \mathbf{M}\bar{\mathbf{C}}\right), \quad \mathbf{D}_{MKP} = \begin{bmatrix}\mathbf{C} & \mathbf{0} & \mathbf{0}\end{bmatrix}\mathbf{M}.$$

The KP can then be implemented using its state-space representation (47), or its matrix transfer function which is given by

$$\mathbf{Y}_p(z) = \mathcal{Z}\{\mathbf{y}_p(k+D|k)\} = \mathbf{S}_{MKP}(z)\mathbf{U}(z) + \mathbf{F}_{MKP}(z)\mathbf{Y}(z), \tag{48}$$

where

$$\mathbf{S}_{MKP}(z) = \mathbf{C}_{MKP}(z\mathbf{I} - \mathbf{A}_{MKP})^{-1}\bar{\mathbf{B}}, \tag{49}$$

$$\mathbf{F}_{MKP}(z) = \mathbf{C}_{MKP}(z\mathbf{I} - \mathbf{A}_{MKP})^{-1}\mathbf{L} + \mathbf{D}_{MKP}. \tag{50}$$

4.4 KALMAN PREDICTOR ANALYSIS

In Chapter 3, where the Kalman Predictor was introduced for SISO systems, it was shown that the KP implicitly defines an FSP which allows the direct use of all analysis tools of the FSP as presented in Section 4.1. This relation is still valid for the MKP, as stated in Theorem 4.1.

Theorem 4.1. *The Modified Kalman Predictor (MKP), defined by (47) and (48) for MIMO square processes, implicitly defines a MIMO Filtered Smith Predictor (FSP) (described by (36)) where the prediction-error filter* $\mathbf{F}_{FSP}(z)$ *is equal to* $\mathbf{F}_{MKP}(z)$ *from (50).*

Proof. See Lima; Lima; Normey-Rico (12), Appendix B. ■

This theorem makes the closed-loop analysis of the MKP in the frequency domain much more simple. Since the MIMO FSP and the MKP are equivalent, all the properties of the MIMO FSP given in Section 4.1 are valid for the MKP. It is only necessary to obtain $\mathbf{F}_{MKP}(z)$, the implicit prediction-error filter of the MKP, given by (47).

Generally, for step disturbances, the $\mathbf{F}_{FSP}(z)$ static gain is chosen such that $\mathbf{F}_{FSP}(1) = \mathbf{I}$. This is necessary to guarantee that the prediction $\mathbf{y}_p(k+D|k)$ will converge to the real output $\mathbf{y}(k)$ in steady-state. From the MIMO FSP prediction equation (30),

$$\mathbf{Y}_p(1) = (\mathbf{I} - \mathbf{F}_{FSP}(1)\mathbf{L}(1))\,\mathbf{G}(1)\mathbf{U}(1) + \mathbf{F}_{FSP}(1)\mathbf{Y}(1)$$

and since, by definition, $\mathbf{L}(1) = \mathbf{I}$, the only way for $\mathbf{Y}_p(1) = \mathbf{Y}(1)$ is if $\mathbf{F}_{FSP}(1) = \mathbf{I}$, which results in

$$\begin{aligned}\mathbf{Y}_p(1) &= (\mathbf{I} - \mathbf{I})\,\mathbf{G}(1)\mathbf{U}(1) + \mathbf{Y}(1)\\ \mathbf{Y}_p(1) &= \mathbf{0}\,\mathbf{U}(1) + \mathbf{Y}(1)\\ \mathbf{Y}_p(1) &= \mathbf{Y}(1).\end{aligned}$$

This property is satisfied by the MKP as shown in Corollary 4.1.

Corollary 4.1. *Given Theorem 4.1, the implicit prediction error filter* $\mathbf{F}_{MKP}(z)$ *defined by the Modified Kalman Predictor (47) for MIMO square systems has unitary static gain, that is:*

$$\mathbf{F}_{MKP}(1) = \mathbf{I}. \tag{51}$$

Proof. See Lima; Lima; Normey-Rico (12), Appendix C. ■

Another important property of the MKP has to do with its application to unstable and integrative processes. For the reasons explained in Section 4.1, the MIMO FSP must be tuned following some specific rules, given by (34), to avoid internal and closed-loop instability. Note that each undesirable pole will add another condition that $\mathbf{F}_{FSP}(z)$ must

satisfy making the tuning procedure more involved depending of the size of the system and number of undesirable poles. The MKP, however, inherently does this independent of tuning and of the system's characteristics. This is shown in Corollary 4.2:

Corollary 4.2. *Given the detectability of system (44), Theorem 4.1 and the MKP defined by (47) and (48), independent of tuning, the obtained observer is stable, thus, the resulting implicit* $\mathbf{S}_{MKP}(z)$ *will never have any integrative or unstable open-loop poles of* $\mathbf{P}(z)$.

Proof. By Theorem 4.1, the MKP implicitly implements a MIMO FSP given by (36). Thus, if $\mathbf{F}_{FSP}(z) = \mathbf{F}_{MKP}(z)$, then $\mathbf{S}_{FSP}(z) = \mathbf{S}_{MKP}(z)$. In the MIMO FSP, the internal stability is obtained if $\mathbf{S}_{FSP}(z)$ is stable. However, by the definition (48) of $\mathbf{S}_{MKP}(z)$, its poles are the eigenvalues of $\mathbf{A}_{MKP} = \bar{\mathbf{A}} - \mathbf{L}\bar{\mathbf{C}}$, and, as the observer is designed to be stable, its closed-loop poles are all inside the unitary circle. Therefore, $\mathbf{S}_{MKP}(z)$ is stable which guarantees the internal stability of the KP. ■

Remark 4.1. Note that the poles of the filter and of the S_{MKP} are defined with the choice of the MKP tuning parameters and, in practice, could be slow poles when robustness is required or fast poles when improved disturbance rejection is desired (as it is done in the classical MIMO FSP).

Another interesting point is that the KP automatically adjust the elements out of the main diagonal of the prediction error filter to obtain the desired result, whereas traditional approaches such as the MIMO FSP only use the main diagonal, which can make the tuning procedure complicated given the coupling between the variables. This happens with the MKP because the matrices $\mathbf{L}$ and $\mathbf{M}$, which define the filter $\mathbf{F}_{MKP}(z)$ (47), generally do not have null elements. Hence, $\mathbf{F}_{MKP}(z)$, in general, will have all of its elements nonzero transfer functions, except in very specific cases.

In summary, the advantages of the proposed Modified Kalman Predictor are the following:

a) With the proposed system model, the KP deals with multiple delays and directly estimates the outputs after the minimal delays without the need for integration of the model, as is common with other predictors.

b) The procedure to obtain the KP is independent of the order of the transfers functions, the number of inputs/outputs, and the type of dynamics including unstable, integrative and non-minimum phase.

c) Although the number of tuning parameters increases with the number of inputs/outputs, there is no need for specific guidelines for unstable, integrative or non-minimum phase processes as is the case with some of the other predictors (52, 53, 54, 55).

4.5 SIMULATION RESULTS

In this section, two examples will be shown to highlight some important characteristics of the proposed predictor. It is important to emphasize that, in this work, the problems of control and prediction are considered separately, with a focus on the latter. Hence, in the comparative examples, the same primary controllers are used and the changes will be made to the predictor structures only. This will help highlight the predictor importance in disturbance rejection and robustness. Nonetheless, ideally, these two problems should be dealt together as in Normey-Rico; Camacho (35) and Santos; Franklin; Torrico (69). Moreover, note that, at least in theory, it should be possible to tune different predictor structures to obtain the same closed-loop behavior for a given process. This, however, might not be trivial due to the different approaches used to obtain the predictions. This aspect is important because the main point here is not to show that a given predictor structure will have better or worse response than others, but that it is easier to implement and tune.

4.5.1 Example 1: Disturbance Rejection Tuning

In this first example, an unstable MIMO process will be used to show that the proposed predictor can cope with unstable dynamics and to show tuning guidelines regarding disturbance rejection. This process was presented in Flesch et al. (59), and its continuous transfer matrix is given by

$$\mathbf{P}(s) = \begin{bmatrix} \frac{1.6e^{-4s}}{-2.6s+1} & \frac{0.6e^{-6s}}{2.5s+1} \\ \frac{0.7e^{-5s}}{3s+1} & \frac{1.7e^{-3s}}{-2.2s+1} \end{bmatrix}. \tag{52}$$

Discretizing with a zero-order-hold and a sampling time of $T_s = 0.2$ [s] results in

$$\begin{aligned} \mathbf{P}(z) =& \mathbf{L}(z)\mathbf{G}(z) \\ =& \begin{bmatrix} z^{-20} & 0 \\ 0 & z^{-15} \end{bmatrix} \begin{bmatrix} \frac{-0.12793}{z-1.08} & \frac{0.04613z^{-10}}{z-0.9231} \\ \frac{0.045145z^{-10}}{z-0.9355} & \frac{-0.1618}{z-1.095} \end{bmatrix}. \end{aligned} \tag{53}$$

To control this process, the same control structure and tuning of Flesch et al. (59) will be used (Figure 14), which was a Filtered Smith Predictor with a decoupler and PI controllers aiming at a settling time of about 12 [s] in each process variable and with low coupling. The primary controller $\mathbf{C}(z)$, reference filter $\mathbf{F}_r(z)$ and decoupler $\mathbf{D}(z)$ are given below:

$$\mathbf{C}(z) = \text{diag}\left(\frac{-2.2766(z-0.9615)}{z-1}, \frac{-2.154(z-0.9661)}{z-1} \right), \tag{54}$$

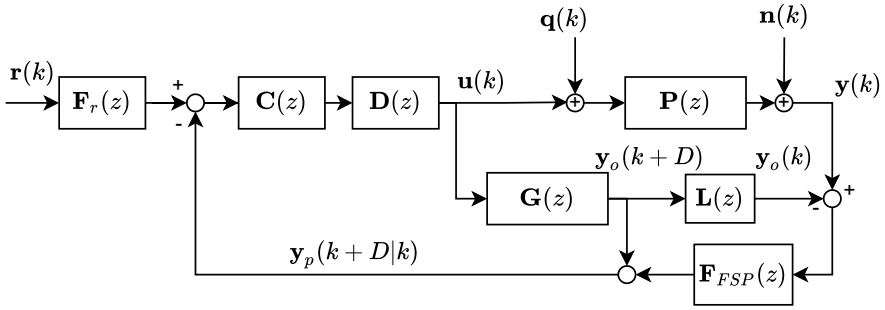
$$\mathbf{F}_r(z) = \text{diag}\left(\frac{0.25641(z-0.85)}{(z-0.9615)}, \frac{0.22599(z-0.85)}{(z-0.9661)} \right), \tag{55}$$

$$\mathbf{D}(z) = \begin{bmatrix} 1 & \frac{0.36058(z-1.08)z^{-10}}{(z-0.9231)} \\ \frac{0.27904(z-1.095)z^{-10}}{(z-0.9355)} & 1 \end{bmatrix}. \tag{56}$$

The filter of the MIMO FSP was tuned to guarantee the internal stability of the predictor, as discussed in Section 4.1, and to obtain an adequate trade-off between robustness and disturbance rejection:

$$\mathbf{F}_{FSP}(z) = \text{diag}\left(\frac{9.232(z-0.9892)}{(z-0.9)}, \frac{5.439(z-0.9908)}{(z-0.95)}\right) \tag{57}$$

Figure 14 – Control Structure of an FSP and a decoupler $\mathbf{D}(z)$.



Source: Diagram from Lima; Lima; Normey-Rico (12) based on Flesch et al. (59).

The simulation scenario is the following: a change in the reference of 0.75 in y_1 is applied at $t = 5\,[\text{s}]$, a change in the reference of 0.35 in y_2 is applied at $t = 40\,[\text{s}]$, and input step disturbances with 0.05 of amplitude are applied at $t = 80\,[\text{s}]$, and $t = 120\,[\text{s}]$ in inputs u_1 and u_2, respectively. Since, in this example, only the nominal case will be dealt with, the reference tracking is always the same, regardless of the predictor and its tuning, as explained earlier. For clarity, the simulation results are divided into two figures. The reference tracking portion of the simulation scenario is shown in Figure 15 and the disturbance rejection responses are shown in figures 16 and 17 for six different cases chosen to highlight and explain the tuning procedure of the Modified Kalman Predictor. For the reference tracking, only one case is shown because the predictors do not change the nominal reference-tracking response. Notice that the response has a settling time of approximately 12 [s] after the dead time.

Each case that will be presented has a different predictor or a different tuning:

a) Case FSP: uses the MIMO Filtered Smith Predictor of Flesch et al. (59) described earlier.

b) Case 0: the base case for comparison. MKP with $\mathcal{R} = \text{diag}(1, 1)$, $\mathcal{Q} = \text{diag}(0.05, 0.05)$.

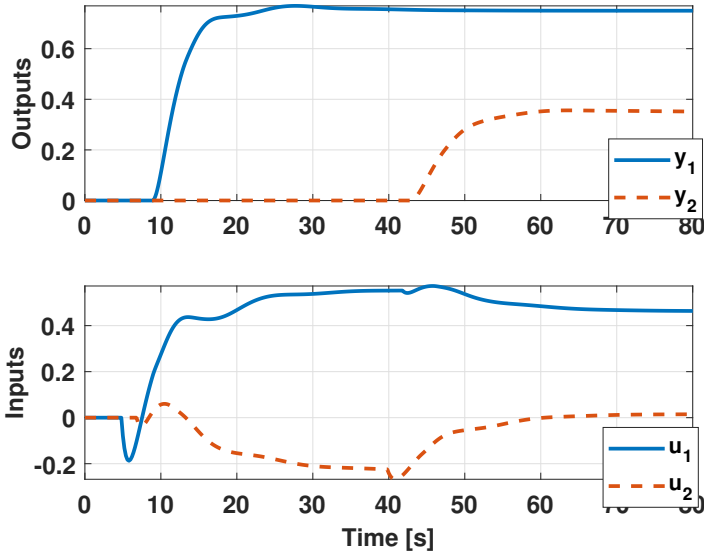
c) Case Y1: worse disturbance rejection for y_1. MKP with $\mathcal{R} = \text{diag}(10, 1)$, $\mathcal{Q} = \text{diag}(0.05, 0.05)$.

d) Case Y2: worse disturbance rejection for y_2. MKP with $\mathcal{R} = \text{diag}(1, 10)$, $\mathcal{Q} = \text{diag}(0.05, 0.05)$.

e) Case Q1: better disturbance rejection for q_1. MKP with $\mathcal{R} = \text{diag}(1, 1)$, $\mathcal{Q} = \text{diag}(5, 0.05)$.

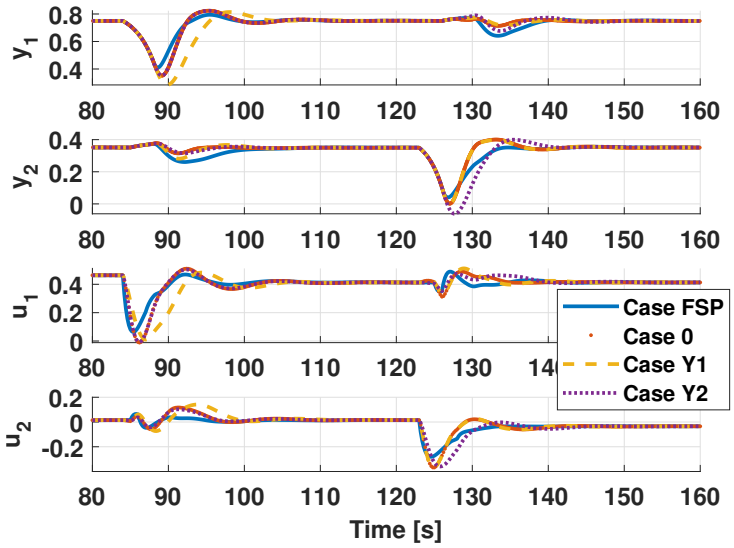
f) Case Q2: better disturbance rejection for q_2. MKP with $\mathcal{R} = \text{diag}(1, 1)$, $\mathcal{Q} = \text{diag}(0.05, 5)$.

Figure 15 – Example 1 - nominal reference tracking for all cases are identical.



Source: Figure from Lima; Lima; Normey-Rico (12).

Figure 16 – Example 1 - nominal reference tracking for all cases are identical. Varying $\mathcal{R}$.



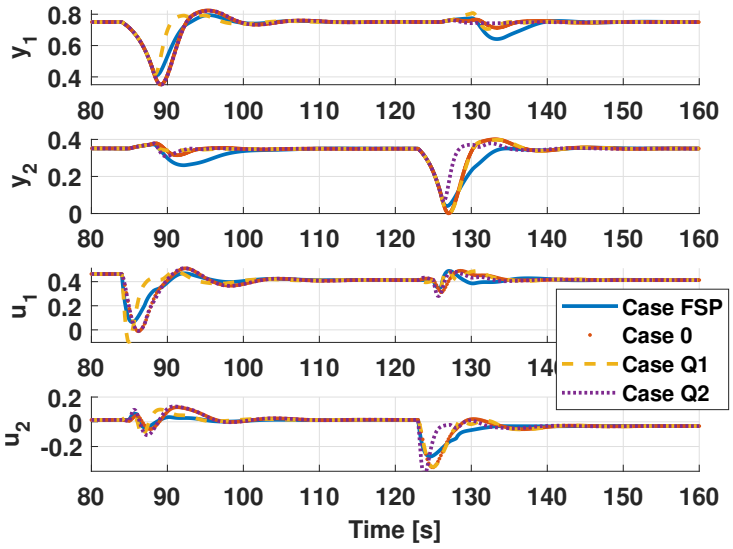
Source: Figure from Lima; Lima; Normey-Rico (12).

To make the comparison easier, the Integral Squared Error (ISE) was computed, which has the following definition for a generic signal x:

$$\delta(t_i, t_f, x) = \int_{t_i}^{t_f} (x_{ref}(t) - x(t))^2 dt, \tag{58}$$

where x_{ref} is the reference for x while t_i and t_f indicate the initial and final time of

Figure 17 – Example 1 - nominal reference tracking for all cases are identical. Varying $\mathcal{Q}$.



Source: Figure from Lima; Lima; Normey-Rico (12).

Table 2 – Normalized Integral Squared Errors (ISE) for Example 1.

Case	δ_{y_1}	δ_{y_2}	δ_{q_1}	δ_{q_2}
Case FSP	0.7470	1.0935	0.7405	1.1041
Case 0	1.0000	1.0000	1.0000	1.0000
Case Y1	**1.8425**	1.0217	1.8589	0.9952
Case Y2	1.0273	**1.9317**	0.9979	1.9781
Case Q1	0.4603	1.0124	**0.4428**	1.0400
Case Q2	1.0014	0.4197	1.0073	**0.4096**

Source: Table from Lima; Lima; Normey-Rico (12).

evaluation, respectively.[1] Considering the ISE index characteristics, a better performance is associated with smaller values of δ, since it indicates smaller errors. The quantitative values based in (58), normalized in relation to Case 0, are shown in Table 2. Thus, δ values smaller than 1 indicate a better performance than Case 0, and the opposite is true for values greater than 1. In the table, $\delta_{y_1} = \delta(80,\ 160,\ y_1)$ and $\delta_{y_2} = \delta(80,\ 160,\ y_2)$ reflect the disturbance rejection of each output while $\delta_{q_1} = \delta(80,\ 120,\ y_1) + \delta(80,\ 120,\ y_2)$ and $\delta_{q_2} = \delta(120,\ 160,\ y_1) + \delta(120,\ 160,\ y_2)$ reflect the disturbance rejection in all outputs for a particular disturbance.

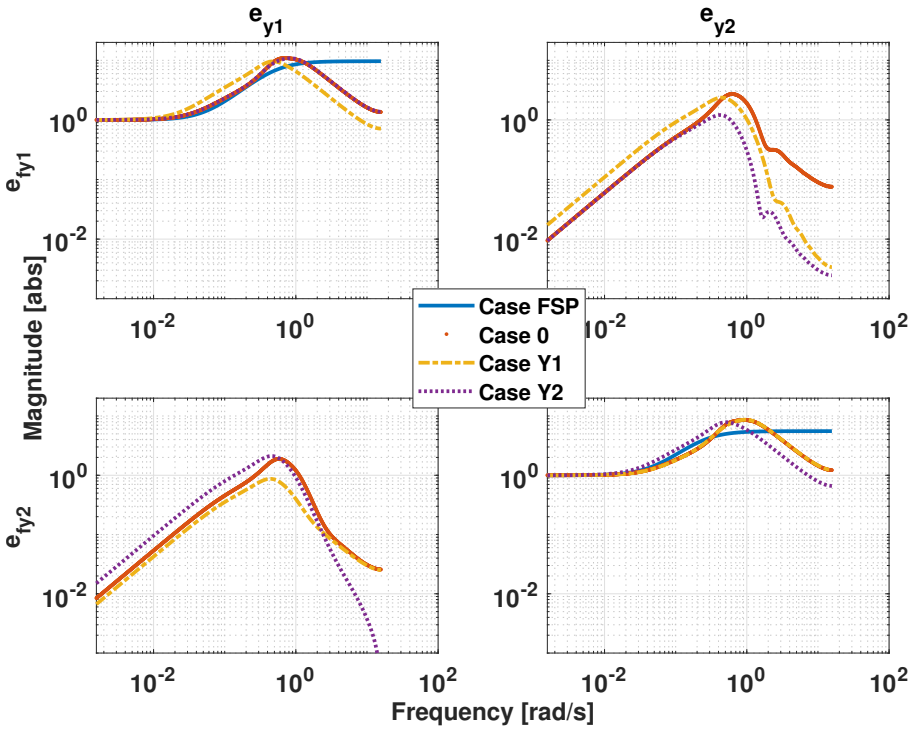
From Figure 16, note that the disturbance rejection for Case Y1 is slower for y_1 in comparison with the base case, whereas for y_2 the response is almost the same. This is also reflected in δ_{y_1} in Table 2, which increases 84 % in relation to Case 0, while δ_{y_2} increases only 2.2 %. The converse is true for Case Y2. Given the coupled nature of the process, a completely decoupled tuning is very hard for any predictor but, with the KP, the coupling

[1] Other performance indices could be used, i.e., Integral Absolute Error (IAE) or Integral Time Absolute Error (ITAE), but, in our example, they all resulted in the same conclusions and were omitted.

effects in the tuning are minimized without any special consideration.

This behavior is reflected in the Bode Diagram of the implicit prediction error filter of the KP which is shown in Figure 18, and where e_{yi} is the i-th input of the filter and the i-th prediction error, and e_{fyi} is the filtered prediction error and i-th output of the filter. With Case Y1, the part of the filter associated with y_1 (e_{y1}-e_{fy1}) becomes less high pass, which indicates a slower dynamics in relation to Case 0, while the part associated with y_2 (e_{y2}-e_{fy2}) almost does not change. The converse is true for Case Y2. The Bode diagram also confirms that the KP automatically adjust the elements of the filter out of the main diagonal to obtain the desired result. It is also important to emphasize that, in the Bode diagram, plots with higher peaks and high magnitude in high frequencies imply faster responses and lower robustness, and vice-versa.

Figure 18 – Example 1 - Bode Diagram of the different prediction error filters. Varying $\mathcal{R}$.

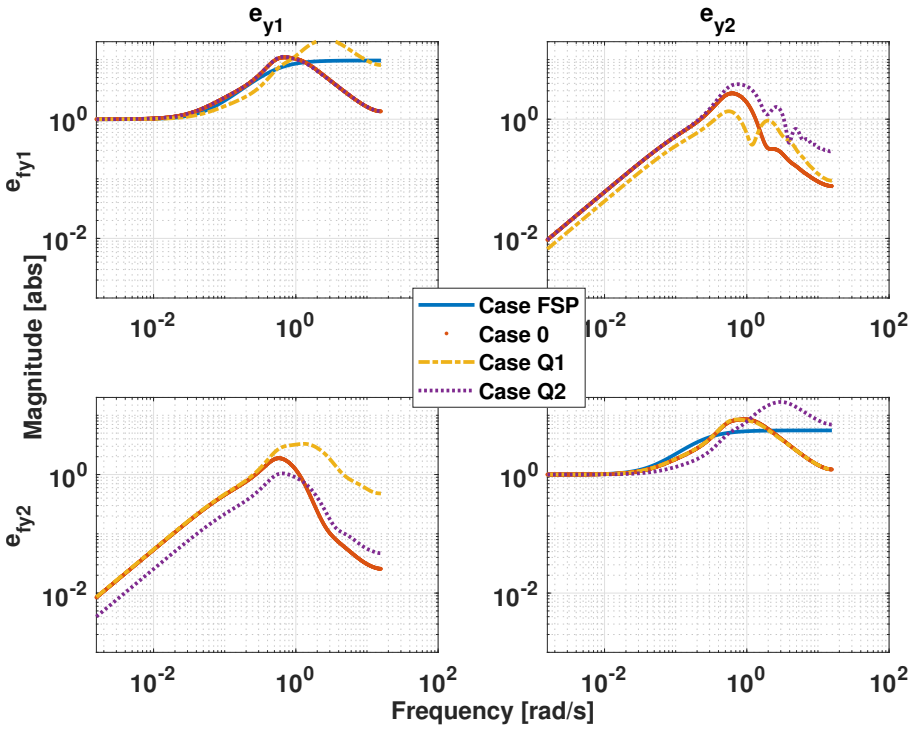


Source: Figure from Lima; Lima; Normey-Rico (12).

In cases Q1 and Q2 the weights associated with the disturbances $\mathcal{Q}$ are varied. In Case Q1, the weight associated with q_1 is increased a hundredfold to 5, indicating a higher variance with this variable, making the predictor react more aggressively in the estimation of q_1. In Case Q2, the same is done but with q_2. The results can be seen in Figure 17. Note that rejection of the disturbance in u_1 ($t = 80\,[\mathrm{s}]$) is faster in both output variables, whereas the rejection of the disturbance in u_2 ($t = 120\,[\mathrm{s}]$) remains practically the same for both outputs. This is also reflected in δ_{q_1} in Table 2 which decreases 56 % in relation to Case 0, whereas δ_{q_2} actually increases a little (4.0 %). The converse is true for Case Q2.

As in cases Y1 and Y2, this behavior is reflected in the Bode Diagram of the implicit prediction error filter of the KP (Figure 19). With Case Q1, the part of the filter associated with y_1 (e_{y1}-e_{fy1}) becomes more high-pass, which indicates a faster dynamics in relation to Case 0, while the part associated with y_2 (e_{y2}-e_{fy2}) almost does not change. The converse is true for Case Q2.

Figure 19 – Example 1 - Bode Diagram of the different prediction error filters. Varying $\mathcal{Q}$.



Source: Figure from Lima; Lima; Normey-Rico (12).

From these results, note that the KP is capable of handling even systems with unstable dynamics without any change to its formulation, which is an important property. Also, its tuning parameters allows the adjustment of the closed-loop response considering each output and input disturbance individually, giving a good number of degrees of freedom to adjust the performance of the closed-loop system. For example, to improve the disturbance rejection for disturbance i, the q_i element of matrix Q must be bigger than the rest of q_j, $\forall j \neq i$. If we intend to worsen the disturbance rejection for output y_i, the r_i element of matrix R must be bigger than the rest of r_j, $\forall j \neq i$. It is important to note that this tuning procedure does not guarantee a perfect decoupling effect, and it is expected that, although all variables are affected by the tuning choices, the selected ones will be the most affected. Finally, the KP tuning does not depend on the order of the system or its dynamics.

Also, from this example, one can draw some conclusion that are useful to tune the Kalman Predictor:

a) Decreasing the weight associated with a given output variable y_i in $\mathcal{R}$ improves the disturbance rejection associated with this variable.

b) Increasing the weight associated with a given disturbance q_i in $\mathcal{Q}$ improves the disturbance rejection associated with the i-th input.

These are heuristics guidelines but, coupled with a frequency domain analysis of the predictor, it is possible to tune the predictor adequately to obtain the desired response. Also, it is important to emphasize that a better disturbance rejection usually results in a worse robustness for the closed-loop system, thus, a careful analysis is important in each specific case, as will be shown in the next example.

4.5.2 Example 2: Robustness Tuning

The example presented in this section is the Shell's Heavy Oil Fractionator, a process widely used in the literature to test control structures (70, 3, 31). This problem will be used to show how the Kalman Predictor can be used to improve the robustness of a control structure.

The controlled variables of this process are: the top (y_1) and side (y_2) product compositions measured by analyzers, and the bottom temperature (y_3). The manipulated variables are: the top draw rate (u_1), the side draw rate (u_2) and the bottom reflux duty (u_3). The upper reflux duty (q_1) is considered an unmeasurable disturbance. Near the operating point, the dynamics of the process can be described by the following model:

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \\ Y_3(s) \end{bmatrix} = \begin{bmatrix} \frac{4.05e^{-28s}}{50s+1} & \frac{1.77e^{-28s}}{60s+1} & \frac{5.88e^{-28s}}{50s+1} \\ \frac{5.39e^{-18s}}{50s+1} & \frac{5.72e^{-14s}}{60s+1} & \frac{6.9e^{-16s}}{40s+1} \\ \frac{4.38e^{-20s}}{33s+1} & \frac{4.42e^{-22s}}{44s+1} & \frac{7.2}{19s+1} \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \\ U_3(s) \end{bmatrix} + \begin{bmatrix} \frac{1.44e^{-27s}}{1+40s} \\ \frac{1.83e^{-15s}}{1+20s} \\ \frac{1.26}{1+32s} \end{bmatrix} Q_1(s). \tag{59}$$

The discrete model was obtained with discretizing with a zero-order-hold and a sampling time of $T_s = 2\,[\text{min}]$.

The simulation scenario is the following: at $t = 0\,[\text{min}]$, the references of y_1, y_2 and y_3 are changed to 0.5, 0.3 and 0.1, respectively; at $t = 100\,[\text{min}]$ the y_1 reference is changed to 0.4; the disturbance q_1 changes its value from 0 to 0.02 at $t = 200\,[\text{min}]$.

For this process, the primary controller will be a GPC with the following tuning parameters (71): weighting factors for error tracking $\delta_1 = \delta_2 = \delta_3 = 1$; weighting factor for control increments $\lambda_1 = \lambda_2 = 0.3$, $\lambda_3 = 0.5$; control horizons $N_{u1} = N_{u2} = N_{u3} = 30$; prediction horizons $N_1 = N_2 = N_3 = 60$ after each minimal output delay, which gives a settling time of about 13 [min] for each output after the minimum dead-time. Three cases will be considered to highlight the robustness properties of the MKP while using the same primary controller based on the GPC:

a) Case GPC: the predictor here is the implicit one defined by the GPC algorithm when the nominal plant $P(z)$ is used (31).

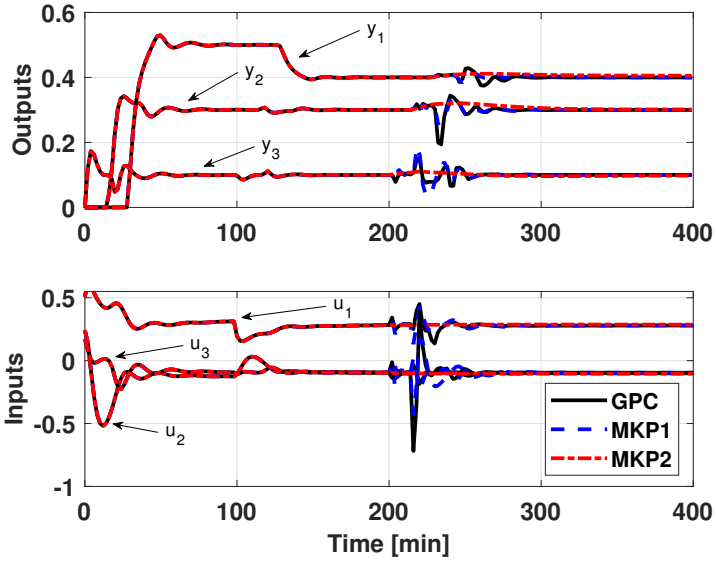
b) Case MKP1: the implicit GPC predictor is substituted by the MKP, which is

tuned to achieve a similar result to the previous case[2]. $\mathcal{Q} = \text{diag}(0.1, 3, 50)$, $\mathcal{R} = \text{diag}(0.1, 0.1, 0.5)$.

c) Case MKP2: the same as Case MKP1, but tuned for robustness. $\mathcal{Q} = \mathbf{I}$, $\mathcal{R} = 13000\mathbf{I}$.

In Figure 20, the nominal case is presented. Note that the reference tracking for all cases is the same, as was expected, and that disturbance rejection of Case GPC and Case KP1 are similar. Case KP2 has a slower disturbance rejection, as expected from the tuning.

Figure 20 – Example 2 - reference tracking and disturbance rejection in the nominal case.



Source: Figure from Lima; Lima; Normey-Rico (12).

To show this quantitatively, the Integral Time Absolute Error (ITAE) was calculated for each case, and then normalized in relation to Case GPC. The ITAE was used here instead of the ISE because it represented more accurately the slow performance of Case MKP2. The ITAE for a vector $\mathbf{x}$ is calculated as:

$$\text{ITAE}(t_i, t_f, \mathbf{x}) = \int_{t_i}^{t_f} t \, |\mathbf{x}_{ref}(t) - \mathbf{x}(t)| \, dt,$$

where $t_i = 200$, $t_f = 400$, so that the performance index is computed only for the disturbance rejection, since the reference tracking is the same. The ITAE values are shown in Table 3, where smaller values indicate a better disturbance rejection.

An SVD analysis of the frequency domain characteristics of the prediction error filters was made and is presented in Figure 21. The filter from Case GPC has higher gain at high frequencies if compared to Case KP1 which presents similar disturbance rejection. In other words, Case KP1 has better robustness with similar performance. Case KP2 has low-pass characteristics which improves robustness at the cost of disturbance rejection performance.

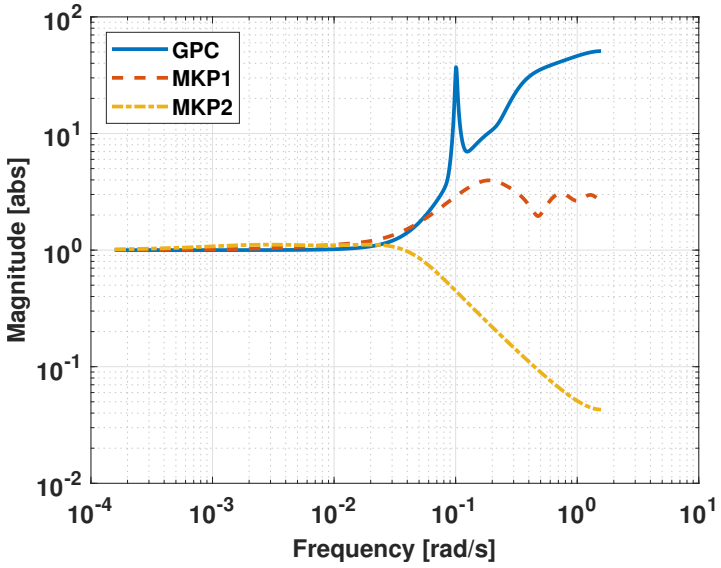
[2] The GPC is tuned using the model of the process without the minimal delays.

Table 3 – Normalized Integral Time Absolute Errors (ITAE) for Example 2.

Case	ITAE
Case GPC	1.0000
Case MKP1	0.9961
Case MKP2	2.7217

Source: Table from Lima; Lima; Normey-Rico (12).

Figure 21 – Example 2 - prediction error filters SVD ($\bar{\sigma}(\cdot)$) analysis.

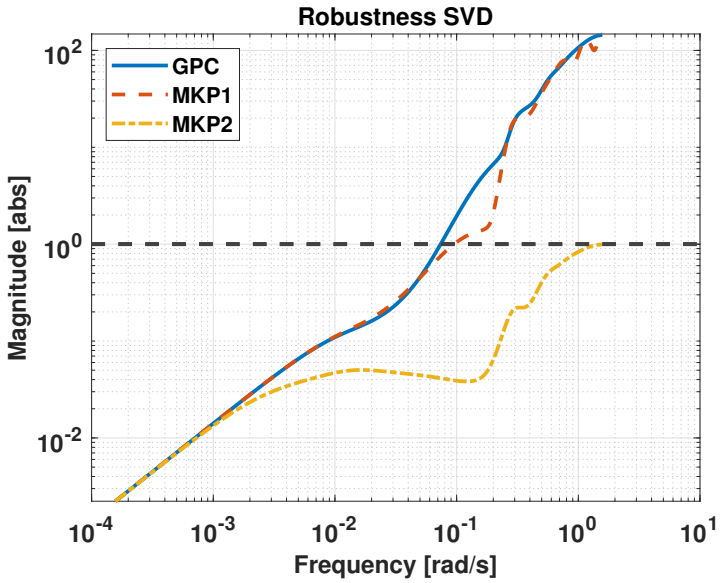


Source: Figure from Lima; Lima; Normey-Rico (12).

To verify the robust performance, consider that there is a model mismatch in the delays of (59). The delay is 30 in element $(1,2)$ of $\mathbf{P}(s)$ instead of 28 used in the model of the predictor and 16 instead of 18 in element $(2,1)$. In both cases this translates in a one sample delay mismatch in the discretized model. Considering an additive modeling of the model mismatch, the control scheme must satisfy the robustness condition (37) which is presented in Figure 22 for all cases, given that $\mathbf{W}_a(z) = \mathbf{W}_2(z)\Delta(z)\mathbf{W}_1$, $\mathbf{W}_2 = \mathbf{I}$ and $\mathbf{W}_1 = \mathbf{P}(z) - \mathbf{P}_n(z)$, where $\mathbf{P}_n(z)$ is the nominal model. From Figure 22, it is clear that Case GPC and Case KP1 do not satisfy the robustness condition, hence, closed-loop stability for this set of uncertainties is not guaranteed. Case KP2 does satisfy the condition, meaning that robust stability is achieved for this set of uncertainties. The simulation results of the uncertain case (Figure 23) corroborate this analysis, only Case KP2 is stable.

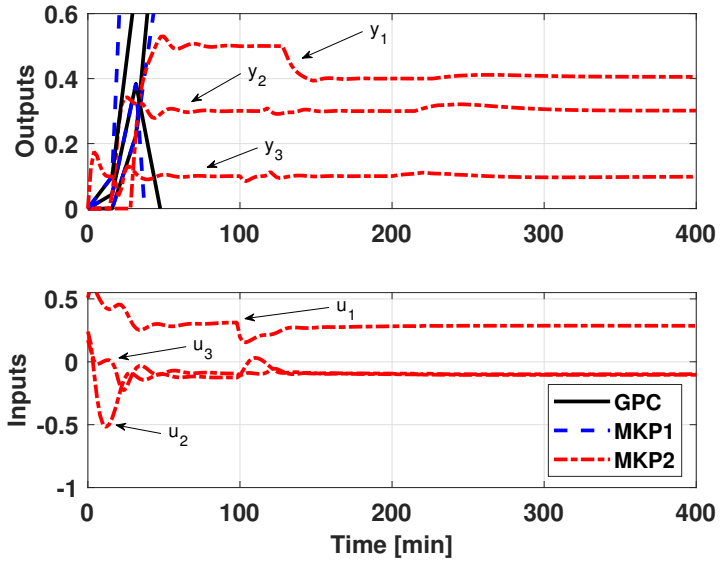
Hence, the results from this example have confirmed that the predictor structure has an important role in the closed-loop robustness. In the MPC context, this was originally addressed in Normey-Rico; Eduardo F. Camacho (31), where it was proven that the implicit predictor of some MPC algorithms provide very little robustness to model uncertainties. But this behavior can be improved by substituting the implicit predictor by an SP or FSP based strategy, just as was made here with the MKP. With the proposed MIMO KP, it

Figure 22 – Example 2 - robustness analysis using (37). Grey dashed line indicates robustness condition limit.



Source: Figure from Lima; Lima; Normey-Rico (12).

Figure 23 – Example 2 - reference tracking and disturbance rejection in the uncertain case. GPC and MKP1 cases are unstable and the simulation of these two cases was stopped at $t = 50$ min approximately and only the process outputs are shown.



Source: Figure from Lima; Lima; Normey-Rico (12).

becomes easier to adjust the system performance without changing the primary controller and without using more complex computations to define the internal stability properties of the predictor, as is the case with other predictors such as the MIMO FSP. Thus, for practitioners, the MKP seems to be an interesting approach for predictor based controllers in MIMO delayed systems. Finally, although the use of a separate tuning strategy for the predictor and primary controller is useful in practice, it is important to emphasize that, in some situations better results can be achieved when both primary controller and predictor

structure are tuned together. This can be a topic of future research in the context of the MKP.

4.6 CONCLUSION

The Modified Kalman Predictor (MKP) presented in this chapter adapts the predictor shown in Lima; Lima; Normey-Rico (11) to multiple-delay-MIMO square systems of any order and dynamics. The MKP has intuitive tuning guidelines and it allows the use of traditional frequency domain analysis tools to verify its closed-loop properties, such as disturbance rejection and robustness, as shown through two examples. Its formulation provides a straight-forward approach to obtain the prediction of the system that does not require integration of the model. Moreover, as its structure is general and does not require complex conditions, its implementation and tuning is easy, allowing to achieve a compromise between robustness and disturbance rejection.

It is important to emphasize that other predictor structures can also be tuned to offer improved disturbance rejection or improved robustness similarly to the MKP, but it is argued that the tuning procedure is not as easy, and many of these structures do not cope with multiple delays.

One of it limitations is that it is dependent of the minimal output delays of the system. If they are small in relation to the remaining delays, the advantages of the MKP are reduced since the order of the fast model $G(z)$ will not be reduced significantly in relation to the full process model $P(z)$. Another issue is that, in its current form, it can only be applied to square systems and only step disturbances are considered in the formulation. Furthermore, for bigger system, there might be some numerical issues in the computation of the Kalman Filter. These limitations will be the subject of future research.

5 MHE OF FAULTS IN RENEWABLE MICROGRIDS

Context. This work was done in 2018 in collaboration with Marcelo M. Morato (graduate student) and Paulo R. C. Mendes (post-doc researcher). At the time, they saw an opportunity to apply an MHE on a industrial process that they were studying. The results summarized in this chapter come from the following published article:

- (13) B. M. Lima, M. M. Morato, P. R. C. Mendes, and J. E. Normey-Rico, "Moving Horizon Estimation of Faults in Renewable Microgrids," *IFAC-PapersOnLine*, vol. 52, no. 1, pp. 311–316, 2019.

Modern technological processes present evermore an increase in complexity and become more and more susceptible to faults. Therefore, the control systems community has been giving attention, for the last two decades or so, to Fault Estimation (FE) methods, including renewable energy systems (72). The system under study is a hybrid energy plant based on a sugarcane processing plant, considering biomass, biogas, solar and wind power energy as primary energy sources (73). In Menezes Morato et al. (9), modern control techniques were applied but the discussion of the effect of faults on this energy system was yet to be researched. This is important because a great portion of the energy production in this system is derived from a biomass-burning boiler and a Combined Heat and Power (CHP) unit, both prone to various kinds of faults. These events can lead the plant to not comply to its operational constraints which results in direct profit loss.

Thus, the main motivation of this work is to estimate faults on an industrial process perceiving possible faults on all subsystems from a supervisory level. To solve this problem, the methods presented in Section 2.2 on optimal estimation were applied, namely EKF and MHE. They are suitable for this process since the equations that describe it become nonlinear when the faults are considered.

The rest of the chapter is divided in three parts: a Brazilian sugarcane energy generation paradigm and possible faults that occur are presented in Section 5.1; an MHE is designed to estimate faults in Section 5.2; and numerical simulations are presented to assess the performance of the proposed MHE in Section 5.3; finally, some final comments are presented in Section 5.4.

5.1 MICROGRID MODEL

The Brazilian sugarcane industry is taken into account which has a great amount of residue and waste that is treated as bio-sources of energy. Brazilian sugar-ethanol plants are good candidates to be considered as hybrid power plants with the use of biomass and biogas for direct-to-electric-energy generation and solar and wind energy generation (73). Most industrial Brazilian sugar cane processing plants have a similar structure with mainly three demands for the processing requirements:

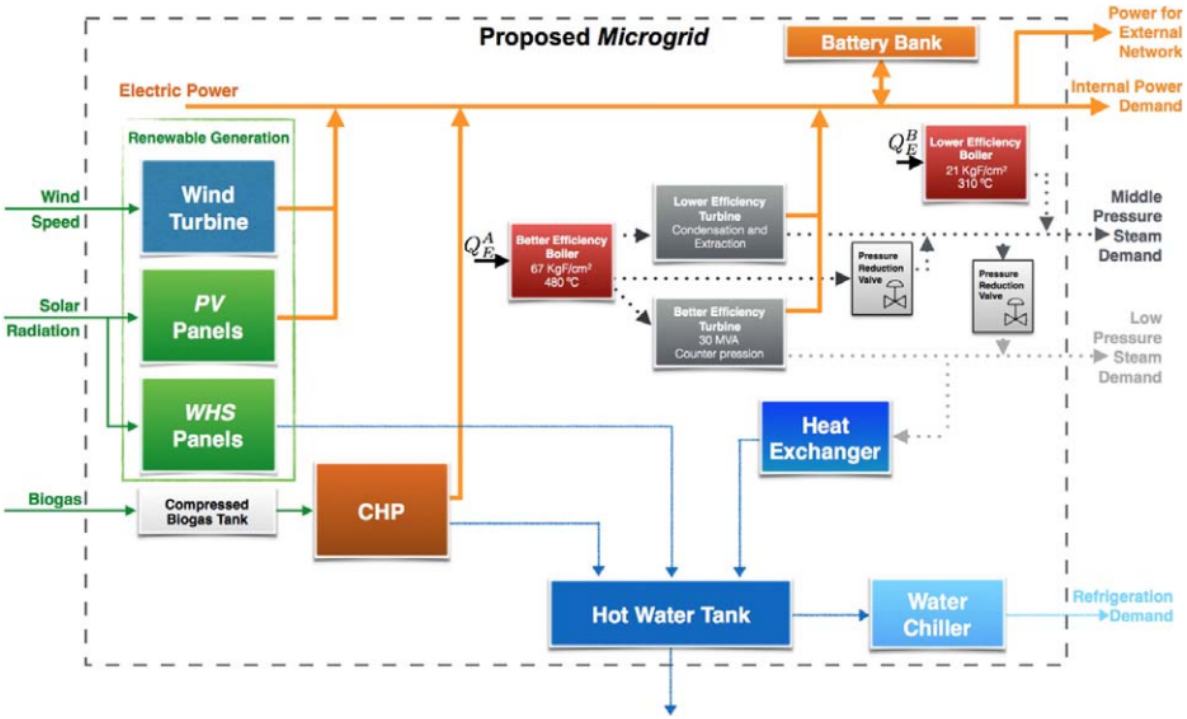
a) Production of steam in different pressures which are used to boost shredders, spray pumps and other equipment;

b) Production of cold water for internal refrigeration needs;

c) Production of electric power to sustain the plant and to sell to a local distribution network.

The microgrid considered here, shown in Figure 24, is composed by: two boilers; two steam turbines; a CHP; a water chiller; a hot water tank; photovoltaic panels; water heating solar panels; a wind turbine; two pressure reduction valves; one heat exchanger; stocks of bagasse, straw and compressed biogas and a battery bank. The use of intermediate storage units in the form of battery banks and biomass stocks allows the system to accumulate energy when the renewable energy generation is high, and use it when it is low. The microgrid's operational constraints are given by the physical requirements of each subsystem such as upper and lower limit bounds of states, controlled outputs and internal setpoints, as well as slew-rate constraints upon the setpoints.

Figure 24 – Proposed generalized hybrid generation system.



Source: Diagram from Morato et al. (73).

For energy management and control goals, these plants can be mathematically modeled as energy hubs (74). A generalized microgrid, considering this modeling framework, is comprised of energy conversion units (e. g. turbines and boilers) and energy storage facilities (e. g. battery and biomass stocks). The control inputs are given by u_j, representing the setpoints to the internal subsystems. The measured outputs y stand for the plant's transmission lines such as electric network, gas pipes and steam lines. A supervisory controller

sets hourly policies u for the plant's internal subsystems such that each global demand is met by z and the energy generation is maximized. This control-oriented hourly-discrete model is given by

$$\Sigma_{MG} \begin{cases} x^+ = A\,x + B_1\omega + B_2 u \\ y = C_1 x + D_{11}\omega + D_{12} u \end{cases} \tag{60}$$

with internal states (x), measured outputs (y), disturbances (ω), and control (u). The supervisory controller is concerned with the controlled outputs (y) which, is this case, are measured. Moreover, the dynamic matrix A is an identity matrix that relates the next $(k+1)$ system state vector with the previous state.

In this model, the state, input, output and disturbance vectors are given by the following expressions:

$$\begin{aligned} x &= (X_{Bat},\, X_{Bag},\, X_{Str},\, X_{Bg},\, X_T) \\ u &= \Big(SP^A_{TU},\, SP^B_{TU},\, Pot_{Net},\, SP^B_C,\, Q^{Out}_V,\, Q^M_{Esc},\, Q^B_{Esc}, \dots \\ &\qquad\qquad SP_{CHP},\, SP_{Ch},\, SP_{TC},\, Pot_{Bat},\, Q^{MB}_V,\, Q^T_{Esc}\Big) \\ y &= \Big(P_{Proc},\, Q^M_V,\, Q^L_V,\, Q_{CW},\, P_{Sale}\Big) \\ \omega &= (Bag_{in},\, Wnd_{in},\, Irrd_{in},\, Str_{in},\, Bg_{in}) \end{aligned} \tag{61}$$

where tx is composed by normalized percentage of each of the plant's storage units: battery bank (X_{Bat}), bagasse stock (X_{Bag}), straw stock (X_{Str}), biogas stock (X_{Bg}) and hot water tank (X_T).

The input vector u stands for zero-order-held setpoints for the conversion and storage units. It is assumed that the sampling period of $\Delta T = 1\,\mathrm{h}$ is large enough so that all these setpoints have been accurately tracked by their respective subsystems. The control vector is composed by: energy boiler's setpoint (SP^B_C), lower efficiency turbine's setpoint (SP^B_{TU}), higher efficiency turbine's setpoint SP^A_{TU}), setpoint of energy flow to/from the battery bank (Pot_{Bat}), CHP's setpoint (SP_{CHP}), water chiller's setpoint (SP_{Ch}), heat exchanger's setpoint (SP_{TC}), high-to-middle pressure reduction valve's setpoint (Q^{Out}_V), middle-to-low pressure reduction valve's setpoint (Q^{MB}_V), hot water escape valve's setpoint (Q^T_{Esc}), middle pressure steam escape flow's setpoint (Q^M_{Esc}), low pressure steam escape flow's setpoint (Q^B_{Esc}) and setpoint of power available to the external network (Pot_{Net}).

The system's controlled outputs are given by: P_{Proc} the electric power produced due to the sugar cane processing demand in kW; Q^M_V the flow of middle pressure steam ($1\,\mathrm{Mgh^{-1}}$), i. e. ton per hour); Q^L_V the flow of low pressure steam ($\mathrm{Mgh^{-1}}$); Q_{CW} the flow of chilled water required by the distillery process ($\mathrm{m^3h^{-1}}$); finally, P_{Sale} represents the electric power made available to the external network in kW.

The disturbances to the system ω are given by: Wnd_{in} the speed of the wind ($\mathrm{kmh^{-1}}$) present in the microgrid's area, used by the wind turbines to generate electric power; $Irrd_{in}$ the amount of solar irradiance ($\mathrm{Mm^{-2}}$) on the microgrid's solar panels (photovoltaic and

water heating); Bag_{in}, Str_{in} and Bg_{in} represent the income of bagasse, straw and biogas to the plant, respectively, in $\mathrm{M\,g\,h^{-1}}$. They are assumed to be measured so they are known by the estimator.

This model was obtained and validated through simulation based on experimental data as described in details in Morato et al. (73). The matrices B_1, B_2 and D_{12} in Equation 60 are given by[1]:

$$B_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \Delta T \eta_{\text{bat}} & 0 & 0 \\ \beta_1 & \beta_3 & 0 & \beta_5 & \beta_7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \beta_2 & \beta_4 & 0 & \beta_6 & \beta_8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \beta_9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \beta_{10} & \beta_{11} & \beta_{12} & 0 & 0 & \beta_{13} \end{bmatrix},$$

$$B_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \varrho_{2,3} & 0 & 0 \\ 0 & 0 & 0 & \varrho_{3,4} & 0 \\ 0 & 0 & 0 & 0 & \varrho_{4,5} \\ 0 & \varrho_{5,2} & 0 & 0 & 0 \end{bmatrix}, \quad D_{11} = \begin{bmatrix} \psi_{1,1} & \psi_{1,2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$D_{12} = \begin{bmatrix} \epsilon_1 & \epsilon_3 & \epsilon_5 & 0 & 0 & 0 & 0 & \epsilon_9 & 0 & 0 & \epsilon_{12} & 0 & 0 \\ 0 & \epsilon_4 & 0 & \epsilon_7 & \epsilon_8 & \epsilon_{15} & 0 & 0 & 0 & 0 & 0 & \epsilon_{13} & 0 \\ \epsilon_2 & 0 & 0 & 0 & 0 & 0 & \epsilon_{16} & 0 & 0 & \epsilon_{11} & 0 & \epsilon_{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \epsilon_{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & \epsilon_6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The operation of sugar-ethanol plants varies during the year. For 200 days of the year during the harvest, the operation is non-stop; while for the remaining months, the whole systems is manually put in rest for revision, conservation and maintenance services. Because of that, most faults that occur on the plant are only corrected after the whole harvest period, without being detected and solved while the energy is generated. Because of that, the considered faults are mainly those that lead to a subsystem's gain decrease and that would only be noted in the maintenance stage. If detected early on, the subsystem's loss of efficiency could be accounted for to optimize the whole plant or it could trigger an early maintenance.

The possible faulty situations that might occur in this sugarcane processing plant and each subsystem has been explored in other another work and they are summarized below (75):

a) Steam rotates the turbines which produce electric energy. Sometimes, the turbine's shovels may get stuck on the inner wall. In practice, if the shovels are not released quickly and the sheer stress does not decrease, the turbines stop their operation. This kind of event is rare but, once a shovel begins to halt its movement, the energy generation rapidly ends (at most, two hours later);

b) The debris and the residues from the burnt biomass accumulate on the burning reservoir. This prevents the total income volume of biomass and leads to a biomass-to-steam gain decrease. The volume of debris and residues that gets stuck on the reservoir can lead up to a 25% decrease on the boiler's gain;

[1] For nominal value and unit for each parameter, the reader is referred to Morato et al. (73).

c) Gas leakages might happen on the CHP unit. Gradually, this implies in a biogas-to-power gain decrease. These leakages do not represent a completely opened duct, but small losses around 10%.

To mathematically represent these faults on a given subsystem j, a multiplicative fault representation is taken, with a time-varying factors $\lambda_j[k]$. This framework has been introduced in Hernández-Alcántara et al. (76) and, in short, it is assumed that the real faulty actuation given by $u^f[k]$ is proportionally dependent on the expected production $u[k]$. The loss of effectiveness factors of $\lambda_j = 1$ stand for faultless situations, while $\lambda_j = 0$ represents that subsystem j has a complete failure or breakdown. With this representation, the whole energy system's state-space representation in Equation (60) changes to:

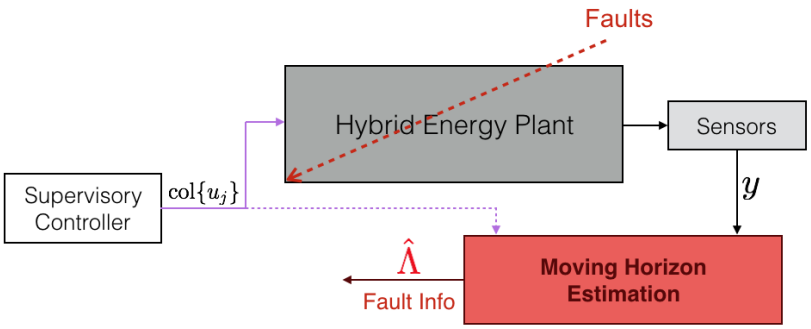
$$\Sigma_{MG}^{f} \begin{cases} x^{+} = A\,x + B_1\,\omega + B_2\,\Lambda\,u \\ y = C_1\,x + D_{11}\,\omega + D_{12}\,\Lambda\,u \end{cases} \tag{62}$$

where the faulty actuation of subsystem j is given by $u_j^f = \lambda_j u_j$ and Λ is a diagonal matrix with the λ term for each subsystem. The main goal of the estimator is to continuously and accurately estimate the term Λ, which gives a valid notion of whether the plant's subsystems are healthy or faulty.

5.2 MHE FOR FAULT ESTIMATION

Considering the problem of a microgrid subject to possible faults on its subsystems, the term Λ can be estimated with the use of an FE scheme that has access to the past control inputs u and past measured output y at each sampling time. An illustration of this fault estimation paradigm is given in Figure (25).

Figure 25 – MHE Fault Estimation Paradigm.



Source: Diagram from Lima et al. (13).

Based on past information of y and u, the MHE solves an optimization problem at

each sampling instant which is defined as:

$$\begin{aligned}
\min \quad & \ell_{ac}\left(\bar{x}_a\right) + \sum_{i=k\text{-}T}^{k-1} \|w_i\|^2_{Q^{-1}} + \sum_{i=k\text{-}T}^{k} \|v_i\|^2_{R^{-1}} \\
\text{s.t.} \quad & x_{i+1} = A\, x_i + B_1\, \omega_i + B_2\, \hat{\Lambda}_i\, u_i \\
& \hat{\Lambda}_{i+1} = \hat{\Lambda}_i + w_i \\
& y_i = C_1 x_i + D_{11}\, \omega_i + D_{12}\, \hat{\Lambda}_i\, u_i + v_i \qquad \forall i \in \mathbb{Z}_{k-T:k-1} \\
& \hat{x}_{\text{lb}} \leq \hat{x}_j \leq \hat{x}_{\text{ub}} \\
& \hat{\Lambda}_{\text{lb}} \leq \hat{\Lambda}_j \leq \hat{\Lambda}_{\text{ub}} \qquad \forall j \in \mathbb{Z}_{k-T:k}
\end{aligned} \tag{63}$$

with u, ω, and y being known variables, k is the current time, T is the estimation horizon, w and v being state and output noises modeled as zero-mean Gaussian with diagonal covariance matrices given by Q and R. Notice that the fault term Λ was modeled as a random-walk.

The arrival cost term $\ell_{ac}(\cdot)$ penalizes the distance between the estimate at the start of the horizon $(k-T)$ and a previous estimate $\bar{x}_a$. This term summarizes the past information outside the estimation horizon. This cost is given by:

$$\ell_{ac}(\bar{x}_a) = \|\hat{x}_a(k-T) - \bar{x}_a\|^2_{P^{-1}}\,,$$

where the augmented state is given by $x_a = [\hat{x}\,\hat{\Lambda}]^T$ and P is a positive definite penalty matrix. One way to compute this estimate $\ell_{ac}(\cdot)$ and the penalty weight P is to use a recursive estimator such as the Kalman Filter or a nonlinear version such as the Extended Kalman Filter (EKF).

At each sampling time k, the optimization problem defined in Equation 63 is solved and a current estimation for the states and faults, i. e. $\hat{x}_k$ and $\hat{\Lambda}_k$, becomes available for control and monitoring purposes. For the following iteration, $\ell_{ac}(\cdot)$ is updated by an EKF and the horizon T is shifted.

It is interesting to notice that, despite a bilinear term $u^f = \Lambda\, u$ appearing in model in Equation 62, the past control actions u are known at every sampling instant k. Because of that, the model is linear in the optimization variables and (63) becomes a convex quadratic problem.

5.3 RESULTS AND ANALYSIS

Numerical simulation results are now presented to illustrate the operation of the proposed MHE method. The following results were obtained using a high-fidelity hybrid model validated and detailed in Morato et al. (73). Results were obtained using MATLAB and YALMIP software packages (77). A realistic 10-days scenario is considered (75):

a) The lower turbine ($j = 2$) has sudden stuck shovels;

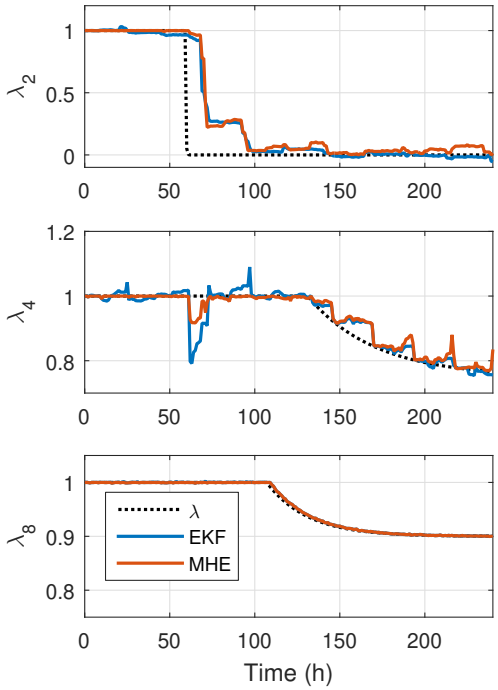
b) The energy boiler ($j = 4$) suffers from continuous accumulation of debris and residues, which leads to a gain decrease of up to 25%;

c) A gas leak occurs in the CHP ($j = 8$), which leads to a gain decrease of up to 5%. This happens at the same time of disturbances w_1 taken from real data of the state of Paraná, Brazil (78).

The plant is simulated with noise in the states and outputs. The MHE is tuned with noise variance of 0.15^2. The variance upon the fault parameters is taken as 0.1^2, which is fairly high considering the range of the loss of effectiveness faults between 0 and 1. This allows a fast convergence of the estimation $\hat{\Lambda}$. The MHE has a horizon of 10 samples and uses an EKF to approximate the arrival cost. They are both initialized with zero estimation error and a small covariance matrix P which represents the where the system is just out of maintenance. The estimation of all the unknown fault terms $\hat{\lambda}_j$ start at 1, which means that all subsystems are fully operational.

The results obtained by the MHE in this scenario are given in Figure 26. These results are compared to those obtained solely by an EKF. Variables λ_2, λ_4 and λ_8 are, respectively, related to the physical outputs of the lower turbine, energy boiler and CHP unit.

Figure 26 – Estimation of λ_2, λ_4, λ_8 for the EKF and MHE. Variables λ_2, λ_4 and λ_8 are, respectively, related to the physical outputs of the lower turbine, energy boiler and CHP unit.



Source: Lima et al. (13).

At $t = 60\,\text{h}$, the lower turbine ($j = 2$) starts to fail due to a stuck shovel, which means all its effectiveness (steam-to-energy gain) is lost in around 2 hours. Both the MHE and the EKF present similar estimation for $\hat{\lambda}_2$, but the MHE presents less overall error.

One can notice that, due to the physical nature of the problem, the estimation of λ_2 and λ_4 is coupled. This is due to the fact that they both directly and indirectly consume biomass mixture (bagasse and straw) from the same stocks. Therefore, when λ_2 changes

due to the stuck shovel, a disturbance appears in $\hat{\lambda}_4$ which is quickly rejected. At first, the estimators are not sure where the measured effects come from. Compared to the EKF, the MHE always respect the upper bound of λ.

The CHP unit fails at $t = 110\,\text{h}$ and the estimation by both MHE and EKF are equally good. The boiler ($j = 4$) starts to fail due to accumulation of debris from $t = 140\,\text{h}$. Estimation of faults by the MHE is a bit smoother with less overall error. The estimation of the other $\hat{\lambda}_j$ terms of the faultless subsystems stay close to 1, as expected. This is equally true for the state estimations $\hat{x}$ which yield good results. Both are not presented herein for brevity.

To further analyze the achieved results, Table 4 presents the Mean Squared Error (MSE) of the fault estimations for both techniques. Most of the other subsystem fault terms that are not shown in Figure 26 have, in general, small estimation errors. The variable that has the biggest error is λ_2, as expected, given that it has a sharp step-kind of fault event (breakdown due to stuck shovels).

Table 4 – MSE of the estimation. The other terms have values close to zero.

	λ_1	λ_2	λ_4	λ_5	λ_{11}	λ_{12}
EKF	0.0036	0.0427	0.0014	0.0003	0.0018	∼0.0
MHE	0.0012	0.0483	0.0007	0.0011	∼0.0	0.0012

Source: Lima et al. (13).

5.4 CONCLUSION

The estimation of faults provided by the MHE can quantify well the state of the monitored plant. With this estimation scheme, one knows in real-time if a given subsystem presents faults and, overall, how much does the fault in a given device lead to a loss of its effectiveness. Moreover, the MHE is able to directly compute all faults terms, by solving a single constrained optimization problem. There is no need for redundant sensors or a bank of observers as in Morato et al. (75). In comparison to the EKF, the MHE always respected the bounds of fault terms, which makes it coherent with the purpose of this work.

6 EARLY FAULT DETECTION IN WINE FERMENTATION

Context. The results from this chapter were achieved during an internship in the engineering research group of the Center for Research and Innovation of Concha y Toro Winery in Talca, Chile. The results were compiled to a scientific article titled "Advancing Wine Fermentation: Extended Kalman Filter for Early Fault Detection" written by B. M. Lima, R. Luna, D. M. Lima, J. E. Normey-Rico, and J. R. Pérez-Correa. Currently, the manuscript is under review in Food and Bioproducts Processing.

Alcoholic fermentation, which involves the conversion of grape sugars to ethanol and carbon dioxide by yeasts, is a crucial step in winemaking. This complex microbial process requires optimal conditions to complete successfully; otherwise, it can lead to problematic fermentations with significant economic losses. Factors that affect fermentation include toxicity due to high ethanol, low pH, extreme temperatures, and the limitation of key nutrients such as nitrogen and oxygen (79).

The most common problems encountered in industrial wineries are *stuck* fermentations, characterized by a high level of residual sugar remaining at the end, and *sluggish* fermentations, marked by a low rate of sugar consumption (80). These problems result in reduced productivity and, in the worst-case scenario, all of the raw materials are lost. These consequences highlight the need to develop techniques for early detection of problematic batches and investigate the best intervention strategies.

Online monitoring is seen as a promising method for improving the winemaking process, as it can support the enologist in decision making during fermentation and allows the adoption of novel control techniques (10). This requires, first of all, the implementation of sensors to measure variables related to fermentation rate such as density or CO_2 production.

There are several data-driven methods for monitoring in the process industry such as Principal Component Analysis (PCA), Neuro-Fuzzy Systems, and Artificial Neural Networks (ANN) (81). This class of methods requires a large amount of data for building black-box predictive models, but limited knowledge about the underlying process is needed. In contrast, model-driven methods are based on first principles equations describing the chemical and physical phenomena of the process (82). Although these methods usually assume several simplifying conditions, they are useful for describing the internal variables of the process that are not directly measured. These two approaches can also be used together in the development of hybrid models that could result in better prediction and extrapolation using the same amount of training data (83). For instance, Luna et al. (84) developed a hybrid model that integrates a mechanistic model for alcoholic fermentation with a machine learning algorithm for predicting the extraction of polyphenols in red wines. Another example is Coleman; Fish; Block (85) which models the wine fermentation with Michaelis-Menten style kinetics, but uses polynomials for the reaction rate terms to fit

the data.

Data-driven approaches for monitoring and fault detection of wine fermentation can be found in the literature. Urtubia et al. (86) employed PCA and cluster analysis (K-means) of data from 24 industrial Cabernet Sauvignon fermentations to detect anomalous behaviors in advance, using data on sugar, alcohol and organic acids. In this case study, it was possible to predict the likelihood of the process ending normally using only data from the first three days of fermentation. In later studies, the authors compared the prediction provided by multiway PCA, multiway partial least squares, and ANN on the same data, achieving 100 % accuracy (87). Despite achieving good results, the methods rely on measurements that are difficult to acquire in an industrial setting, such as alcohol and organic acids, limiting their applicability.

Among model-driven methods, nonlinear observers and filters, such as the Extended Kalman Filter (EKF), are among the most popular (20). In Bárzaga-Martell et al. (88), the authors combine an EKF with an asymptotic observer for the estimation of biomass and ethanol from high-density yeast cultures using measurements of dissolved oxygen, CO_2 and glucose. The proposed observer was able to accurately predict the switch between different metabolic regimes, which is an important sign of overflow metabolism. Sipos (89) develop an EKF for wine fermentation considering measurements of temperature and CO_2 to estimate biomass and sugar concentrations. However, different models were used for the decay and exponential phases, increasing the complexity of the implementation.

Kalman filtering is also used for *fault detection*, where the fault is incorporated into the model equation and estimated along with the states (90). For example, the process can be monitored for operational problems or component degradation. When coupled with some kind of uncertainty propagation, the EKF can also be used for *fault prognosis*, where the future status of the process is predicted (22). When the EKF equations are used for propagation of the mean and covariance of the estimation, it corresponds to a linear approximation of the "true" uncertainty, which can be innacurate for nonlinear systems (91). Nevertheless, it has a low computational load and its implementation is simple compared to other methods such as Monte Carlo simulations and polynomial chaos expansion.

This work proposes a monitoring strategy for wine fermentation with two components: an EKF for online estimation; and a predictor of faulty fermentations at the end of the batch. Unlike previous studies, the proposed methodology combines the advantages of EKF with a first-principles model to obtain a robust algorithm for monitoring and fault prediction in wine fermentation processes. The goal is to provide accurate information in real time for the winemaker to support his decision making. The proposed method, alongside the simulation studies, result in the following contributions:

a) Application of EKF for early detection of problematic wine fermentations based on final sugar prediction;

b) Accurate prediction available online using only temperature and density measurements.

c) Analysis of the main measurements affecting the prediction of the final sugar concentration;

d) Analysis of the performance and reliability of the proposed method based on Monte Carlo scenarios.

This chapter is organized as follows: in Section 6.1, the fermentation process and EKF for fault detection are explained in details; in Section 6.2.1 a case study on white wine fermentation is presented, followed by Monte Carlo scenarios on white and red wine in Section 6.2.2; in Section 6.2.2.1 the effects of measuring initial nitrogen is analyzed; finally, the results are discussed in sections 6.2.3 and 6.3.

6.1 METHODS

After some notation, the wine fermentation process and the dynamic model considered in this work are discussed in detail. Then, the proposed state estimation and fault detection algorithms are presented.

6.1.1 Preliminaries and Notation

A continuous dynamic nonlinear model can be represented by:

$$\begin{aligned} \dot{x} &= f_c(x(t),\, u(t),\, p) \\ y(t) &= h(x(t)) \end{aligned}$$

which has dimensions $x \in \mathbb{R}^{n_x}$ for the states, $u \in \mathbb{R}^{n_u}$ for the control inputs, $p \in \mathbb{R}^{n_p}$ for the parameters, and $y \in \mathbb{R}^{n_y}$ for the measured outputs. $f_c(\cdot)$ and $h(\cdot)$ represent nonlinear functions for the states and outputs, respectively. This model can be transformed to discrete time by an integration method or by an ODE solver, resulting in:

$$\begin{aligned} x(k+1) &= f(x(k),\, u(k),\, p) \\ y(k) &= h(x(k)) \end{aligned} \tag{64}$$

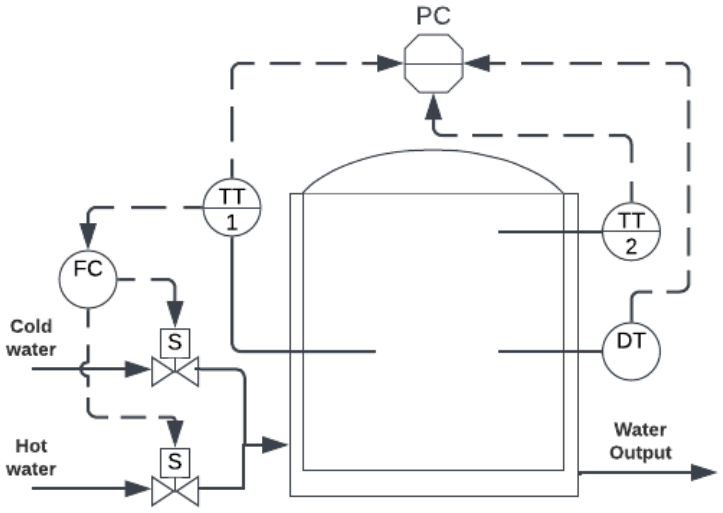
where $t = t_s k$ with t_s as the sampling time and k as the discrete sample. This notation for discrete-time variables can be abbreviated to $x_k = x(k)$. Finally, in the estimation context, $\hat{x}$ represents the estimate of x and $\hat{x}_{k|k\text{-}1}$ represents $\hat{x}_k$ given the information at time $k\text{-}1$. For matrices, A' represents the transpose of A.

6.1.2 Wine Fermentation

The fermentation starts when the tank is inoculated with yeast, usually *Saccharomyces cerevisiae*, and ends when there is no sugar left, i. e. the wine is dry. This takes

around 6-10 days operating around 25 °C for red wines and 12-20 days at 15 °C for white wines (92). A diagram of the fermentation tank can be seen in Figure 27 which shows the temperature control using service water and sensors for temperature and density. Specifically for red wines, the juice is fermented in contact with the solids in a process called fermentative maceration which finishes when the must is pressed and the wine is separated (draining point). This process can continue long after the wine is dry, i. e. extended maceration, depending on the style of wine (93).

Figure 27 – P&ID of the tank fermentation. TT1 and TT2 are temperature sensors for different sections of the tank. DT is a density sensor.



Source: Source: the author.

The management of nitrogen additions at the beginning and during the fermentation process is key since it is the major limiting nutrient for yeast growth (94). A suitable dosage during the process helps to synthesize secondary metabolites such as esters that contribute to the aromatic wine quality (95). Furthermore, the amount of nitrogen in the must is an important factor on fermentation kinetics: musts with higher nitrogen content lead to faster fermentations, whereas lower concentration results in slower fermentations. Inadequate levels of nitrogen can cause stuck or sluggish fermentations.

There are several sources of nitrogen available in grapes and musts, however, only some of them are easily assimilable by yeast such as ammonium and amino acids. Yeast assimilable nitrogen (YAN) is important because it is responsible for cell growth and metabolism, although it is expensive to measure accurately during fermentation (96). In addition, there may be significant variability in YAN in vineyards, increasing the uncertainty of the process (80).

To ensure the success of wine fermentation, most industrial wineries have invested in automation technologies such as temperature control and automatic pump-overs, among others. In the last decade, affordable and reliable density sensors have been developed, which allow online monitoring of the fermentation progress (97). Density measurements

based on differential pressure or production of CO_2 can be used to infer the sugar content in the fermenting must (98). In this work, it is assumed that sugar (S) and density (ρ) are related by the following linear equation:

$$\rho = \frac{100}{250} S + 1000 \tag{65}$$

with sugar in g/L and density in kg/m^3. Furthermore, temperature (T) is assumed constant and known.

6.1.3 Fermentation Model

Most wine fermentation models are described by a set of nonlinear ODEs with specific assumptions and purposes, as reviewed in Miller; Block (99). In Coleman; Fish; Block (85), a simple lumped parameter model, assuming nitrogen as the primary limiting substrate and temperature-dependent kinetic parameters, was developed. This model is the one used in this work, with a few adjustments.

The continuous model equations are described by:

$$\dot{X} = \mu X - k_d\, E X, \qquad \mu = \frac{\mu_{max} N}{K_N + N}, \tag{66a}$$

$$\dot{N} = -\mu \frac{X}{Y_{X/N}}, \tag{66b}$$

$$\dot{E} = \beta X, \qquad \beta = \frac{\beta_{max}\, S}{K_S + S}, \tag{66c}$$

$$\dot{S} = -\beta \frac{X}{Y_{E/S}}, \tag{66d}$$

where the state variables are given by the biomass (X), nitrogen (N), ethanol (E) and sugar (S), while the other terms are given by the maximum specific growth rate (μ_{max}), maximum specific rate of sugar utilization (β_{max}), half-velocity constant for nitrogen (K_N), half-velocity constant for sugar (K_S), death rate by ethanol (k_d), the yield coefficient for cell mass grown per mass of nitrogen ($Y_{X/N}$) and yield coefficient for ethanol produced per sugar consumed ($Y_{E/S}$). These terms related to the kinetics, summarized in Table 5, are modeled as exponentials of polynomials up to order 2 dependent on temperature T or initial nitrogen N_0.

The nominal values for the polynomial coefficients and their standard deviations are shown in Table 6. The standard deviations for the parameters were taken from the 90 % credible region (CR) reported in Coleman; Fish; Block (85) divided by four, i. e. it is assumed that the distribution is normal and centered around the mean with $\sigma = \mathrm{CR}/4$.

6.1.4 Extended Kalman Filter

Extended Kalman filter (EKF), is a recursive algorithm that uses the past measurements and inputs to estimate the current state based on a statistical model. For estimation,

Table 5 – Model parameters for each term in (66).

Parameter	Polynomial
$\log \mu_{max}$	$p_0 + p_1 T$
$\log K_N$	p_2
$\log k_d$	$p_3 + p_4 T + p_5 T^2$
$\log Y_{X/N}$	$p_6 + p_7 N_0$
$\log Y_{E/S}$	p_8
$\log \beta_{max}$	$p_9 + p_{10} T$
$\log K_S$	p_{11}

Source: Source: Coleman; Fish; Block (85).

Table 6 – Nominal values ($\bar{p}$) and standard deviations ($\bar{\sigma}$) of the parameters. Values in bold are different from Coleman; Fish; Block (85) since they appear to be typos.

	$\bar{p}$	$\bar{\sigma}$
p_0	-3.92	0.11
p_1	7.82 e-2	0.45 e-2
p_2	-4.73	0.14
p_3	-9.81	0.42
p_4	**-1.08 e-1**	0.40 e-1
p_5	4.78 e-3	0.86 e-3
p_6	3.50	0.098
p_7	**-3.61 e-3**	0.36 e-3
p_8	-5.98 e-1	0.19 e-1
p_9	**-2.98**	0.12
p_{10}	7.71 e-2	0.55 e-2
p_{11}	2.33	0.16

Source: Adapted from Coleman; Fish; Block (85).

the model given by (64) is augmented with noises on the state and output equations

$$\begin{aligned} x_{k+1} &= f(x_k,\, u_k,\, p,\, w_k) \\ y_k &= h(x_k) + v_k \end{aligned} \tag{67}$$

with $w \in \mathbb{R}^{n_w}$ and $v \in \mathbb{R}^{n_y}$ being random vectors modeled by zero-mean Gaussian noises with covariance matrices Q and R, respectively. For simplicity of the design, they are often modeled as non-correlated variables resulting in diagonal matrices containing the variance for each term. Furthermore, $h(x_k)$ corresponds to the sugar measurement, which is simply the fourth state plus v_k.

Specifically for the fermentation model in (66), the noise w_k is modeled as an additive term in the parameters as in $\bar{p} + w_k$, where $\bar{p}$ and Q defined by the values in Table 6. With this design choice, the model used for estimation in this work is as similar as possible as the one reported in Coleman; Fish; Block (85). It is common practice to include additive noise in the states when designing Kalman filters. However, they were not included in this work since it would require the addition of arbitrary tuning parameters for the EKF in the form of variance for each noise term.

The filter is initialized by two components: the expected mean $\hat{x}_0$ and the expected covariance P_0. After that, the filter performs two steps at each sampling time: the *prediction* step and the *update* step. At first, an *a priori* state estimate is calculated by (67). At the same time, the state noise propagates through the model resulting in an *a priori* covariance $P_{k|k\text{-}1}$. The prediction equations are given by

$$\hat{x}_{k|k\text{-}1} = f(\hat{x}_{k\text{-}1|k\text{-}1},\, u_{k\text{-}1},\, \hat{p}) \tag{68a}$$

$$P_{k|k\text{-}1} = A_{k\text{-}1}\, P_{k\text{-}1|k\text{-}1} A'_{k\text{-}1} + G_{k\text{-}1} Q\, G'_{k\text{-}1} \tag{68b}$$

where $A_{k\text{-}1}$ and $G_{k\text{-}1}$ are linearization of the nonlinear model as in

$$A_{k\text{-}1} = \left.\frac{df}{dx}\right|_{\hat{x}_{k\text{-}1},\, u_{k\text{-}1},\, \bar{p}}, \qquad G_{k\text{-}1} = \left.\frac{df}{dp}\right|_{\hat{x}_{k\text{-}1},\, u_{k\text{-}1},\, \bar{p}}.$$

Notice that $G_{k\text{-}1}$ is a derivative to $\bar{p}$ and not w_k since, in this work, the noise appears as $\bar{p} + w_k$. Furthermore, there is no time index in $\bar{p}$ since it is assumed to be constant and equal to the nominal values.

Next, the update step is performed. First, the Kalman gain K_k is calculated using $P_{k|k\text{-}1}$, C_k and R, where C_k is the linearization of $h(\hat{x}_{k|k\text{-}1})$. Then, using K_k, the estimates $\hat{x}_{k|k\text{-}1}$ and $P_{k|k\text{-}1}$ are updated according to the following equations:

$$K_k = P_{k|k\text{-}1} C'_k (C_k\, P_{k|k\text{-}1} C'_k + R)^{-1} \tag{69a}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k\text{-}1} + K_k \left(y_k - h(\hat{x}_{k|k\text{-}1})\right) \tag{69b}$$

$$P_{k|k} = (I - K_k C_k) P_{k|k\text{-}1} \tag{69c}$$

Real-world process constraints add additional challenges to the design of state estimators for nonlinear systems. For instance, all the states in model (66) must be nonnegative. In the EKF algorithm, there are two steps which can result in invalid values: the prediction of $\hat{x}_k$ in (68a), and the correction of the estimation in (69b). In this paper, it is used a simple strategy of clipping negative values to zero (5).

Remark 6.1. As can be seen in Table 5, the yield coefficient $Y_{X/N}$ depends on the initial value of N. Hence, for a proper use of the model, the EKF should include online estimation of initial nitrogen, i. e. $\hat{N}_{0|k}$. Nevertheless, in order to simplify the design of the filter, this value is fixed at $\hat{N}_{0|0}$ in this work.

6.1.5 Uncertainty Propagation

One of the challenges in nonlinear systems is that there is no easy way to propagate the covariance matrix P through the model. The EKF attempts to solve this using the nonlinear model for the estimate of $\hat{x}$ and using a linearized model to calculate the uncertainty matrix P and the Kalman gain K at each sampling time, as can be

seen in (68b) and (69c). Because of the linearized model for the covariance, EKF might have poor performance for highly nonlinear systems (5). Nevertheless, the EKF is a straightforward algorithm to generate predictions of the states $\hat{x}_{t_f|k}$ and their uncertainties $\hat{P}_{t_f|k}$ by sequentially applying (68) until t_f.

Although useful, EKF is not accurate for estimation or uncertainty propagation, specially for severe nonlinearities in the model. Therefore, a more accurate method, such as Monte Carlo (MC) simulations, is required for validation (22). For the fermentation model, the uncertainty at each sampling time is given by the covariance matrix $P_{k|k}$ for the current state estimate $\hat{x}_{k|k}$, and the covariance matrix Q for the parameter vector p. Using the MC method, a large enough number of scenarios (M) are generated with $\hat{x}_{k|k}$ and p as central points, and $P_{k|k}$ and Q as covariance matrices. Next, each scenario given by a set of x_k^i and p^i, is simulated by ODE solver until t_f resulting in $\mathbf{X}_{t_f} = [x_{t_f}^0, x_{t_f}^1, \ldots, x_{t_f}^M]$, while statistical properties can be calculated such as the median, mean $\mu_{t_f} = \mathrm{E}[\mathbf{X}_{t_f}]$ or covariance $P_{t_f} = \mathrm{cov}(\mathbf{X}_{t_f}, \mathbf{X}_{t_f})$. Then, this information can be compared to the result of the EKF algorithm in order to assess its accuracy.

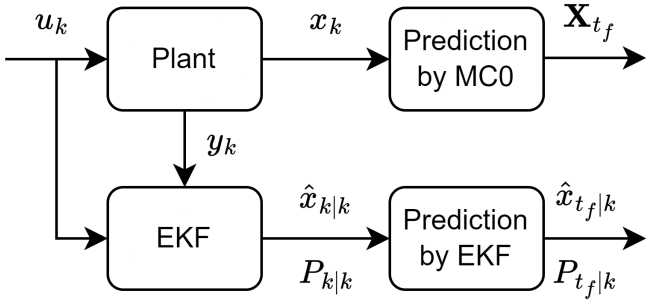
A variation of the MC method applied in this work considers only the pair (p, Q) for the scenario generation and assumes knowledge of the current state x_k. This method, referred here as MC0, captures the uncertainty in the prediction caused only by the parameters of the model, removing the effects due to the estimation error of $\hat{x}_k$.

6.1.6 Proposed Method for Fault Prediction

The proposed method for monitoring and fault prediction, visualized in Figure 28, considers an EKF as an online observer and uses its prediction equations for uncertainty propagation. At each sampling time, the EKF uses the current measurement y_k and the input u_k to update the prediction for $\hat{x}_{k|k}$ and its uncertainty $P_{k|k}$. With this information, the prediction equations for EKF are repeated until t_f resulting in $\hat{x}_{t_f|k}$ and $P_{t_f|k}$. Consequently, at each sampling interval t_s, it is possible to evaluate the probability of a failure occurring at t_f from those values. At the same time, for validation, the MC0 method is applied to the x_k considering only the uncertainty in p. The result from MC0 represents the best achievable performance for the EKF since it removes the effects of estimation error in $\hat{x}_k$.

Using the information provided by the above methods, it is possible to evaluate, at each t_s, the likelihood of a fault occurring in the future. For the purpose of this paper, which focuses on stuck and sluggish fermentations, a fault is defined as a sugar concentration at t_f above a desired value. For example, a fault could be defined as $S(t_f) \geq 2\,\mathrm{g/L}$ which corresponds to a dry wine, or could be higher for musts that are drained before the end of alcoholic fermentation, which is common for varietal wines (14). With this definition, an alarm for a possible fault could be raised at any time when the prediction is above the threshold.

Figure 28 – Block diagram summarizing the proposed method for monitoring and fault prediction based on the EKF, and MC0 running in parallel for validation.



Source: Source: the authors.

6.2 RESULTS AND DISCUSSION

Simulation studies written in Python are performed to evaluate the performance of the proposed method for wine fermentation. The software CVODES is used as the ODE solver interfaced by CasADi for simulation and for calculating the sensitivities required by the EKF (100, 101).

The first case study focuses on the performance of the EKF for a specific scenario by considering the nominal parameter values. The second case study is composed of numerous MC scenarios where the overall performance of the algorithm is evaluated. Then, the MC scenarios are repeated considering measurement for initial nitrogen. Finally, the results are discussed in further detail in the last section.

6.2.1 Illustrative Scenario

This simulation study, implements the EKF for fault detection in the nominal case for a white wine scenario with validation of the prediction by the MC0 method.

The sampling time used is $t_s = 6\,\text{h}$ with $t_f = 240\,\text{h}$ or 10 days. The desired sugar concentration at t_f is $10\,\text{g/L}$ with a final value higher than that representing a problematic fermentation. The fermentation temperature of $15\,^{\circ}\text{C}$ is constant and known. It is assumed that measurement for sugar is available, and that it is disturbed by a zero-mean Gaussian noise with $\sigma_y = 5^2$. The initial values for the states are $0.2\,\text{g/L}$ for biomass, $N_0 = 150\,\text{mg/L}$, $S_0 = 265\,\text{g/L}$ and zero for ethanol. Both the system and the estimator have the same nominal parameters given by $\bar{p}_i$ and $\bar{\sigma}_i$ (Table 6). This scenario results in $S(t_f) = 26.4\,\text{g/L}$ which is a sluggish fermentation.

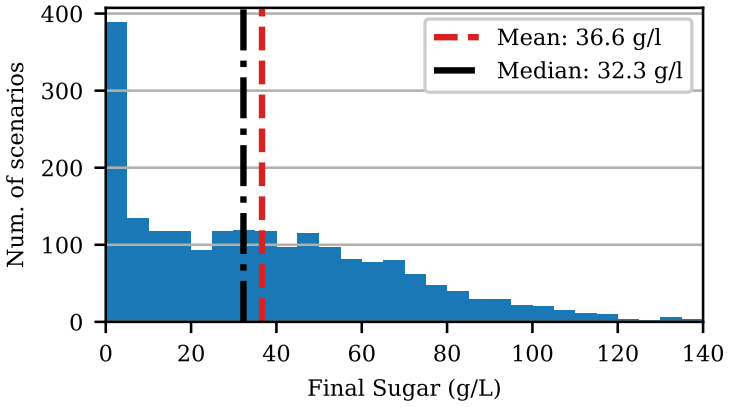
Meanwhile, the initial estimate for the EKF is the same as the process with the exception of nitrogen, which is significantly different with $\hat{N}_0 = 150\,\text{mg/L}$. The initial uncertainty P_0 considered in the estimator is given by

$$P_0 = \text{diag}\left(0.025^2,\, 0.050^2,\, 0,\, 5^2\right), \tag{70}$$

with units in grams per liter. Notice that the standard deviation of 50 $^{mg}/_{L}$ for nitrogen in the second term is large considering the nominal value, and σ for ethanol is zero, since it is certain that there is no alcohol at t_0.

Figure 29 shows an histogram of the prediction of the sugar at the end of the fermentation process, $S(t_f)$, using the MC0 method with $M = 2049$. Although MC0 does not consider the uncertainty in the states, the simulated scenarios result in a wide range for $S(t_f)$, showing that this system is highly sensitive to the parameters. A large portion of the scenarios end successfully with almost no sugar left, but there is a significant tail which increases the mean and median. The histogram is not bell-shaped and, because of that, the standard deviation and the mean of the scenarios may be biased.

Figure 29 – Histogram of $S(t_f)$ using the MC0 method for the nominal model with $n_{MC} = 2049$ and no uncertainty in $x(0)$. For this scenario, $S(t_f) = 26.4\,^{g}/_{L}$.



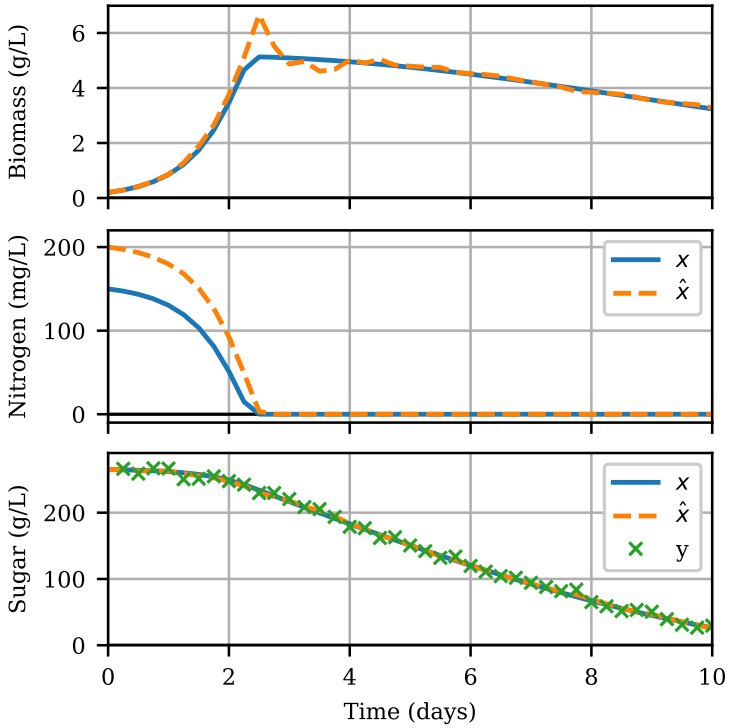
Source: Source: the authors.

Figure 30 shows the estimation of the states by the EKF. At first, there is a large error in initial nitrogen, but after the lag phase of the fermentation is over, the filter is able to correct the estimated states, resulting in an error close to zero in steady-state.

Figure 31 shows the prediction of final sugar $S(t_f)$ by the EKF and MC0 method during the 10 days of fermentation. The EKF starts with an optimistic initial estimate of 200 $^{mg}/_{L}$ for nitrogen and, because of that, the prediction for $\hat{S}(t_f)$ is zero for the first 60 hours. After the lag phase of the fermentation ends and the measurement for sugar starts changing, the filter quickly updates the prediction. This sudden increase in the predicted value for $S(t_f)$ way above the threshold of 10 $^{g}/_{L}$ could be interpreted as an alarm of problematic fermentation. After a few more iterations, the prediction converges to the final value around 26 $^{g}/_{L}$ confirming the problematic fermentation.

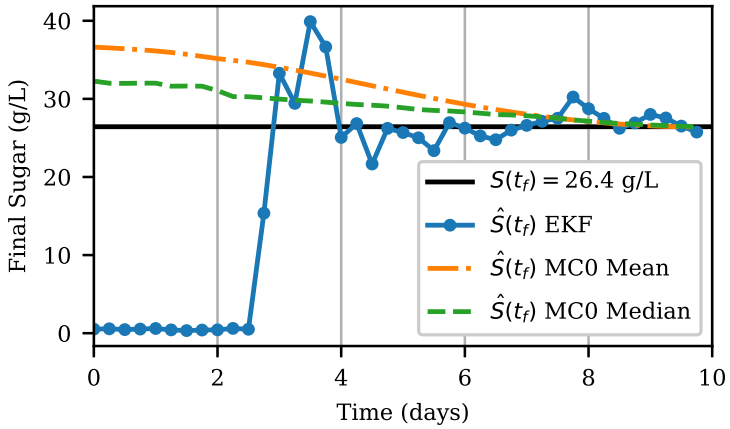
Figure 31 also shows two values for the predictions by MC0: the mean and the median. It can be seen that the mean of the MC scenarios is less reliable since the distribution is not normal, resulting in an overprediction of the final sugar. Overall, the MC0 method using the median is more accurate compared to the EKF. However, it still has a considerable error in $S(t_f)$ in the first few days, even though no uncertainty is considered for x_k. This

Figure 30 – Estimation of the states by EKF for a white wine scenario resulting $S(t_f) = 26.4\,\text{g/L}$. Estimation for ethanol is omitted for brevity.



Source: Source: the authors.

Figure 31 – Prediction of the sugar at t_f by EKF and by MC0 (median and mean) compared to the final value of $26.4\,\text{g/L}$.



Source: Source: the authors.

shows that the best achievable performance for the EKF, represented by MC0, can still be quite inaccurate. Despite that, the EKF was still able to generate an alarm in a reasonable time.

6.2.2 Monte Carlo Scenarios

This section shows Monte Carlo analysis of the proposed EKF for two cases: White Wine (WW) and Red Wine (RW). Each scenario is simulated as described in Section 6.2.1, but using different initial conditions and parameters for the model. Each scenario is defined by the variables described in Table 7 with $t_f = 10$ days for WW and 5 days for RW. The desired sugar at t_f for both cases is 10 g/L. For each scenario, the temperature is constant for the whole batch with 12-18 °C for whites and 22-28 °C for reds. Scenarios were generated using the Sobol method from SciPy with $M = 1024$ plus the nominal scenario, totaling 2050 simulations for both cases. Meanwhile, the EKF is initialized with the same information: nominal N_0 and $\bar{p}$ from Table 7, and P_0 from (70). For 2050 scenarios, 1026 scenarios presented faults considering the threshold of 10 g/L, with 876 being sluggish fermentation and 150 being stuck.

Table 7 – Variables' range for each scenario.

Variable	Nominal	Distribution
N_0	200 mg/L	Uniform [50, 350]
S_0	265 g/L	Uniform [250, 280]
Temp. WW	15 °C	Uniform [12, 18]
Temp. RW	25 °C	Uniform [22, 28]
p	$\bar{p}$	Normal (see Table 6)

Source: Source: the authors.

As shown in the study case of Section 6.2.1, the filter takes a few days to converge when the initial estimation error is large. This happens around the same time the fermentation leaves the lag phase and starts the exponential phase with a faster conversion of sugar to ethanol. Since EKF depends heavily on sugar measurements, it is expected that it will only update its model after the process provides meaningful information, which happens after the fermentation leaves the lag phase. This effect results in a large prediction error for the final sugar on the first days for both RW and WW.

Because of that, it is important to ignore the alarms generated before some minimum time has passed such as $t_{min} = 24$ h for RW and 48 h for WW. With this criteria, the *first* alarms produced by EKF are summarized in Table 8.

Table 8 – Confusion matrix for the first alarm generated by EKF for both WW and RW cases.

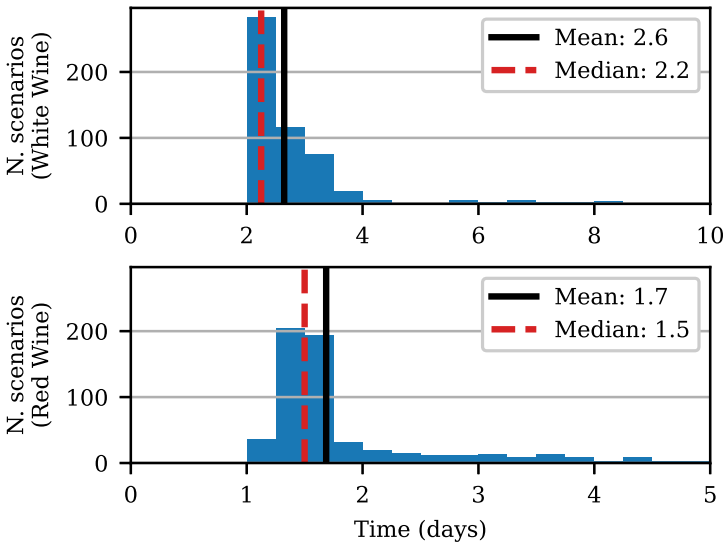
Total: 2050	Alarm: 1103	No Alarm: 947
Fault: 1026	True Pos.: 1010	False Neg.: 16
No Fault: 1024	False Pos.: 93	True Neg.: 931

Source: Source: the authors.

The algorithm was successful in 1941 scenarios (94.7 %), where 1010 were scenarios with faults and 931 with no faults or alarms. There were 93 false alarms, which is where the algorithm expects a fault that will not occur. However, since only the first one is analyzed in each scenario, it is expected that the following alarms were more accurate in the prediction. More concerning are the 16 scenarios where the EKF failed to generate an alarm for a faulty batch. In spite of that, the final sugar for these cases was close to the threshold with a median of 11.7 g/L and a maximum of 17.6 g/L.

Figure 32 shows histograms of the time of the *first* alarm generated by the EKF for each scenario. For the WW case, most alarms were generated in the first 2-3 days with few outliers. Meanwhile, for the RW case, most alarms were generated between the first and second days with a heavier tail. This histogram shows that it is harder to predict the fermentation for RW probably because of the short duration. However, in both cases, this histogram shows that the algorithm is quite reliable in providing early alarms to the enologist in time for acting on the process.

Figure 32 – Histogram of time of the first alarm generated by EKF in each scenario, separated by WW (top graph) and RW (bottom graph) cases.



Source: Source: the authors.

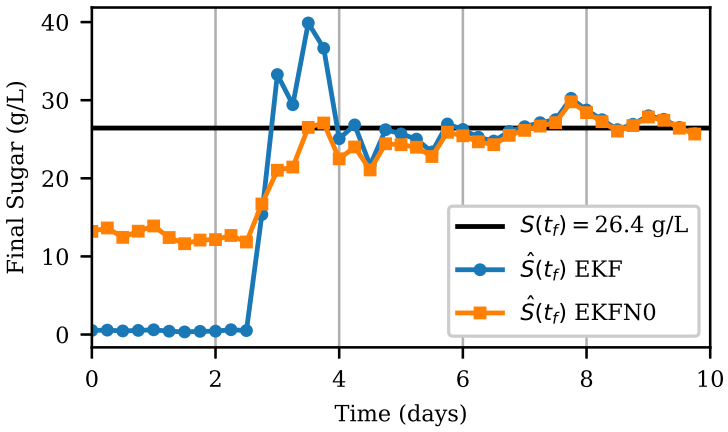
For example, in industrial wineries it is common to add nitrogen a second time to ensure a successful process in terms of the productivity and quality. This decision is carried out typically when the density reaches the range of 1080-1050 kg/m^3 that coincides with the time range of the first alarm generated (14). Since the number of false positives and false negatives is small, the information is reliable enough to decide how much nutrient should be added.

6.2.2.1 Measuring Initial Nitrogen

This section explores the effects of measuring nitrogen at the beginning of fermentation. All of the MC scenarios of the last section are repeated with two differences. First, the initial variance for nitrogen in P_0 is reduced from 0.050^2 to 0.015^2 which corresponds to a higher confidence on $\hat{N}_0$. This represents a measurement error for nitrogen around $\pm 30\,^{\mathrm{mg}}/_{\mathrm{L}}$ which is still significant considering the range for the variable. Second, at the beginning of each scenario, $\hat{N}_0$ is initialized as N_0 plus a zero-mean Gaussian noise. This new algorithm is named EKFN0.

Figure 33 shows the prediction of $\hat{S}(t_f)$ by EKFN0 for the scenario of Section 6.2.1 with $\hat{N}_0 = 165\,^{\mathrm{mg}}/_{\mathrm{L}}$ compared to EKF with $\hat{N}_0 = 200\,^{\mathrm{mg}}/_{\mathrm{L}}$. When measurement for initial nitrogen is included, the prediction error for $\hat{S}(t_f)$ decreases significantly. Despite being far from the real value, the prediction by EKFN0 in the first two days is much more accurate. In this scenario, an accurate alarm would be available even before the fermentation starting (considering the threshold of $10\,^{\mathrm{g}}/_{\mathrm{L}}$).

Figure 33 – Prediction of the sugar at t_f by EKF and EKFN0 compared to the final value of $26.4\,^{\mathrm{g}}/_{\mathrm{L}}$.



Source: Source: the authors.

The performance of EKFN0 in generating alarms is very similar to that of EKF, as reported in the confusion matrix of Table 8. Almost all of the scenarios with false positives and false negatives for EKF resulted the same for EKFN0. The main difference was the time of the first alarm for EKFN0. While the EKF gives the results shown in the histogram in Figure 32, EKFN0 generated 87 % of the alarms for WW and 80 % for RW on the first day of fermentation, with a few outliers. As summarized in Table 9, the mean time of the first alarm generated was reduced in 73 % for WW and 53 % for RW. This is a significant improvement which could justify the investment in nitrogen measurement in some cases.

Table 9 – Time for the first alarm generated by EKF and EKFN0 for both WW and RW cases.

		First Alarm (days)	
Case		Mean	Median
EKF	WW	2.6	2.2
EKFN0	WW	0.7	0.0
EKF	RW	1.7	1.5
EKFN0	RW	0.6	0.0

Source: Source: the authors.

6.2.3 Discussion

The white wine case study in Section 6.2.1 showed that the EKF is able to adequately estimate unmeasurable states, such as active biomass. This can help the winemakers on deciding how to act on the process. For example, if the amount of biomass is not sufficient, it can be helpful to add more nitrogen so that it can grow. But if there is enough biomass but the fermentation is slow, maybe the temperature can be increased.

There are two main sources of uncertainty that lead to estimation error. First of all, there is the problem of uncertainty in the models' parameters that is inherent to the system. The histogram in Figure 29, with the predicted sugar concentration ($S(t_f)$) at the end of fermentation, highlights the sensitivity of the system to the parameters. This aspect can only be improved by developing models that better represent the underlying phenomena and by performing experiments to collect data. This can reduce the uncertainty in the model and would improve its prediction accuracy.

The second aspect is the use of effective algorithms for estimation and prediction. It is known that EKF can show poor performance in some cases, especially for high nonlinearities and constraints (5). One reason for that is the use of a first-order method for uncertainty propagation which decreases accuracy and can lead to a poor estimation (102). Although the EKF performed well in this work, it is expected that more advanced methods for uncertainty propagation such as the sigma points or polynomial chaos expansion will lead to better estimation and fault prediction (22).

It is possible to use different methods for estimation and fault prediction. The EKF for estimation could be substituted by the Unscented Kalman Filter for better performance or by the Moving Horizon Estimation for the use of constraints, although the design process for these methods is more involved. Meanwhile, the uncertainty propagation for faults could be done by the MC method, with the downside of increased computational cost. These possibilities are worth exploring.

Even though the proposed algorithm can be improved, it performed well on the MC scenarios showing robustness under diverse conditions. Providing an accurate prediction of final sugar proved to be a challenge on the first few days of fermentation. The decrease in prediction error around 36 to 48 hours corresponded to the transition from the lag phase to the exponential phase, indicating the EKF's reliance on meaningful sugar measurements

for effective updates. Still, it was early enough for the process to be acted upon.

The histogram of the time of the first alarm of each scenario (Figure 32) provided insight into the reliability of the EKF in providing early alarms. Most of the alarms are concentrated around the average which means that, if an alarm will be generated, it is expected to occur around the end of the lag phase. However, for shorter fermentations, more outliers are expected.

The first alarm generated by the algorithm is fairly accurate with 1010 scenarios (91.6 %) being true positive and 93 (8.4 %) false positive, as seen in Table 8. To further increase the reliability of these alarms, one option is to act on the system only after a few alarms are generated in sequence, which decreases the likelihood of a false alarm. Furthermore, the sampling time could be reduced so that more measurements are available. This happens because the EKF has more information available for a better estimation and prediction. Another important aspect is to use the uncertainty $P_{t_f|k}$ alongside $\hat{x}_{t_f|k}$ in the fault definition. With this, an alarm would only be generated when $\hat{S}(t_f)$ is above a threshold and the uncertainty is low enough.

The last case study shows the benefits of measuring initial nitrogen. At least for the model considered, total nitrogen is crucial since it is directly related to peak biomass which, in turn, is necessary for successfully converting sugar to ethanol. Measuring nitrogen at the beginning, even with a fairly large error, improves the reliability of the proposed algorithm, although it leads to cost in materials and labor. However, the cost could be avoided by first running the proposed algorithm without measuring N_0 and analyzing the results. If the prediction leads to a successful fermentation with a high degree of certainty, then the measurement could be skipped for that batch.

6.3 CONCLUSION

In this study, a monitoring strategy based on EKF for state estimation and early fault detection is proposed in order to provide accurate information during fermentation. This model-based method can support the enologist in detecting problematic batches and designing intervention strategies, while also enabling advanced automatic control which often requires knowledge of all the states of the process. The algorithm was tested in diverse simulation scenarios in order to access performance and usefulness.

Analysis based on MC simulations showed how the process is sensitive to uncertainty on model parameters and initial conditions, especially for nitrogen. It also showed how the estimation is dependent on meaningful measurements for sugar which are not available during the lag phase of the fermentation. Despite these challenges, the main goal of generating accurate early alarms is achieved.

There are at least two aspects of the proposed algorithm which can be improved. First, more accurate models will result in better estimation, which leads to better fault prediction. Another one is the use of a more accurate uncertainty propagation method for

faults. Further analysis could also indicate if effort should be concentrated on designing better estimation and prediction methods, or if it is better to gather more data for developing accurate models.

Apart from the technical aspect, it is important to improve the performance analysis of the algorithm since the study was conducted only on simulations. For example, the EKF could be applied to data from industrial batches to check if the conclusions drawn here still hold.

The presented findings contribute to the ongoing efforts in leveraging advanced control and monitoring techniques for optimizing wine fermentation processes (10). In an industrial setting, the estimation could be used alongside an automatic system for nutrient addition which would autonomously decide how much nitrogen to add in order to avoid problematic batches. It also enables the use of online model-based controllers such as predictive control. The proposed EKF approach, along with the conducted simulation studies, pave the way for future research in winemaking and analogous processes.

7 OPTIMAL CONTROL OF RED WINE MACERATION

Context. This chapter summarizes the first published results in optimal control of red wine maceration from the engineering research group of the Center for Research and Innovation of Concha y Toro Winery in Talca, Chile. The results were published in:

- (14) R. Luna, B. M. Lima, J. Cuevas-Valenzuela, J. E. Normey-Rico, and J. R. Pérez-Correa, "Optimal Control Applied to Oenological Management of Red Wine Fermentative Macerations," *Fermentation*, vol. 7, no. 2, p. 94, Jun. 2021.

In red winemaking, the must is fermented in contact with grape pomace and seeds to extract several important phenolic compounds such as anthocyanins and tannins that define the color and mouthfeel properties of red wine, respectively (103). This process is known as fermentative maceration and generally begins with the grape crushing process (104). The extraction kinetics of these phenolic compounds depend on several variables such as the solid-liquid phase's contact area, their molecular structure (skins and seeds), maceration duration, temperature, and solvent composition (105). Therefore, the alcoholic fermentation operation is a significant factor contributing to wine quality.

At industrial-scale wineries only some of these variables can be controlled. The fermentation temperature is usually controlled by a closed-loop control manipulating the refrigerant's flow rate that circulates through the tank's jacket. Winemakers control phenolic extraction comprising homogenization of must and skins by defining the maceration duration and the pump-over program. The maceration process finishes when the solids are separated from the fermenting juice known as the digging-out point. Overall, winemakers define this point when a specific sugar content or must density is achieved. This criterion is handled through wine taste, wine color, grapes quality and the management of the winery capacity.

The operation strategy should be adapted season to season to the uncertainties and vintage shortening caused by climate change (106). Global warming has led to problems in the planning, management and productivity of the winery. In particular, the winery's management is limited to the fermentation tanks' capacity and availability on a specific date. Consequently, winemakers frequently change the original fermentation recipe and take new decisions during the fermentation such as reducing the maceration duration or increasing the fermentation rate by adding diammonium phosphate (DAP). Optimal handling of maceration and DAP additions could improve wineries' capacity and reduce production costs without sacrificing wine´s quality and typicity.

Optimal control and MPC can achieve optimal fermentation operating strategies that simultaneously optimize efficiency, economic indexes and product quality. The winery operation can be adapted to unexpected events during the vintage season by applying these techniques providing faster and more reliable solutions than experimental trial and

error (107). The process model's reliability and predictive capacity are crucial to guarantee a robust performance of these techniques.

Most wine fermentation models are described by a set of nonlinear differential and algebraic equations (DAEs) with specific assumptions and purposes, as reviewed in Miller; Block (99). In Coleman; Fish; Block (85), it was developed a simple lumped parameter model assuming nitrogen as the primary limiting substrate and temperature-dependent kinetic parameters. In Cerda-Drago; Agosin; Pérez-Correa (108), it was considered a spatial distribution to describe oxygen incorporation and dissolution during the white wine fermentation. In Zenteno et al. (109), it was developed a compartment model to simulate the temperature gradients at an industrial scale. It was also developed a three-dimensional industrial red wine fermentation model using the finite element method to simulate the temperature gradient and their influence on anthocyanins and tannins extraction rates (110, 111). The cited studies focused on replicating experimental fermentation operating under similar conditions. Rarely new operating conditions are explored by optimization techniques. Model based-design could allow exploring the impact of new operating conditions such as fermentation temperature profile or nutrients additions during fermentation.

Overall, MPC has been widely applied in the pharmaceutical and biochemical industries. Nevertheless, the applications of MPC to fermentation tanks in the wine industry are scarce. One of the most recent work developed a MPC to minimize the cooling energy in white wine fermentation (112). The MPC strategy reduced the energy consumption by 52% compared to an industrial controller without decreasing the wine quality.

Model-based optimal operation has been increasingly applied in the process industries since the arrival of the Industry 4.0 paradigm (113). The higher levels of automation and digitalization allow mathematical models to transform process data into valuable information for online automatic high-level decision-making (114). The beverage and food industry have adopted these technologies at a slower rate than other process industries such as mineral, chemical and pharmaceutical production plants (115). To the author's knowledge, the application of MPC and the development of optimal strategies for industrial red wine fermentations have not been tackled yet.

There are very few applications of model-based optimal decision-making in the alcoholic beverage industry. For instance, in Rodman; Gerogiorgis (116) it is applied a dynamic Multi-Objective Optimization (MOO) for the beer fermentation process. The optimization problem looked for an optimal temperature profile that simultaneously maximize the final ethanol concentration and minimize the fermentation time. In Luna; López; Pérez-Correa (117), it was developed a dynamic MOO problem to design optimal distillation recipes for muscat wine spirits. Moreover, they used Multi-Criteria Decision-Making (MDCM) algorithms to select different kinds of muscat spirits recipes base on objective prioritization. These applications made evident that model-based optimal operation and MDCM techniques could be valuable tools for managing the wine process.

Hence, the hypothesis is that using an model-based optimal control and MCDM

techniques it is possible to improve process efficiency and costs compared with standard operation without sacrificing products quality. In this work, an Optimal Control Problem (OCP) for red wine fermentation is developed using a reparametrized fermentation kinetic model (118). The OCP considered two oenological objectives: i) sugar target concentration at the digging-out point and ii) minimization of the total DAP additions throughout fermentation. The decision variables were the fermentation temperature profile and the DAP addition during the fermentative maceration.

A dynamic OCP was formulated and solved by applying a weighting approach. Two nutrition strategies were evaluated: free nutrition on each sampling time and traditional nutrition that considers two additions, one at the beginning and another at the middle of the fermentation process. Then, the optimal strategies were selected using MCDM algorithms.

The main contributions of this work are:

a) Optimal control for managing industrial bulk red winemaking;

b) Weighting approach for solving OCP with oenological requirements;

c) MCDM algorithms for helping the decision-making process.

This chapter is organized as follows: A general explanation of the dynamic fermentation model is introduced in Section 7.1. The optimal control problem formulation, the nutrition strategies and MCDM algorithms are presented in Section 7.2. Finally, the results and the conclusions are highlighted in Section 7.3 and Section 7.4, respectively.

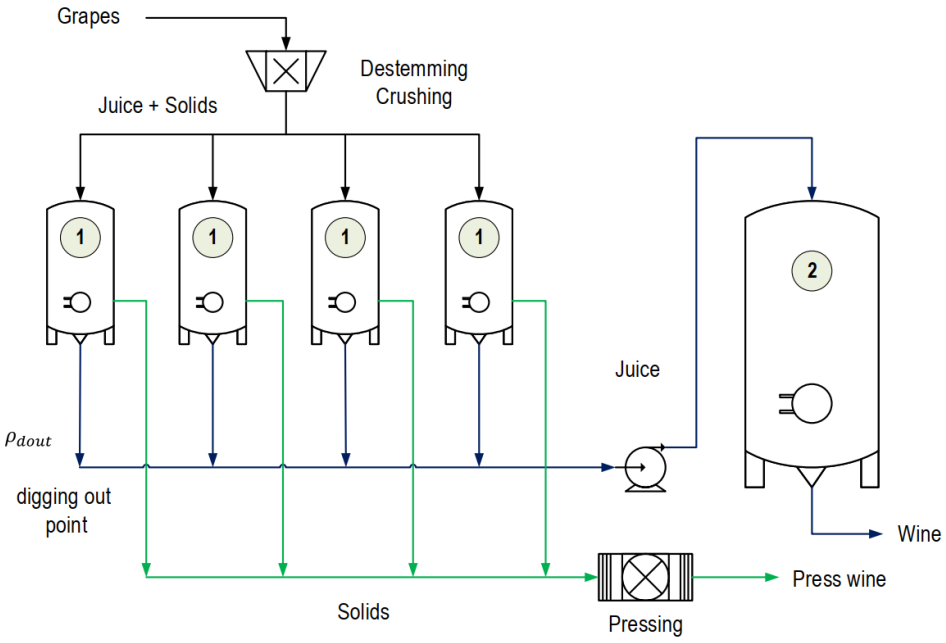
7.1 FERMENTATION MODEL

Red wine fermentations involve simultaneous fermentation and solid-liquid extraction, hence, it is also called fermentative maceration. In commercial wineries, the maceration duration is typically regulated to achieve a specified level of phenolics. Short macerations are preferred for varietal or bulk wines while longer macerations (extended macerations) are favored for premium wines (119). The focus of this work is the former where the maceration finish before the fermentation is completed.

Figure 34 shows the process scheme for industrial red winemaking for bulk wines. The process is divided into fermentative maceration (FM) and fermentation completion (FC). The FM stage is carried out in first set of tanks until the digging-out point, where the specific sugar content is achieved. At this point, the solids are separated from the fermenting must and the liquids from the different tanks are transferred into a bigger tank to complete the fermentation.

The kinetic fermentation model developed in Zenteno et al. (109) was used to design the optimal operation strategies. The model was simplified by removing the compartment equations so that the temperature gradients were not considered (118). The nitrogen addition and temperature setpoint were added as model inputs since they are the decision

Figure 34 – Scheme for industrial bulk red winemaking: fermentative maceration (1) and alcoholic fermentation completion (2). The blue line represents the juice fed at the digging-out point to complete the alcoholic fermentation, and the green line represents the solids (pomace) fed to produce press wines.



Source: Luna et al. (14).

variables of the formulated optimal control problem. The CO_2 state variable was removed from the model since it did not affect other state variables. The model equations are represented by:

$$\begin{aligned} \frac{dx}{dt} &= f\left(x(t),\, u(t),\, \theta\right) \\ \rho(t) &= h\left(G(t),\, F(t)\right) \end{aligned} \tag{71}$$

where x, u and θ correspond to the state variables, model input vector and model parameters vector, respectively. The state variables are biomass X, nitrogen N, glucose G, fructose F, and ethanol E. Total sugar S includes glucose and fructose. The model inputs u are the nitrogen addition using DAP (N_{dap}) and the fermentation temperature T. The function ρ corresponds to the must density and was calculated through a correlation with total sugar S obtained using experimental fermentation data. Table 10 shows the initial values of the state variables.

The model parameters θ were estimated using a specifically designed robust reparametrization method for red wine fermentations which yields an optimal model structure characterized by given free and fixed parameters. The model parameters values and model equations are available in Torrealba et al. (118) and Zenteno et al. (109),

Table 10 – Initial conditions for states variables.

Variable	Symbol	Value
Biomass	$X(0)$	$0.200\,g/L$
Nitrogen	$N(0)$	$0.150\,g/L$
Glucose	$G(0)$	$105\,g/L$
Fructose	$F(0)$	$105\,g/L$
Ethanol	$E(0)$	$0\,g/L$

Source: Luna et al. (14).

respectively. The differential equations are given by:

$$\frac{dX}{dt} = [\mu(T) - k_d(T,\, E)]\, X \tag{72a}$$

$$\frac{dN}{dt} = -\frac{\mu(T)}{Y_{XN}} X \tag{72b}$$

$$\frac{dG}{dt} = -\left[\frac{\mu(T)}{Y_{XG}} + \frac{\beta_G(T,\, G,\, E)}{Y_{EG}} + \frac{G}{G+F} m(T)\right] X \tag{72c}$$

$$\frac{dF}{dt} = -\left[\frac{\mu(T)}{Y_{XF}} + \frac{\beta_F(T,\, G,\, F,\, E)}{Y_{EF}} + \frac{F}{G+F} m(T)\right] X \tag{72d}$$

$$\frac{dE}{dt} = [\beta_G(T,\, G,\, E) + \beta_F(T,\, G,\, F,\, E)]\, X \tag{72e}$$

where the parameter dependency on the states were highlighted. The specific growth rate μ is given by

$$\mu(T) = \mu_{max}(T) \frac{N}{N + K_N(T)}, \qquad \mu_{max}(T) = \mu_0 \exp\left(E_{ac} \frac{T - 300}{300\, R\, T}\right).$$

The biomass decay rate k_d is given by

$$k_d(T,\, E) = k_{d0} \exp\left(C_{de}\, E + E_{td} \frac{T - 305.65}{305.65\, R\, T}\right),$$

where it is valid only when a thermal death threshold T_D is passed,i. e. $T > T_D(E)$, with

$$T_D(E) = -0.0001\, E^3 + 0.0049\, E^2 - 0.1279\, E + 315.89.$$

On the glucose and fructose equations, the specific cell maintenance rate m is given by

$$m(T) = m_0 \exp\left(E_{am} \frac{T - 293.3}{293.3\, R\, T}\right)$$

and the ethanol production rate β for each one are given by

$$\beta_G(T,\, G,\, E) = \beta_{Gmax}(T) \frac{G}{G + K_G(T)} \frac{K_{IE}(T)}{E + K_{IE}(T)},$$

$$\beta_F(T,\, G,\, F,\, E) = \beta_{Fmax}(T) \frac{F}{F + K_F(T)} \frac{K_{IG}(T)}{G + K_{IG}(T)} \frac{K_{IE}(T)}{E + K_{IE}(T)},$$

with maximum ethanol production rates β_{max} given by

$$\beta_{imax}(T) = \beta_{i0} \exp\left(E_{afe} \frac{T - 296.15}{296.15\, R\, T}\right), \qquad i = G,\, F.$$

The inhibition parameters K_i present in many terms are given by

$$K_i(T) = K_{i0} \exp\left(E_{aki}\frac{T - T^*}{293.3\,R\,T}\right), \qquad i = N,\, G,\, F,\, IG,\, IE.$$

The other parameters not defined have their valued fixed.

7.2 OCP AND MCDM FORMULATION

The objective function used considers the process efficiency and costs during fermentative maceration. The process efficiency is associated with oenological requirements such as sugar level at the end of the fermentative maceration stage (digging-out point). The total amount of DAP added along the maceration defines this stage's operating costs. The cost function J is given by:

$$J = \alpha\frac{\left(S(t_{dout}) - S_{sp}\right)^2}{100} + (1-\alpha)\frac{\sum_{i=1}^{n} N_{DAP,i}}{0.200} \tag{73}$$

where α is a relative weight (positive scalar ≤ 1) defining the objective's priority and $N_{dap,i}$ is the nitrogen added at time i considering a sample time T_s of 6 h. This approach is known as the *weighting method* that turns the multi-criteria objective into a traditional single-objective problem (120, 121). The first objective is the desirable sugar concentration at the digging-out point. Here, S_{sp} is the sugar setpoint defined by the winemaker at the digging-out point and t_{dout} is the digging-out time. The second objective is the minimization of the total nitrogen addition.

Both objectives were normalized to get comparable values. The sugar set point weight was defined by trial and error, and the second term weight ($0.200^{\mathrm{g}}/_{\mathrm{L}}$) corresponds to approximately the maximum DAP dose of $96^{\mathrm{g}}/_{\mathrm{hL}}$ allowed by the Chilean law, considering that it is composed of 21 % nitrogen (122). Finally, grid values for α were defined ($\alpha = [0,\, 0.02,\, 0.04, ...,\, 1]$) resulting in 51 optimization problem runs.

In this study the digging-out point is fixed as five fermentation days ($t_{dout} = 120$h) with $S_{sp} = 11^{\mathrm{g}}/_{\mathrm{L}}$ of sugar concentration (equivalent to $995^{\mathrm{kg}}/_{\mathrm{m}^3}$).

The Optimal Control Problem (OCP) finds the manipulated variables profile (nitrogen addition and fermentation temperature) that minimizes the objective function and satisfies the constraints. The optimization problem was discretized using a direct collocation method which simultaneously solves the simulation and optimization problem (123). This method discretizes the control and state variables over a time horizon using collocation points on finite elements (124). Then, the OCP was reformulated as a nonlinear

programming problem (NLP) by:

$$\begin{aligned}
\min \quad & \alpha\frac{(S(t_{dout})-S_{sp})^2}{100} + (1-\alpha)\sum_{i=1}^{n}\frac{N_{DAP,i}}{0.200} \\
\text{s.t.} \quad & x_{i,j} = x_{i-1} + h_i\sum_{j=1}^{m}\Omega_j(\tau_j)\frac{dx}{dt_{i,j}} \\
& \frac{dx}{dt_{i,j}} = f(x_{i,j},\, y_{i,j},\, u_{i,j},\, \theta) \\
& 0 = g(x_{i,j},\, y_{i,j},\, u_{i,j},\, \theta) \\
& u_L \leq u_{i,j} \leq u_U \\
& x_L \leq x_{i,j} \leq x_U \qquad\qquad \forall i = 1, ..., n \\
& y_L \leq y_{i,j} \leq y_U \qquad\qquad \forall j = 1, ..., m
\end{aligned} \tag{74}$$

where n is the number of finite elements, m is the number of collocation points, h_i is the length of the finite element (T_s = 6h) and Ω is the interpolation polynomial functions for each finite element. 20 finite elements over a time horizon of 120h and five Legendre collocation points were used. The simultaneous simulation/optimization problem was coded in Python using the CasADi package with the optimization solver IPOPT (101, 125).

Two control strategies were compared. The first considers the fermentation temperature variation and nitrogen additions on each T_s as decision variables. This strategy has more degrees of freedom and, because of that, is called free nutrition (FN). The second strategy, called traditional nutrition (TN), considers temperature variation on each sampling time but with only two nitrogen additions; at the beginning of the process (t = 0h) and at t = 48h, which corresponds to the start of the yeast stationary growth phase (1070-1050$^{kg}/_{m^3}$) (126, 127). In both cases, the fermentation temperature path was constrained to the interval 20-30$°C$ while total nitrogen addition should be lower than 0.200$^{g}/_{L}$.

The solution of the multi-criteria problem is a set known as non-dominated solutions or Pareto Front, which is represented as an objective matrix f with n rows (one row for each Pareto solution) and m columns (one column for each objective). Here, the dimension of the objective matrix is n = 21 and m = 2. The next step is to select one of the non-dominated solutions where MCDM algorithms have been developed for this purpose (128).

In this work, simple MCDM algorithms were evaluated to select an optimal solution: TOPSIS, LINMAP, SAW, MEW and FUCA. These algorithms require a weight w for each objective defined by the user. For this application, a compromise among the objectives are needed. These solutions are known as "knee" points where a significant improvement in some objectives means insignificant degradation in the other objectives (129, 130). Therefore, the weights of the MCDM algorithms were set with equal priority (w = [0.5, 0.5]) for both objectives. Moreover, the optimal solutions selected by the MCDM algorithms were compared with the knee point to evaluate the performance. The knee point was calculated as the minimum euclidean distance between each optimal solution d_i and the

utopian solution as shown on:

$$F_{i,j} = \frac{f_{i,j}}{\max_{i \in n}(f_{i,j})}, \qquad d_i = \sqrt{\sum_{j=1}^{m} \left(F_{i,j} - F_j^*\right)^2},$$

where $F_{i,j}$ is the normalized objective matrix and F_j^* are the coordinates of the utopian solution for each objective. Since both objectives correspond to a minimization criterion, the utopian point was zero, i. e. $F_{1,2}^* = [0, 0]$. The knee point is defined as the solution with the smallest d_i.

7.3 SIMULATION RESULTS

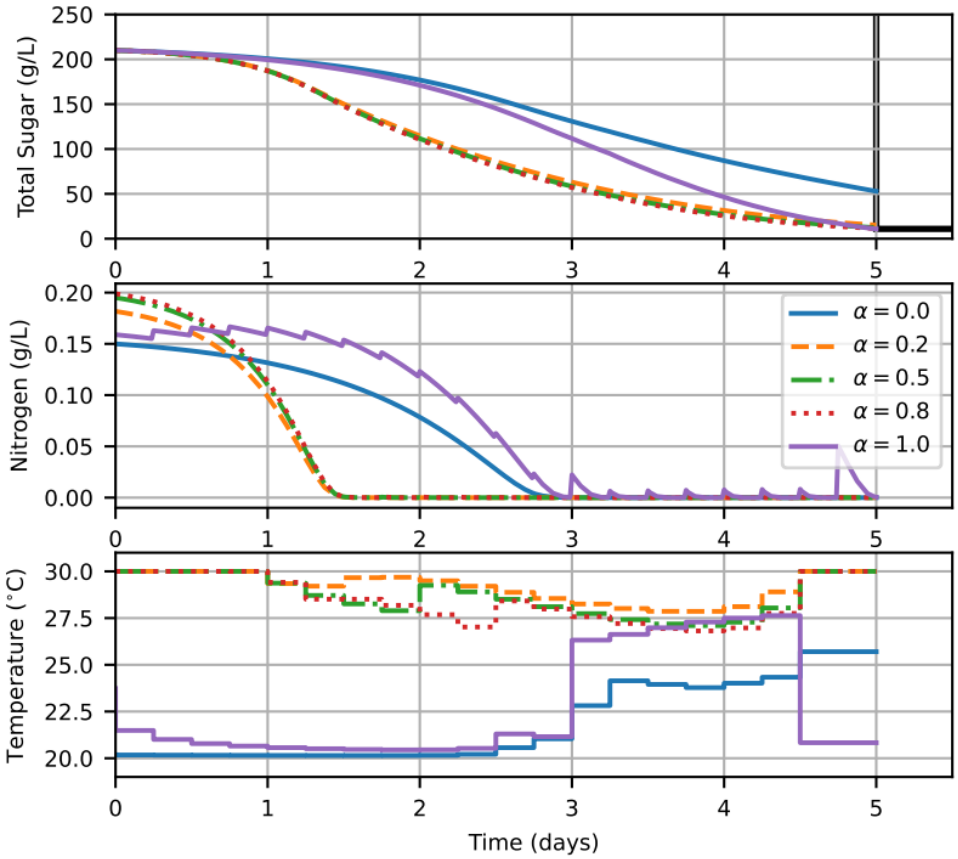
This section presents the obtained results and it is divided in three parts: (i) the free nutrition case study; (ii) the traditional nutrition case study, and (iii) comparative results of the most suitable strategies.

The simulations of optimal control for free nutrition strategies are shown in Figure 35 for different values of α. As expected, the strategy for $\alpha = 0$ (priority for minimization of nitrogen additions) did not add any DAP, but the sugar level at the end of the maceration process was far from the desired digging-out point (42 g/L sugar difference). On the other hand, the strategy with $\alpha = 1$ achieved at the end of the maceration the sugar concentration of the desired digging-out point and DAP was added at each sampling time, obtaining 94.3 g/hL of total DAP dose which near the allowed limit. Overall, the fermentation temperature profile was low for both strategies for the first three days and then increased near the end.

Intermediate α values (0.2, 0.5, 0.8) yielded similar strategies, characterized by nitrogen additions concentrated at the beginning of the fermentation with DAP doses of 15, 22, and 25 g/hL, respectively. Besides, the sugar differences from the desired digging-out point were small with 4.3, 1.45 and 0.43 g/L, respectively, representing an error smaller than 5%. In all cases, temperature profiles throughout the fermentation were characterized by high values ($> 27°C$).

Figure 36 shows the Pareto front composed of 51 solutions, the selected solutions by MCDM and the knee point. The solution chosen by TOPSIS and LINMAP was the same and coincided with the knee point. These algorithms are very similar and usually recommend similar or the same solution (128, 117). This optimal solution corresponds to the strategy with $\alpha = 0.08$. SAW and MEW are also similar approaches that often recommend the same solution (128, 117). Here, the solution chosen by SAW and MEW is the neighbor of the knee point ($\alpha = 0.06$). This solution added 3.6 g/hL of DAP at the beginning of the fermentation and the error from the digging-out point was 11.3 g/L. Finally, the FUCA method selected a solution ($\alpha = 0.02$) far from the knee point that prioritizes the minimization of nitrogen additions (no DAP) with 14 g/L of sugar difference from the desired digging-out point.

Figure 35 – Optimal control Simulations for free nutrition strategies. Black line in total sugar graph corresponds to the desired digging-out point.

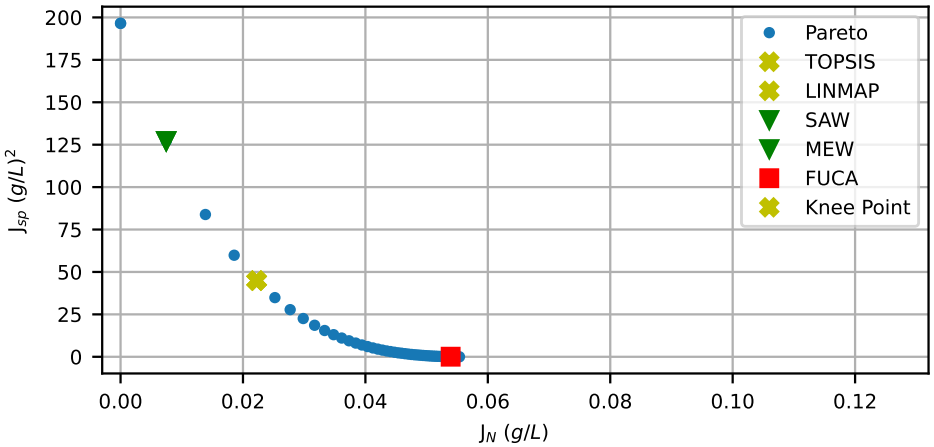


Source: Luna et al. (14).

Figure 37 shows the simulation of optimal control for traditional strategies for different values of α. Like the FN strategy, the TN strategy for $\alpha = 0$ did not add DAP and the sugar concentration achieved 53.4 g/L, i. e. 43.4 units of sugar difference from the desired digging-out point. Contrarily, the TN strategy with $\alpha = 1$ achieves the desired digging-out point (11 g/L of sugar) while the total DAP addition was 11.3 g/hL, (35.3 g/hL at the beginning and 23.4 g/hL at $t = 48h$). For both TN strategies, the temperature profile was very similar. The first 2.5 days the temperature was low (near the lower bound of $20°C$) and then linearly increased to $25°C$.

Contrarily to the FN strategies with α values 0.2, 0.5 and 0.8, the TN strategies with α in this range added nitrogen only at time $t = 48$h, with DAP doses of 44.6, 45.4 and 60.2 g/hL, respectively. Total DAP additions in TN strategies were higher than total DAP additions in FN strategies. Nevertheless, the sugar differences from the desired digging-out point (11 g/L) were similar to the FN strategies with 5.5, 1.7 and 0.5 g/L for $\alpha = [0.2, 0.5, 0.8]$, respectively. This difference is present because the fermentation temperature profile for TN strategies started with a lower temperature level than FN strategies and then increased until $30°C$ from day three. Instead, the temperature profiles for FN strategies were high all along the process. The TN strategies with $\alpha = [0.2, 0.5, 0.8]$

Figure 36 – Pareto front for free nutrition strategies, optimal solutions selected by MCDM algorithms and knee point. Points corresponding to $\alpha = 0$ and $\alpha = 1$ are omitted.



Source: Luna et al. (14).

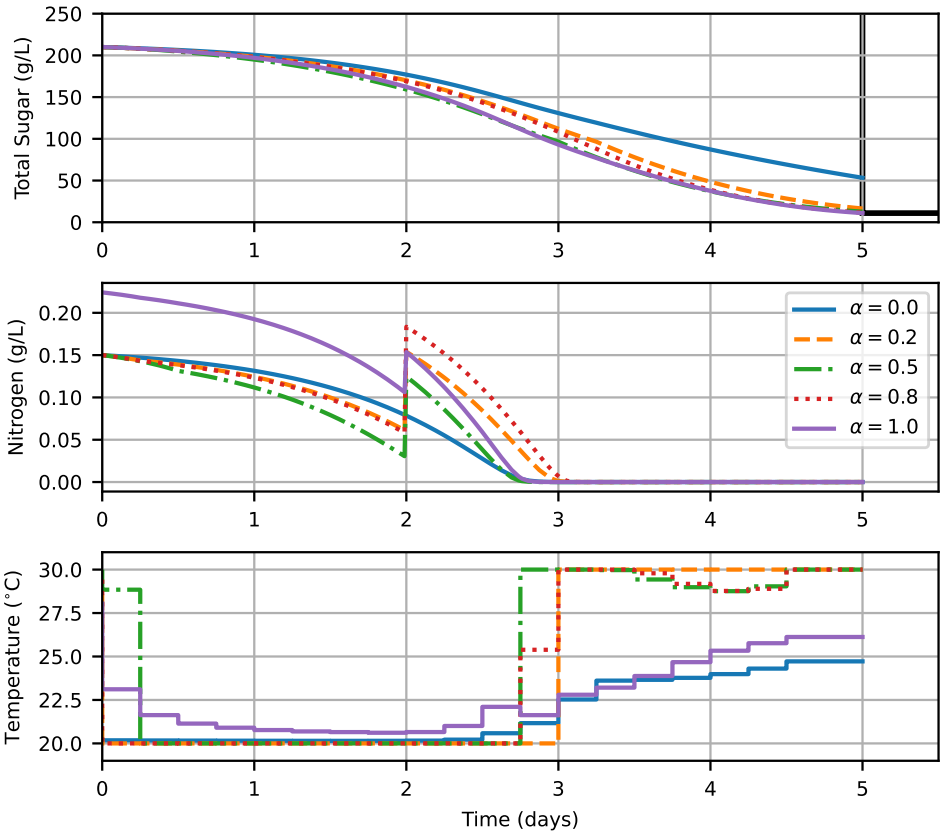
are ideal for white wines. In Pérez et al. (126), it is reported that lower fermentation temperatures and nitrogen additions in the middle of the fermentation obtained more aromatic wines with more concentration of higher alcohol acetates.

As was done in FN strategies, MCDM algorithms and the knee point were used to select a suitable compromise between both objectives. Figure 38 shows the Pareto front and the selected solutions by MCDM for TN strategies. The Pareto front shape was different from the obtained by the FN strategies (Figure 36). Here, the Pareto front shape was disconnected, whereas for FN strategies were convex (131). In addition, the solutions for TN strategies were more sensitive to intermediate α values, i. e. the solutions were more separated from each other.

Again, TOPSIS and LINMAP algorithms recommended the same solution which it coincides with the knee point. This optimal solution corresponds to the strategy with $\alpha = 0.12$. Meanwhile, SAW and MEW choose identical solutions with ($\alpha = 0.06$) different from the knee point, prioritizing the minimization of nitrogen additions (3.6 g/hL of DAP) with 11.3 g/L of sugar distance from the desired digging-out-point. Finally, the solution recommended by the FUCA algorithm ($\alpha = 0.88$) was far from the knee point. This alternative prioritized the minimization of the distance (0.28 g/L) from the sugar concentration at the digging-out point and added 50.6 g/hL of DAP.

Figure 39 shows the simulation of the optimal control solutions for FN and TN strategies on the knee point. Despite obtaining different Pareto front for FN and TN strategies, the solutions on the knee point for both strategies were very similar. The nitrogen addition was concentrated only at the beginning of the fermentation with DAP doses of 6.6 and 10.6 g/hL for FN and TN, respectively. Similarly, the sugar differences from the desired digging-out point were small with 9.16 and 6.7 g/L for FN and TN, respectively. Moreover, the fermentation temperature path was very similar for both

Figure 37 – Optimal control simulations for traditional nutrition. Black line in total sugar graph corresponds to the desired digging-out point.



Source: Luna et al. (14).

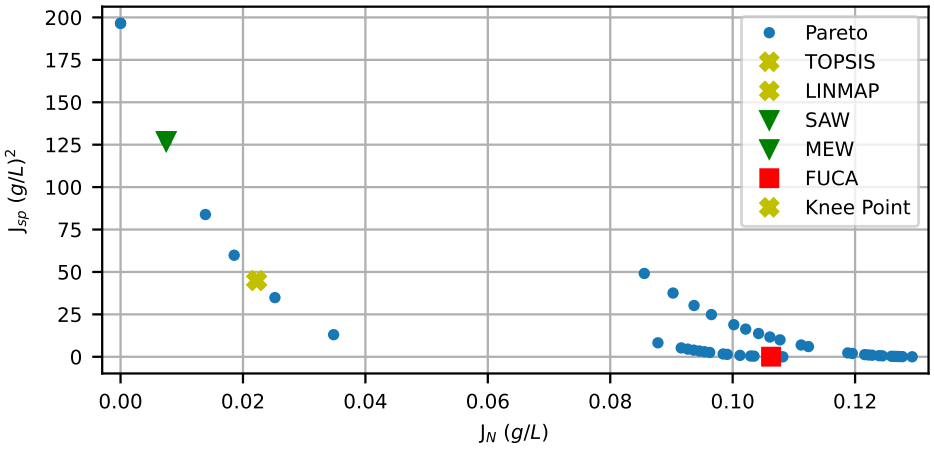
strategies characterized by high values ($> 27°C$) along the whole process. These strategies are ideal for red wine since high temperatures favor the extraction of phenolic compounds (105, 104).

At the knee point, both strategies converge to similar solutions with DAP addition at the beginning of the fermentation. Nevertheless, an analysis of the Pareto Fronts points to another direction. The TN has less degrees of freedom because it can only add nutrition at two opportunities. Because of this, it is harder to optimize the process which leads to more scattered solutions and higher DAP use in most cases, as can be seen in Figure 38. In contrast, the FN has a smoother Pareto Front and lower nitrogen use in most cases. These differences indicates that the FN strategy probably is more robust to changes in parameters of the model or the current objective function used. The robustness of the strategies could be properly analyzed in future works.

7.4 CONCLUSION

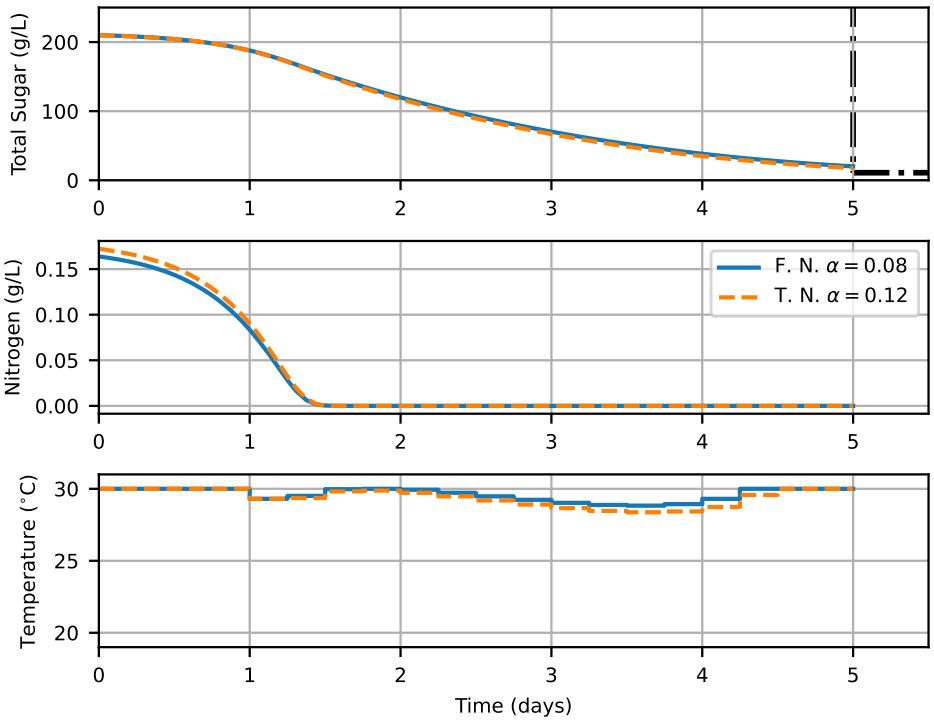
An optimal control was developed to manage the digging-out point to improve the process efficiency and costs simultaneously. A trade-off between the minimization of the sugar concentration error at the digging-out point and the minimization of nitrogen

Figure 38 – Pareto front with DAP addition at $t = 0$h and $t = 48$h. Total of 50 points with 0.05 step. The points corresponding to $\alpha = 0$ and $\alpha = 1$ are omitted.



Source: Luna et al. (14).

Figure 39 – Comparison of the chosen optimal control strategies. Black line in total sugar graph corresponds to the desired digging-out point.



Source: Luna et al. (14).

additions using DAP was identified. Then, five MCDM algorithms were assessed to choose an optimal strategy from the Pareto front using equal priorities for each objective. The selected solutions by MCDM were compared with the knee point. TOPSIS and LINMAP algorithms recommended identical solutions and coincided with the knee point for both free and traditional strategies.

Despite that Pareto fronts being different for FN and TN strategies, the solutions on the knee point were similar. The most suitable option was adding nitrogen only at the beginning of the fermentation (6.6-10.6 g/hL of DAP) and high fermentation temperatures (close to $30°C$), yielding the desired digging-out point with a small error (6-9 g/L). Thus, these strategies are a good option for red winemaking because they favor phenolic extraction.

The methods described in this study can be extended to include quality indexes such as aroma production in white wines or phenolic extraction in red wines and also adapted to any fruit wine fermentation. Finally, it is planned for future works to integrate a phenolic extraction model with the fermentation model used here. Then, a multi-objective optimization problem could be formulated including a quality index for the decision-making process.

8 CONCLUSION

This thesis presents contributions on predictors for time-delay systems, fault estimation and optimal control, with applications for industrial processes, including renewable microgrids and wine fermentation. The underlying theme is the use of model-based algorithms for monitoring and optimal operation of industrial processes.

The first contribution of this thesis is a predictor based on the Kalman filter for time-delay systems. The Kalman Predictor demonstrated advantages in handling various types of system dynamics and disturbances with its closed-loop stability and robustness analyzed using tools on the frequency domain. The results shown are significant but limited, since only linear SISO and square MIMO cases were studied. This work could be continued by exploring how the proposed predictor can handle general MIMO systems and nonlinear models. Some results for the nonlinear case were already shown in Fiori et al. (132), but they require further analysis.

In this work, model-based estimation was used for faults in two industrial processes. First, in Chapter 5, an MHE scheme for fault detection in renewable microgrids was designed considering multiple points of failure. The well established EKF was compared to the MHE with the latter showing better performance and robustness. The proposed method was further analyzed in Morato; Mendes; Normey-Rico (133) and an MPC using the MHE was presented in Bernardi et al. (134). Currently, further improvement of this industrial process is carried out by other researchers.

The second application of estimation in Chapter 6 showed that, apart from being able to detect problems that occurred in the past, it can also be used for estimating the likelihood of a fault occurring in the future. In this case, it was achieved by estimating the latent states of the fermentation in real-time and using them for predicting the residual sugar at the end of the batch. The usefulness of this scheme lies on providing the enologists with valuable information about the process in order to avoid problematic fermentations. The use of observers for this kind of bioprocess is crucial since many of the key variables of the model are not easily measurable. Because of that, estimation is an important component in any advanced control strategy based on first principles models.

One example of advanced control for red wine maceration is presented in Chapter 7 where a multi-objective optimization was proposed for handling the digging-out point and nutrient addition. Using numerous simulation scenarios, it was possible to try different cost functions which were latter analyzed using multi-criteria decision-making techniques. This study highlights the advantage using a representative model since it can be used for exploring innovative ways of operating the process with virtually no cost. This is important for winemaking since each fermentation batch last for several days and can only be done in a few months each year.

In future works, the integration of online monitoring and optimal control of wine

fermentation could be explored. With information provided by the observer in real-time, an MPC could be designed in order to act on the process without requiring intervention from the enologist. This would result in faster action in order to avoid stuck and sluggish batches. Furthermore, the same optimization framework of MPC could be used for including cost terms related to productivity and efficiency.

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